

Generalized Linear and Additive Models

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Practical informations

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Office hours: TBA.

Schedule:

Course: Tue 12h15 - 14h, room M2160

Exercises: Wed 12h15 - 14h, room M5290.

Exam:

Written exam, open book, 2 hours.

Software: R

Freely available from <http://www.r-project.org>.

Packages needed:

`mgcv`, `MASS`, `nnet`, `lme4`, `gee`, `robustbase`, `brglm`,
`ggplot2`, `aod`, `pscl`, `MuMIn`.

`mgcv`, `MASS`, `nnet`, `lme4` come with the default R installation.
The others have to be added manually.

Main bibliography

Generalized Linear Models (GLM)

- ▶ Dobson, A. (2002) *An Introduction to Generalized Linear Models*, Chapman & Hall.
- ▶ Lindsey, J. K. (1997) *Applying Generalized Linear Models*, Springer-Verlag.
- ▶ McCullagh, P. and Nelder, J. (1989) *Generalized Linear Models*, Chapman & Hall, 2nd ed.
- ▶ Faraway, J. (2006) *Extending the Linear Model with R*, Chapman & Hall.

Robust Statistics

- ▶ Hampel, F., Ronchetti, E., Rousseeuw, P. and Stahel, W. (1986) *Robust Statistics: The Approach Based on Influence Functions*, Wiley.
- ▶ Maronna, R., Martin, D. and Yohai, V. (2006) *Robust Statistics: Theory and Method*, Wiley
- ▶ Heritier, S., Cantoni, E., Copt, S. and Victoria-Feser M.-P. (2009) *Robust Methods in Biostatistics*, Wiley.

Longitudinal Data Analysis

- ▶ Diggle, P. J., Heagerty, P., Liang, K.-Y. and Zeger, S. L. (2002) *Analysis of Longitudinal Data*, Oxford University Press, 2nd ed.

Nonparametric Regression

- ▶ Green, P. J. and Silverman, B. W. (1994) *Nonparametric regression and Generalized Linear Models: A Roughness Penalty Approach*, Chapman & Hall.
- ▶ Hastie, T. and Tibshirani, R. (1990) *Generalized Additive Models*, Chapman & Hall.
- ▶ Simonoff, J. S. (1996) *Smoothing Methods in Statistics*, Springer-Verlag.
- ▶ Wood, S. (2017) *Generalized additive models: an introduction with R*, 2nd edition, Chapman & Hall.
- ▶ Keele, L. (2008) *Semiparametric regression for the social sciences*, Wiley

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Modeling excess of zeros in count data

Generalized Estimating Equations

Generalized linear mixed models (GLMM)

Robust GLM

Robust GEE

Robust hurdle model

Generalized additive models (GAM)

“Statistics is often wrongly perceived as a set of tools instead of a set of problems”

from J. H. Friedman (2001) “The Role of Statistics in Data Revolution” *International Statistical Review*, 69, 5-10.

Example 1 - Boston Housing

The Boston Housing Data have been used to describe the relationship between housing values in suburbs of Boston and different attributes.

The sample size is 506.

The data is originally from Harrison and Rubinfeld (1978), used in Belsley, Kuh, and Welsch (1980), available from the University of California at Irvine Repository Of Machine Learning Database (www.ics.uci.edu/~mllearn/MLRepository.html).

medv	median value of owner occupied homes in \$1000's
crim	per capita crime rate by town
zn	proportion of residential land zone for lots over 25'000 sq.ft
indus	proportion of non-retail business acres per town
nox	nitric oxides concentration (parts per 10 million)
rm	average number of rooms per dwelling
age	proportion of owner-occupied units built prior to 1940
dis	weighted distances to five Boston employment centres
rad	index accessibility to radial highways
tax	full-value property-tax rate per \$10'000
ptratio	pupil-teacher ratio by town
b	$1000(B_k - 0.63)^2$, where B_k is the proportion of blacks by town
lstat	proportion of the population that is lower status
chas	Charles River dummy variable (=1 if tract bounds river; 0 otherwise)

Table: Boston Housing Data description.

Model (1):

$$\begin{aligned}\log(\text{medv}) = & \beta_0 + \beta_1 \text{crim} + \beta_2 \text{zn} + \beta_3 \text{indus} + \\ & + \beta_4 \text{nox} + \beta_5 \text{rm} + \beta_6 \text{age} + \beta_7 \text{dis} + \beta_8 \text{rad} \\ & + \beta_9 \text{tax} + \beta_{10} \text{ptratio} + \beta_{11} \text{b} + \beta_{12} \text{lstat} \\ & + \beta_{13} \text{chas} + \epsilon.\end{aligned}\tag{1}$$

```
> linearfit <- lm(log(medv)~chas+crim+zn+indus+nox+rm+age+dis+  
  rad+tax+ptratio+b+lstat , data=BostonHousing)  
> summary(linearfit)
```

Coefficients :

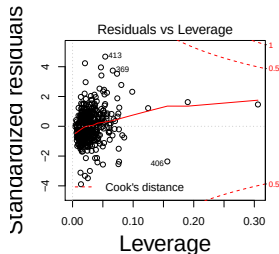
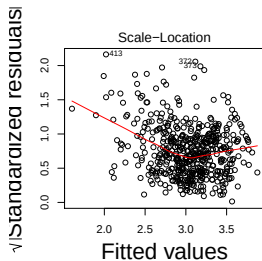
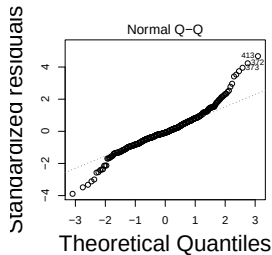
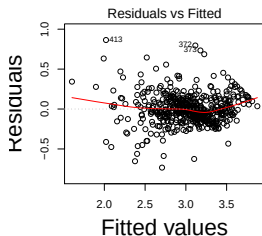
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.1020423	0.2042726	20.081	< 2e-16	***
chas	0.1008876	0.0344859	2.925	0.003598	**
crim	-0.0102715	0.0013155	-7.808	3.52e-14	***
zn	0.0011725	0.0005495	2.134	0.033349	*
indus	0.0024668	0.0024614	1.002	0.316755	
nox	-0.7783993	0.1528902	-5.091	5.07e-07	***
rm	0.0908331	0.0167280	5.430	8.87e-08	***
age	0.0002106	0.0005287	0.398	0.690567	
dis	-0.0490873	0.0079834	-6.149	1.62e-09	***
rad	0.0142673	0.0026556	5.373	1.20e-07	***
tax	-0.0006258	0.0001505	-4.157	3.80e-05	***
ptratio	-0.0382715	0.0052365	-7.309	1.10e-12	***
b	0.0004136	0.0001075	3.847	0.000135	***
lstat	-0.0290355	0.0020299	-14.304	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1899 on 492 degrees of freedom

Multiple R-squared: 0.7896, Adjusted R-squared: 0.7841

F-statistic: 142.1 on 13 and 492 DF, p-value: < 2.2e-16



Model (2) (inspired by Belsley, Kuh, and Welsch 1980):

$$\begin{aligned}\log(\text{medv}) = & \beta_0 + \beta_1 \text{crim} + \beta_2 \text{zn} + \beta_3 \text{indus} + \\ & + \beta_4 \text{nox}^2 + \beta_5 \text{rm}^2 + \beta_6 \text{age} + \beta_7 \log(\text{dis}) + \beta_8 \log(\text{rad}) \\ & + \beta_9 \text{tax} + \beta_{10} \text{ptratio} + \beta_{11} \text{b} + \beta_{12} \log(\text{lstat}) \\ & + \beta_{13} \text{chas} + \epsilon.\end{aligned}\tag{2}$$

```
> linearfit2 <- lm(log(medv)~chas+crim+zn+indus+l(nox^2)+  
  l(rm^2)+age+log(dis)+log(rad)+tax+ptratio+b+log(lstat),  
  data=BostonHousing)  
> summary(linearfit2)
```

Coefficients :

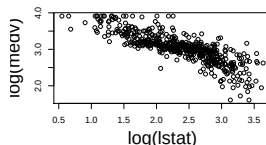
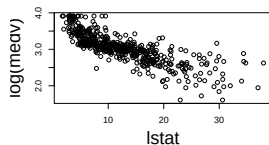
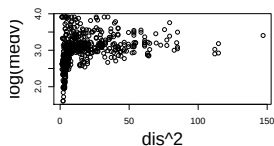
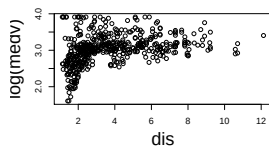
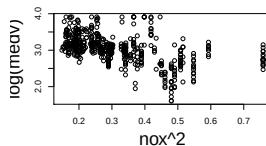
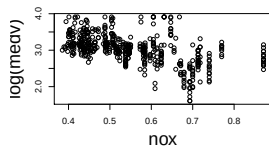
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.558e+00	1.544e-01	29.512	< 2e-16	***
chas	9.140e-02	3.320e-02	2.753	0.006129	**
crim	-1.186e-02	1.245e-03	-9.532	< 2e-16	***
zn	8.016e-05	5.056e-04	0.159	0.874105	
indus	2.395e-04	2.364e-03	0.101	0.919318	
l (nox^2)	-6.380e-01	1.131e-01	-5.639	2.88e-08	***
l (rm^2)	6.328e-03	1.312e-03	4.823	1.89e-06	***
age	9.074e-05	5.263e-04	0.172	0.863179	
log (dis)	-1.913e-01	3.339e-02	-5.727	1.78e-08	***
log (rad)	9.571e-02	1.913e-02	5.002	7.91e-07	***
tax	-4.203e-04	1.227e-04	-3.426	0.000664	***
p ratio	-3.112e-02	5.013e-03	-6.208	1.14e-09	***
b	3.637e-04	1.031e-04	3.527	0.000460	***
log (lstat)	-3.712e-01	2.501e-02	-14.841	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1

Residual standard error: 0.1825 on 492 degrees of freedom

Multiple R-squared: 0.8059, Adjusted R-squared: 0.8008

F-statistic: 157.1 on 13 and 492 DF, p-value: < 2.2e-16

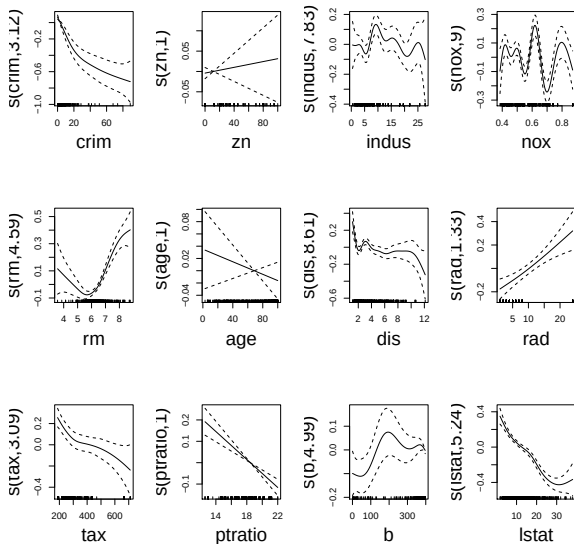


Model (3), nonparametric:

$$\begin{aligned} \log(\text{medv}) = & \alpha + f_1(\text{crim}) + f_2(\text{zn}) + f_3(\text{indus}) + \\ & + f_4(\text{nox}) + f_5(\text{rm}) + f_6(\text{age}) + f_7(\text{dis}) + \\ & + f_8(\text{rad}) + f_9(\text{tax}) + f_{10}(\text{ptratio}) \\ & + f_{11}(\text{b}) + f_{12}(\text{lstat}) + \beta \text{chas} + \epsilon. \end{aligned}$$

```
> library(mgcv)
> BHnpfit.new <- gam(log(medv)~chas+s(crim)+s(zn)+s(indus)+
  s(nox,k=20)+ s(rm)+s(age)+s(dis,k=20)+s(rad,k=8)+s(tax)+s(ptratio)+
  s(b)+s(lstat), data=BostonHousing)
> plot(BHnpfit.new, scale=0, page=1)
```

★ Variable selection by nonnegative garrote, see Breiman (1995) and Cantoni, Flemming, and Ronchetti (2011).



Example 2: Breast cancer diagnostic

Breast cancer diagnosis through breast cytology (fine needle aspiration instead of invasive surgery).

Breast cancer Wisconsin data, available from the University of California at Irvine Repository Of Machine Learning Database
www.ics.uci.edu/~mlearn/MLRepository.html.

Outcome: 2 classes: benign (coded 0) or malignant (coded 1).

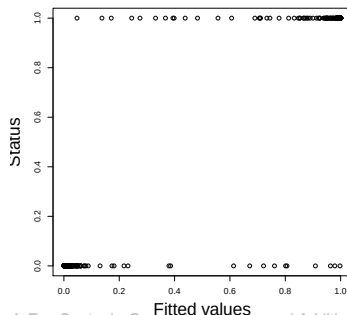
There are 699 patients on the study, 16 of which have one or more missing value(s) in the predictors.

Available predictors:

- ▶ Clump Thickness (1-10)
- ▶ Uniformity of Cell Size (1-10)
- ▶ Uniformity of Cell Shape (1-10)
- ▶ Marginal Adhesion (1-10)
- ▶ Single Epithelial Cell Size (1-10)
- ▶ Bare Nuclei (1-10)
- ▶ Bland Chromatin (1-10)
- ▶ Normal Nucleoli (1-10)
- ▶ Mitoses (1-10)

We consider a logistic linear model.

```
> breast.fit <- glm(status~clump+cellsize+cellshape+adhesion+
+ singlesize+nuclei+chromatin+nucleoli+mitoses , data=breast ,
+ family=binomial , na.action=na.omit)
> fitted.breast <- predict(breast.fit , type="response")
> plot(fitted.breast , na.omit(breast)$status ,
+ xlab="Fitted values" , ylab="Status")
```



Example 3: Air Pollution Study

Data from the National Morbidity, Mortality and Air Pollution Study Database (NMMAPS) available at <http://www.ihapss.jhsph.edu> for the 88 largest metropolitan areas in the United States.

Outcome is the number of death on a day (`death`), that we model as a Poisson variable.

Potentially interesting explanatory variables:

- ▶ particulate matter less than $10\text{ }\mu\text{m}$ in aerodynamic diameter, 10%-trimmed mean over all monitors in a county (pm10tmean)
- ▶ average temperature (over 24 hours) (tmean)
- ▶ time (day)
- ▶ dew point temperature (dptp)
- ▶ day of the week (dow : 1=Sat, 2=Sun, ...)

Main interest is in the effect of pm10tmean (pollution), that enters linearly in the model.

San Francisco, 1st Jan., 1987 to 31st Dec, 2000, $n = 15342$

```
sanFfit <- gam(death~pm10tmean+s(tmean)+s(day)+s(dptp)+factor(dow),
data=sanF, family=poisson)
```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.7820648	0.0245328	72.640	< 2e-16	***
pm10tmean	0.0017404	0.0005446	3.196	0.00139	**
factor(dow)2	-0.0261762	0.0349350	-0.749	0.45369	
factor(dow)3	0.0194925	0.0343809	0.567	0.57074	
factor(dow)4	-0.0207042	0.0348386	-0.594	0.55232	
factor(dow)5	-0.0013170	0.0346407	-0.038	0.96967	
factor(dow)6	-0.0002978	0.0346081	-0.009	0.99314	
factor(dow)7	-0.0140022	0.0345012	-0.406	0.68486	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

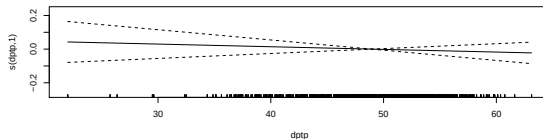
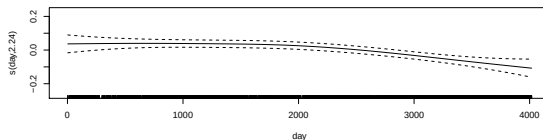
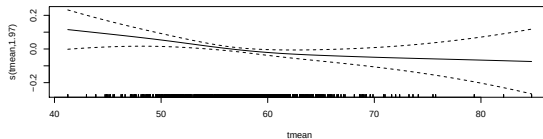
Approximate significance of smooth terms:

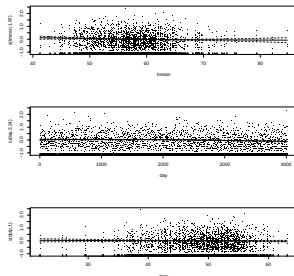
	edf	Ref.df	Chi.sq	p-value	
s(tmean)	1.973	2.568	7.642	0.0388	*
s(day)	2.241	2.796	24.476	2.14e-05	***
s(dptp)	1.003	1.006	0.494	0.4830	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.0145 Deviance explained = 2.03%

UBRE = 0.74149 Scale est. = 1 n = 1986





Open questions:

- ▶ Are all the terms significant?
- ▶ Other factors impacting the number of deaths.
(R^2 and explained deviance low because few explanatory variables.)
- ▶ ★ Different models for different age groups (see Dominici, Samet, and Zeger 2000).

Example 4: Cost of stay for back problems

100 patients hospitalized at the *Centre Hospitalier Universitaire Vaudois (CHUV)* in Lausanne (Switzerland) during 1999 for “medical back problems” (APDRG 243).

Data provided by A. Marazzi, see Marazzi and Yohai (2004).

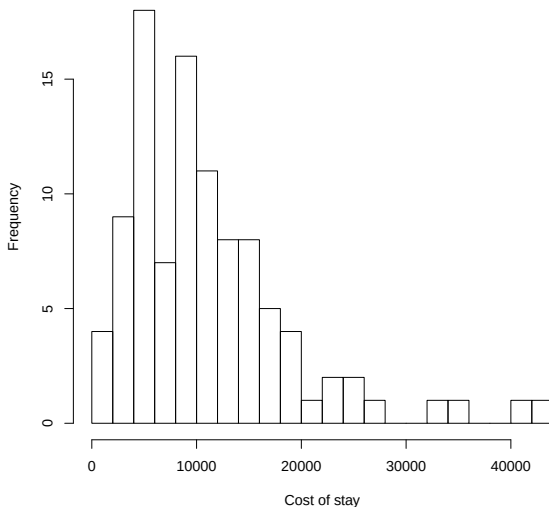
The outcome is the cost of stay (C_{ouTot} in Swiss francs)

Explanatory variables:

- ▶ length of stay (`LOS`, in days)
- ▶ admission type (`Typadm`: 0=planned, 1=emergency)
- ▶ insurance type (`TypAss`: 0=regular, 1=private)
- ▶ age in years (`age`)
- ▶ sex (`Sexe`: 0=female, 1=male)
- ▶ discharge destination (`dest`: 1=home, 0=another health institution)

Goals:

- ▶ Identify factor impacting cost of stay
- ▶ Prediction



Gamma model with logarithmic link.

Coefficients :

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	7.2338121	0.1468364	49.264	< 2e-16	***
log (LOS)	0.8222203	0.0279641	29.403	< 2e-16	***
Typadm	0.2136148	0.0500028	4.272	4.67e-05	***
Typass	0.0932763	0.0791100	1.179	0.2414	
age	-0.0005335	0.0012852	-0.415	0.6790	
Sexe	0.0951009	0.0499814	1.903	0.0602	.
dest	-0.1043560	0.0692713	-1.506	0.1353	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 '0'

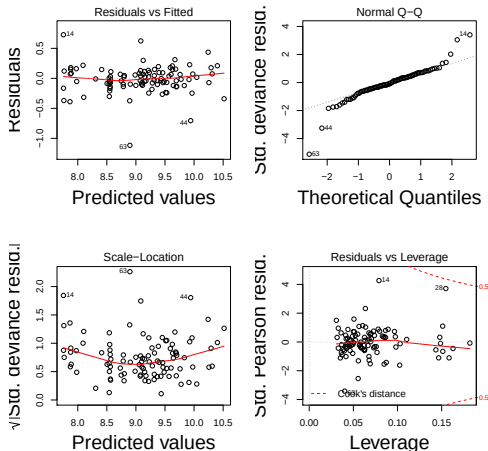
(Dispersion parameter for Gamma family taken to be 0.04961642)

Null deviance: 47.9536 on 99 degrees of freedom

Residual deviance: 5.0718 on 93 degrees of freedom

AIC: 1817.9

Number of Fisher Scoring iterations: 5



Outliers! (see Chapter 6)

Example 5: CARDIA study

Data from the CARDIA study (Coronary Artery Risk Development in young Adults), available from the JASA data archive at <http://lib.stat.cmu.edu/jasadata/>, see Preisser, Galecki, Lohman, and Wagenknecht (2000).

5'115 adults aged 18-30 years were followed and examined up to 5 times from 1986 to 1996 at years 0, 2, 5, 7 and 10.

We consider the subsample of 3693 individuals for which we have full information for the first 4 visits (no missing values).

Outcome (at each visit): self-reported smoking status (yes/no, coded 1/0).

Predictors (at each visit):

- ▶ age (time dependent)
- ▶ birth cohort (1955-1962, 1963-1967) (time independent)
- ▶ education (high school or less, some college, college degree) (time (in)dependent)
- ▶ race_{sex} (black males, black females, white males, white females) (time independent)

★ Missingness issues, see Preisser, Galecki, Lohman, and Wagenknecht (2000).

Call :

```
gee(formula = smoke ~ age + factor(birth) + factor(education) +
    factor(racesex), id = id, data = CARDIA.sub, family = binomial,
    corstr = "exchangeable")
```

Summary of Residuals:

	Min	1Q	Median	3Q	Max
	-0.5506120	-0.3121130	-0.1189440	0.4816228	0.9095992

Coefficients :

	Estimate	Naive S.E.	Naive z	Robust S.E.	Robust z
(Intercept)	-1.33707777	0.54231807	-2.465486	0.54510289	-2.452891
age	0.06480087	0.02646754	2.448315	0.02669744	2.427231
factor(birth)2	-0.19856466	0.14606724	-1.359406	0.14563751	-1.363417
factor(birth)3	-0.40380459	0.23689643	-1.704562	0.23858001	-1.692533
factor(education)2	-0.68712005	0.08313860	-8.264754	0.08490269	-8.093030
factor(education)3	-1.96140806	0.10016299	-19.582164	0.10054908	-19.506972
factor(racesex)2	-0.17668132	0.09763816	-1.809552	0.09900202	-1.784623
factor(racesex)3	-0.13445604	0.10634023	-1.264395	0.10833799	-1.241079
factor(racesex)4	-0.12730906	0.10530815	-1.208919	0.10599195	-1.201120

Estimated Scale Parameter: 1.000362

Number of Iterations: 1

Working Correlation

	[,1]	[,2]	[,3]	[,4]
[1,]	1.000000	0.727139	0.727139	0.727139
[2,]	0.727139	1.000000	0.727139	0.727139
[3,]	0.727139	0.727139	1.000000	0.727139
[4,]	0.727139	0.727139	0.727139	1.000000

Example 6: Birds abundance

Are birds counts decreasing? Abundance of animal species.

The example of 5 species of Seal Island (Nova Scotia, Canada) that winters mostly in the Southern USA: Northern Flicker, Red-wing Blackbird, Rusty Blackbird, Ruby-crowned Kinglet, Yellow-rumped Warbler.

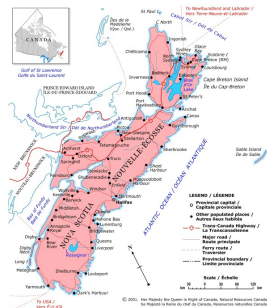
Data courtesy of Prof. I. MacLaren, Dalhousie University.

2245 observations (autumn season, between Aug. 1st and Nov. 14th, over 40 years since 1963), that is 449 days of observations for each bird species.

Generalized Linear and Additive Models

Introduction and motivation

Example 6: Birds abundance



Outcome: Number of birds observed (counts, NUMBER).

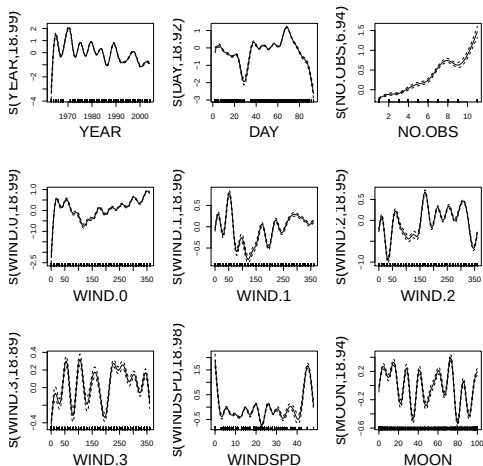
Predictors:

- ▶ year (YEAR) - Main interest.
- ▶ day (DAY)
- ▶ number of observer on the island (NO.OBS)
- ▶ wind direction at midnight the night before (WIND.0)
- ▶ wind direction at noon the day before (WIND.1)
- ▶ wind direction at midnight 2 days before (WIND.2)
- ▶ wind direction at noon 2 days before (WIND.3)
- ▶ the speed of the wind that day (WINDSPD)
- ▶ sky condition (1=clear, 2=cloudy, 3=fog, 4=rain) (SKY)
- ▶ proportion of illuminated moon (MOON)
- ▶ bird species (1-5) (BIRD)

Poisson model with logarithmic link:

$$\begin{aligned} \text{NUMBER} \sim & \alpha + f_1(\text{YEAR}) + f_2(\text{DAY}) + f_3(\text{NO.OBS}) \\ & + f_4(\text{WIND.0}) + f_5(\text{WIND.1}) + f_6(\text{WIND.2}) \\ & + f_7(\text{WIND.3}) + f_8(\text{WINDSPD}) + \text{factor}(\text{SKY}) \\ & + f_9(\text{MOON}) + \text{factor}(\text{BIRD}) \end{aligned}$$

```
> require(mgcv)
> GrA.fit <- gam(NUMBER~s(YEAR,k=20)+s(DAY,k=20)+s(NO.OBS,k=8)+
  s(WIND.0,k=20)+s(WIND.1,k=20)+s(WIND.2,k=20)+s(WIND.3,k=20)+
  s(WINDSPD,k=20)+factor(SKY)+s(MOON,k=20)+factor(BIRD),
  family=poisson,data=GrA,na.action=na.omit)
> plot(GrA.fit,scale=0,pages=1,pers=T,all.terms=T,cex.lab=2)
```



★ Possible excess of zeros. Review: Ridout, Demétrio, and Hinde (1998)

What do Examples 1 to 6 have in common?

Same general model formulation:

$$\begin{aligned} g(E(Y_i)) &= g(\mu_i) = \\ &= \eta_i = \begin{cases} \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} \\ \alpha + f_1(x_{i1}) + \dots + f_p(x_{ip}) \end{cases} \end{aligned}$$

g is called the link function.

η_i is called the linear (additive) predictor.

Notation

Outcome (response) variable:

observations $y_1, \dots, y_n$ for n individuals (or subjects)

corresponding to random variables $Y_1, \dots, Y_n$.

y_i (resp. Y_i) can be multidimensional, in which case we will denote its elements as y_{it} with $t = 1, \dots, n_i$.

Distribution properties of Y_i :

Y_i issued from a distribution F_{θ_i} :

$$Y_i \sim F_{\theta_i}$$

We denote expectation $E(Y_i) = \mu_i$ and variance $\text{Var}(Y_i) = v_i$.

Explanatory variables (predictors):

- for unidimensional y_i :

$$x_i = (x_{i0}, x_{i1}, \dots, x_{ip})^T$$

is a vector of dimension $(p + 1)$. x_{i0} usually equal to 1 (intercept). Combined information:

$$X = \begin{pmatrix} x_1^T \\ \vdots \\ x_n^T \end{pmatrix} = \begin{pmatrix} x_{10} & x_{11} & \cdots & x_{1p} \\ x_{20} & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n0} & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

► for multidimensional y_i :

$$X_i = \begin{pmatrix} X_{i10} & X_{i11} & \cdots & X_{i1p} \\ X_{i20} & X_{i21} & \cdots & X_{i2p} \\ \vdots & \vdots & \vdots & \vdots \\ X_{in_i0} & X_{in_i1} & \cdots & X_{in_ip} \end{pmatrix}$$

Combined information:

$$X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}$$

Background and prerequisite

Familiarity with the following concepts:

- ▶ Estimation (maximum likelihood, least squares, method of moments, etc.)
- ▶ Confidence intervals.
- ▶ Sampling distributions.
- ▶ Hypotheses testing (null and alternative hypothesis, test statistic and its distribution, p-value, significance level, etc.).
- ▶ Linear regression.
- ▶ Calculus and matrix notation.
- ▶ ...

Principles of statistical modelling I

A statistical analysis should include the following steps:

1. **Exploratory data analysis** to check data quality and to help with model formulation. Consider each variable separately and consider the following questions:
 - ▶ What is the scale of measurements?
 - ▶ What is the shape of the distribution?
 - ▶ How is it associated with other variables?
2. **Model formulation**:
 - ▶ Probability distribution of Y .
 - ▶ Link with the explanatory variables.
3. **Parameter estimation**: which method? (maximum likelihood, least squares, method of moments, etc.)

Principles of statistical modelling II

4. Residuals and model checking:

- ▶ Graphical inspections of residuals (independence, approximate normality, zero mean, constant variance).
- ▶ Goodness of fit tests.

5. Inference and interpretation:

Test hypothesis about parameters to obtain a parsimonious model. Then (and only then) interpret parameters.

Generalized Linear Models (GLM)

Class of models unified by Nelder and Wedderburn (1972) (see McCullagh and Nelder 1989).

The well known **linear model** is often written

$$Y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i,$$

with $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ and hence $Y_i \sim \mathcal{N}(\mu_i, \sigma^2)$.

The same model can also be written as

$$E(Y_i) = \mu_i = \mathbf{x}_i^T \boldsymbol{\beta}.$$

Generalized linear models include more general situations:

1. Response variables that have distributions other than the Normal distribution (even discrete or categorical).
2. Relationship between the response and the explanatory variables need not be of the simple linear form above.

The general form of a GLM is:

$$g(E(Y_i)) = g(\mu_i) = \eta_i = \mathbf{x}_i^T \boldsymbol{\beta},$$

where g is called the link function. We denote $\text{Var}(Y_i) = v_i = v(\mu_i)$.

Examples

- ▶ logistic regression:

$$Y_i \sim \text{Bernoulli}(p_i), E(Y_i) = p_i$$

$$\log\left(\frac{p_i}{1-p_i}\right) = \text{logit}(p_i) = \mathbf{x}_i^T \boldsymbol{\beta}.$$

- ▶ Poisson:

$$Y_i \sim \mathcal{P}(\lambda_i), E(Y_i) = \lambda_i$$

$$\log(\lambda_i) = \mathbf{x}_i^T \boldsymbol{\beta}.$$

Limitations

- ▶ the linear component is retained ($\rightsquigarrow$ GAM, Chapter 9)
- ▶ distributions are restricted to the exponential family (see discussion in Section 4)
- ▶ responses must be independent ($\rightsquigarrow$ GEE, Chapter 4 and GLMM, Chapter 5)

The unified theory of GLM is built upon the **exponential family** of distributions, taking advantage of its nice properties.

Definition: Consider Y whose distribution depends on a single parameter θ . A density or probability function is said to belong to **exponential family** if it can be written in the form

$$f(y; \theta, \phi) = \exp \left(A \frac{y\theta - b(\theta)}{\phi} + c(y, \frac{\phi}{A}) \right),$$

for some specific functions $b(\cdot)$ and $c(\cdot)$ and constant A . If ϕ is known, it is an exponential family with **canonical parameter** θ .

A link function g of the GLM specification is said to be canonical if it is such that $\theta(\mu) = \eta$, that is if $g(\cdot) = (b')^{-1}(\cdot)$ (given that $\mu = b'(\theta)$, see (7) on page 57).

Example: The **normal distribution** (where μ is the parameter of interest and σ^2 is regarded as a nuisance parameter) belongs to the exponential family. In fact:

$$\begin{aligned}f(y; \mu) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right) \\&= \exp\left(-\frac{y^2}{2\sigma^2} + \frac{y\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)\right).\end{aligned}$$

We identify $\theta = \mu$, $A = 1$, $\phi = \sigma^2$, $b(\theta) = \mu^2/2 = \theta^2/2$ and

$$c(y, \phi/A) = -\frac{1}{2} \left(\frac{y^2}{\phi} + \log(2\pi\phi) \right).$$

The canonical link is the identity function.

Example: The **Poisson distribution** belongs to the exponential family.

$$\begin{aligned}f(y; \lambda) &= \frac{\lambda^y \exp(-\lambda)}{y!} \\&= \exp(y \log(\lambda) - \lambda - \log(y!)),\end{aligned}$$

where $\theta = \log(\lambda)$, $A = 1$, $\phi = 1$, $b(\theta) = \lambda = \exp(\theta)$ and $c(y, \phi/A) = -\log(y!)$.

The canonical link is the $\log()$ function.

Distribution	A	ϕ	$b(\theta)$	$c(y, \phi/A)$
$\mathcal{N}(\mu, \sigma^2)$	1	σ^2	$\theta^2/2$	$-\frac{1}{2}(y^2/\phi + \log(2\pi\phi))$
$\mathcal{P}(\lambda)$	1	1	$\exp(\theta)$	$-\log(y!)$
$\mathcal{B}(m, p)/m$				
$\Gamma(\mu, \nu)$				

Table: Members of the exponential family.

Properties of the exponential family

Let us first develop general expressions for the expected value and the variance of Y , useful later in this Chapter.

Reminder: If $l(\theta, \phi; y) = \log(f(y; \theta, \phi))$, it holds:

$$E \left(\frac{\partial l(\theta, \phi; Y)}{\partial \theta} \right) = 0, \quad (3)$$

and

$$E \left(\frac{\partial^2 l(\theta, \phi; Y)}{\partial \theta^2} \right) + E \left(\left(\frac{\partial l(\theta, \phi; Y)}{\partial \theta} \right)^2 \right) = 0, \quad (4)$$

if the order of integration and differentiation can be reversed (compact support of the distribution).

For the densities in the exponential family, we have

$$\frac{\partial l(\theta, \phi; y)}{\partial \theta} = A \frac{y - b'(\theta)}{\phi}, \quad (5)$$

and

$$\frac{\partial^2 l(\theta, \phi; y)}{\partial \theta^2} = -A \frac{b''(\theta)}{\phi} \quad (6)$$

From (3) and (5) we obtain

$$0 = E \left(\frac{\partial l(\theta, \phi; Y)}{\partial \theta} \right) = A \frac{E(Y) - b'(\theta)}{\phi}$$

which gives

$$E(Y) = \mu = b'(\theta) \quad (7)$$

Similarly, from (4) and (6), and using (7), we obtain

$$\begin{aligned} 0 &= E \left(-A \frac{b''(\theta)}{\phi} \right) + E \left(\left(A \frac{Y - b'(\theta)}{\phi} \right)^2 \right) \\ &= -A \frac{b''(\theta)}{\phi} + A^2 E \left(\left(\frac{Y - \mu}{\phi} \right)^2 \right) \\ &= -\frac{b''(\theta)}{\phi} + A \frac{\text{Var}(Y)}{\phi^2} \\ &= -b''(\theta) + A \frac{\text{Var}(Y)}{\phi} \end{aligned}$$

which gives

$$\text{Var}(Y) = \frac{b''(\theta)\phi}{A} \quad (8)$$

Three components:

1. Independent response variables $Y_1, \dots, Y_n$ which are assumed to share the same distribution from the exponential family (F_{θ_i}).
2. A set of parameters β and explanatory variables $x_i^T = (1, x_{i1}, \dots, x_{ip})$ for $i = 1, \dots, n$.
3. A monotone and differentiable link function g such that

$$g(\mu_i) = x_i^T \beta.$$

$\eta_i = x_i^T \beta$ is called the linear predictor.

The initial problem has n (=sample size) unknown parameter $\mu_1, \dots, \mu_n$ (or $\theta_1, \dots, \theta_n$) which are reduced to $(p + 1)$ unknown parameters $(\beta_0, \beta_1, \dots, \beta_p)$ by imposing the structure of a GLM model.

The transformation via the link function ensures that the estimated parameter lies in the admissible space of values (for example, $(0, 1)$ for Bernoulli/binomial and $(0, \infty)$ for Poisson).

Normal distribution: $Y_i \sim \mathcal{N}(\mu_i, \sigma^2)$

The normal distribution is used to model continuous data that have a symmetric distribution. Examples: height or blood pressure of people. The corresponding link function is the identity function $g(\mu_i) = \mu_i$ (no restriction on the possible values for μ_i , a linear model).

Remark: This model is very popular:

- ▶ many phenomena well described by the normal distribution
- ▶ average or total of a random sample approximated by the normal distribution (central limit theorem)
- ▶ a lot of theory available for this model (explicit, analytical results).

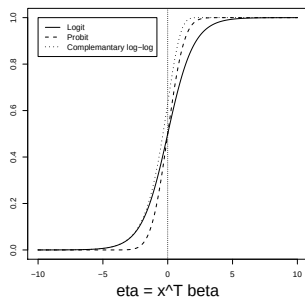
Binomial distribution: $Y_i \sim \mathcal{B}(m_i, p_i)$

Models a process with binary outcomes. Examples: the number of candidates who pass a test (outcome: pass or fail), the number of patients with some disease who are alive at a specified time since diagnosis (outcome: alive or dead), the smoking status of a person.

For GLM we prefer to consider the proportion of “success” $P_i = Y_i/m_i$, for which $E(P_i) = \mu_i = p_i$.

The link functions commonly used for the binomial distribution are (restrict the domain of μ_i within $(0,1)$):

- ▶ **logit**: $g(\mu_i) = g(p_i) = \log\left(\frac{p_i}{1-p_i}\right)$ (canonical link)
- ▶ **probit**: $g(\mu_i) = g(p_i) = \Phi^{-1}(p_i)$
- ▶ **complementary log-log**: $g(\mu_i) = g(p_i) = \log(-\log(1 - p_i))$



Which one? Difficult to test from data, because a large amount of data is needed for small p .

- ★ General families of link functions, see Aranda-Ordaz (1981)
- ★ Nonparametric link for GLM, see Weisberg and Welsh (1994)
- ★ Goodness of fit test for the link, see Pregibon (1980)

The values $o_i = \frac{p_i}{1-p_i}$ are called the odds.

An odd of 2/3 means that there are 2 chances (out of 5) for success against 3 chances (out of 5) for failure.

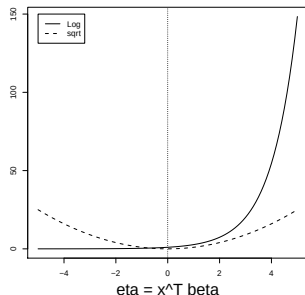
Note: $p_i = \frac{o_i}{1+o_i}$.

The logit link has the advantage to allow interpretation on the odds scale.

Poisson distribution: $Y_i \sim \mathcal{P}(\lambda_i)$

Used to model count data, the number of occurrences of some event in a defined time period or space. Examples: the number of medical conditions reported by a person, the number of tropical cyclones during a season, the number of birds observed on a particular day, the number of spelling mistakes on the page of a newspaper, the number of incoming calls in 4 hours.

Here the link guarantees that $\lambda > 0$. Common choices are: $g(\mu_i) = g(\lambda_i) = \log(\lambda_i)$ (canonical link) or $g(\mu_i) = \sqrt{\lambda}$.



Remark: the number of events out of a total is appropriately modelled by a binomial distribution. But, if success probabilities are small and the totals are large, the Poisson is a good approximation. (For small p , $\text{logit}(p) = \log\left(\frac{p}{1-p}\right) \simeq \log(p)$.)

Gamma distribution: $Y_i \sim \Gamma(\mu_i, \nu)$

Used to model asymmetric continuous behaviors. For example, income or cost of hospital stay.

Here the link guarantees that $\mu_i > 0$. A common choice is: $g(\mu_i) = \log(\mu_i)$. The canonical link is $1/\mu_i$ but is more rarely used. We note that the canonical link does not guarantee $\mu_i > 0$.

Remark: the exponential distribution $\mathcal{E}(\lambda_i)$ is a particular case where $\nu = 1$.

GLM Extensions

In the spirit of GLM, some close models can be considered.

Multinomial distribution:

Generalization of the binomial distribution where there are K possible outcomes (instead of two), with associated probabilities $\pi_1, \dots, \pi_K$. Denote by Y_i the random variable that takes values over categories $1, \dots, K$.

Need to distinguish between **nominal** data (no natural order) and **ordinal** data.

Nominal data

Examples: eye color, party affiliation.

Denote by Y_{ik} the number of observations falling into category k for individual i (if only one measure, then only one of $Y_{i1}, \dots, Y_{iK}$ is equal to one and the other are zero). Associated probabilities are $p_{ik} = P(Y_i = k)$.

Link the predictors x_i to the probabilities through:

$$\log \left(\frac{p_{ik}}{p_{i1}} \right) = x_i^T \beta_k, \quad k = 2, \dots, K$$

under the constraint that $\sum_{k=1}^K p_{ik} = 1$, that is,
 $p_{i1} = 1 - \sum_{k=2}^K p_{ik}$.

Ordinal data

Examples: food-testing, classification of radiographs, determination of physical and mental well-being.

The cumulative response probabilities

$$\gamma_1 = \pi_1, \gamma_2 = \pi_1 + \pi_2, \dots, \gamma_K = 1,$$

that is $\gamma_k(x_i) = P(Y_i \leq k \mid x_i)$ are modelled.

For example (logit-link model)

$$\log \left(\frac{\gamma_k(x_i)}{1 - \gamma_k(x_i)} \right) = \alpha_k - x_i^T \beta$$

for $k = 1, \dots, (K - 1)$. The negative sign on the linear predictor is a convention ensuring that large values of $x_i^T \beta$ lead to an increase in probability in the higher numbered categories.

Extension: allow for β_k instead of β .

This model is called the **proportional-odds** model because the ratio of the event $Y \leq k$ at $x_i = x$ and $x_i = \tilde{x}$ is

$$\frac{\text{odds}(Y \leq k \mid x)}{\text{odds}(Y \leq k \mid \tilde{x})} = \frac{\gamma_k(x)/(1 - \gamma_k(x))}{\gamma_k(\tilde{x})/(1 - \gamma_k(\tilde{x}))} = \exp(-(x - \tilde{x})^T \beta),$$

independent of α_k . In particular, if x_i is an indicator for 2 treatments groups, T_1 and T_2 ,

$$\frac{\text{odds}(Y \leq k \mid T_1)}{\text{odds}(Y \leq k \mid T_2)} = \exp(-\Delta),$$

where Δ measures the treatment effect.

Alternatively one can use a complementary log-log link

$$\log(-\log(1 - \gamma_k(x_i))) = \alpha_k - x_i^T \beta,$$

which produces the **proportional hazard** model.

Note: in all of these models we need $\alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_{K-1}$ to ensure that the probabilities are non-negative.

Parameters estimation: maximum likelihood (ML) I

The likelihood function for the **canonical exponential family** is

$$\begin{aligned} L(\theta_1, \dots, \theta_n; y_1, \dots, y_n) &= \prod_{i=1}^n \exp \left(A \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \frac{\phi}{A}) \right) \\ &= \exp \left(\frac{A}{\phi} \sum_{i=1}^n y_i \theta_i - \frac{A}{\phi} \sum_{i=1}^n b(\theta_i) + \sum_{i=1}^n c(y_i, \frac{\phi}{A}) \right) \end{aligned}$$

and the log-likelihood function is given by

$$l(\theta_1, \dots, \theta_n; y_1, \dots, y_n) = \frac{A}{\phi} \sum_{i=1}^n y_i \theta_i - \frac{A}{\phi} \sum_{i=1}^n b(\theta_i) + \sum_{i=1}^n c(y_i, \frac{\phi}{A}).$$

Parameters estimation: maximum likelihood (ML) II

The relationship $g(\mu_i) = x_i^T \beta$ is used in the above likelihood function. Then the function is maximized with respect to β (by solving the score equations) to obtain $\hat{\beta}$.

No closed-form analytical expression for $\hat{\beta}$ is available in general.

Example: if Y_i is a binary response, that is, $Y_i \sim \text{Bernoulli}(p_i)$, the likelihood is

$$L(p_1, \dots, p_n; y_1, \dots, y_n) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i},$$

and the log-likelihood is

$$l(p_1, \dots, p_n; y_1, \dots, y_n) = \sum_{i=1}^n y_i \log(p_i) + \sum_{i=1}^n (1 - y_i) \log(1 - p_i). \quad (9)$$

The logit link implies that $\log\left(\frac{p_i}{1-p_i}\right) = x_i^T \beta$, and therefore that

$$p_i = \frac{\exp(x_i^T \beta)}{(1 + \exp(x_i^T \beta))},$$

which gives the log-likelihood

$$\begin{aligned}l(\beta; y) &= l(\beta_0, \beta_1, \dots, \beta_p; y_1, \dots, y_n) = \\&= \sum_{i=1}^n y_i \log \left(\frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right) + \sum_{i=1}^n (1 - y_i) \log \left(\frac{1}{1 + \exp(x_i^T \beta)} \right) \\&= \sum_{i=1}^n y_i x_i^T \beta - \sum_{i=1}^n y_i \log(1 + \exp(x_i^T \beta)) \\&\quad - \sum_{i=1}^n (1 - y_i) \log(1 + \exp(x_i^T \beta)) \\&= \sum_{i=1}^n y_i x_i^T \beta - \sum_{i=1}^n \log(1 + \exp(x_i^T \beta))\end{aligned}$$

The score equations (differentiation with respect to β) are:

$$\sum_{i=1}^n \left(y_i - \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right) x_i = 0,$$

which have to be solved numerically.

Newton-Raphson procedure in the univariate case

For x univariate, we would like to solve $h(x) = 0$.

For small distances $x^{(m)} - x^{(m-1)}$, the slope is

$$h'(x^{(m-1)}) = \frac{h(x^{(m)}) - h(x^{(m-1)})}{x^{(m)} - x^{(m-1)}}. \quad (10)$$

If $x^{(m)}$ is the required solution so that $h(x^{(m)}) = 0$, then (10) can be rearranged to give

$$x^{(m)} = x^{(m-1)} - \frac{h(x^{(m-1)})}{h'(x^{(m-1)})}. \quad (11)$$

This is the Newton-Raphson formula for solving $h(x) = 0$ iteratively.

If we apply (11) to maximize a log-likelihood $l(\beta; y)$, we have

$$\beta^{(m)} = \beta^{(m-1)} - \frac{U^{(m-1)}}{U'^{(m-1)}},$$

where $U = \frac{dl(\beta; y)}{d\beta}$.

For maximum likelihood estimation it is common to approximate U' by $E(U') = -\mathcal{J}$ (the Fisher information), to obtain

$$\beta^{(m)} = \beta^{(m-1)} + \frac{U^{(m-1)}}{\mathcal{J}^{(m-1)}}. \quad (12)$$

This is called the [method of scoring](#).

The multivariate equivalents of (11) and (12) are

$$\beta^{(m)} = \beta^{(m-1)} - [\mathcal{U}'^{(m-1)}]^{-1} \mathcal{U}^{(m-1)},$$

and respectively

$$\beta^{(m)} = \beta^{(m-1)} + [\mathcal{J}^{(m-1)}]^{-1} \mathcal{U}^{(m-1)}, \quad (13)$$

where β and $\mathcal{U}$ are now vectors of dimension $(p+1)$ and $\mathcal{J}$ is a $(p+1) \times (p+1)$ matrix.

Iterative Weighted Least Squares (IWLS) algorithm

Major steps of the **Iterative Weighted Least Squares (IWLS)** algorithm for maximizing the log-likelihood for GLM (see McCullagh and Nelder (1989), p. 40-42 for additional details)

To solve:

$$\frac{\partial l(\beta; y)}{\partial \beta} = \begin{pmatrix} \partial l(\beta; y) / \partial \beta_0 \\ \vdots \\ \partial l(\beta; y) / \partial \beta_p \end{pmatrix} = 0$$

We have that

$$\frac{\partial l}{\partial \beta_j} = U_j = \sum_{i=1}^n \frac{\partial l_i}{\partial \beta_j} = \sum_{i=1}^n \frac{\partial l_i}{\partial \theta_i} \cdot \frac{\partial \theta_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \beta_j}. \quad (14)$$

First

$$\frac{\partial l_i}{\partial \theta_i} = A \frac{y_i - b'(\theta_i)}{\phi} \stackrel{(7)}{=} A \frac{y_i - \mu_i}{\phi}.$$

Secondly $\frac{\partial \theta_i}{\partial \mu_i} = 1 / (\frac{\partial \mu_i}{\partial \theta_i})$ and

$$\frac{\partial \mu_i}{\partial \theta_i} \stackrel{(7)}{=} b''(\theta_i) \stackrel{(8)}{=} A \frac{\text{Var}(Y_i)}{\phi}$$

Finally

$$\frac{\partial \mu_i}{\partial \beta_j} = \frac{\partial \mu_i}{\partial \eta_i} \cdot \frac{\partial \eta_i}{\partial \beta_j} = \frac{\partial \mu_i}{\partial \eta_i} x_{ij}.$$

Putting everything together, (14) can be written as

$$\frac{\partial l}{\partial \beta_j} = U_j = \sum_{i=1}^n \frac{y_i - \mu_i}{\text{Var}(Y_i)} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)$$

We use the method of scoring (see (13) on page 79) which implies the computation of $\mathcal{J}$, whose element (j, k) is

$\mathcal{J}_{jk} = E(U_j U_k)$ and is given by

$$\begin{aligned}\mathcal{J}_{jk} &= E \left\{ \sum_{i=1}^n \frac{y_i - \mu_i}{\text{Var}(Y_i)} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) \cdot \sum_{l=1}^n \frac{y_l - \mu_l}{\text{Var}(Y_l)} x_{lk} \left(\frac{\partial \mu_l}{\partial \eta_l} \right) \right\} \\ &= \dots (\text{some extra computations}) \\ &= \sum_{i=1}^n \frac{x_{ij} x_{ik}}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2\end{aligned}$$

$\mathcal{J}$ can be written as $X^T W X$, where W is a $n \times n$ diagonal matrix with elements

$$w_{ii} = \frac{1}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2$$

The method of scoring is

$$\beta^{(m)} = \beta^{(m-1)} + [\mathcal{J}^{(m-1)}]^{-1} \mathcal{U}^{(m-1)},$$

and implies

$$\mathcal{J}^{(m-1)} \beta^{(m)} = \mathcal{J}^{(m-1)} \beta^{(m-1)} + \mathcal{U}^{(m-1)}.$$

The right-hand side of the above can be written in our case (computations omitted) as $X^T W^{(m-1)} z^{(m-1)}$, where $z^{(m-1)}$ is a vector with elements

$$z_i^{(m-1)} = \sum_{k=0}^p x_{ik} \beta_k^{(m-1)} + (y_i - \mu_i^{(m-1)}) \left(\frac{\partial \eta_i}{\partial \mu_i} \right)^{(m-1)},$$

where $\mu_i^{(m-1)}$ and $(\partial \mu_i / \partial \eta_i)^{(m-1)}$ are evaluated at $\beta^{(m-1)}$.

Therefore the method of scoring amounts to solve with respect to $\beta^{(m)}$

$$X^T W^{(m-1)} X \beta^{(m)} = X^T W^{(m-1)} z^{(m-1)}. \quad (15)$$

(15) is a weighted least squares problem to be solved iteratively because $(z^{(m-1)})$ and $W^{(m-1)}$ depend on $\beta^{(m-1)}$. Only needs software that computes weighted least squares.

Remarks:

- ▶ $\frac{\partial \mu_i}{\partial \eta_i}$ depends on the link function used.
- ▶ $\text{Var}(Y_i)$ depends on the model.
- ▶ $\frac{\partial \mu_i}{\partial \eta_i} = b''(\theta) \propto \text{Var}(Y_i)$ when the canonical link is used.

This simplifies the estimating equations:

$$\frac{\partial l}{\partial \beta_j} = U_j = \sum_{i=1}^n (y_i - \mu_i) x_{ij}$$

Consider the breast cancer dataset of Example 2 (p. 19), and fit the simple model that explains the outcome (`status`) as a function of `nuclei` only (plus intercept) with a logit link.

```
Call: glm(formula = status ~ nuclei, family = binomial,
          data = breast, na.action = na.omit)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.52218	0.23205	-15.18	<2e-16 ***
nuclei	0.85935	0.07092	12.12	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 884.35 on 682 degrees of freedom
 Residual deviance: 340.63 on 681 degrees of freedom
 (16 observations deleted due to missingness)
 AIC: 344.63

Number of Fisher Scoring iterations: 6

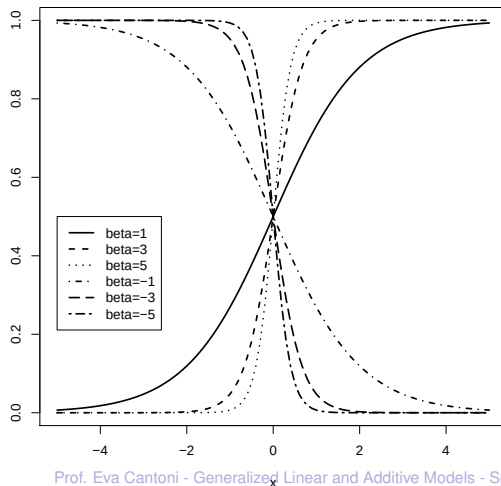
If $p_i = E(\text{status})$, the model is

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \text{nuclei}$$

Interpretation

- ▶ β_1 is related to the slope of the logistic curve (p. 87).
- ▶ The effect of a unit change in `nuclei` is to increase log odds by β_1 . Equivalently, the effect of a unit change in `nuclei` is to increase the odds of a positive response multiplicatively by $\exp(\beta_1)$.
- ▶ When more than one covariate, same interpretation under the condition that the other covariates are kept fixed.
- ▶ The effect of a unit change in `nuclei` on p_i is more complicated to establish. In presence of several explanatory variables, this effect would depend on the other covariates too.

Logistic curves for different values of β_1



We can also fit the same model, but with a probit link.

```
Call: glm(formula = status ~ nuclei, family = binomial(link = probit),
  data = breast, na.action = na.omit)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.96960	0.10934	-18.01	<2e-16	***
nuclei	0.45172	0.03177	14.22	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 884.35 on 682 degrees of freedom

Residual deviance: 342.94 on 681 degrees of freedom

(16 observations deleted due to missingness)

AIC: 346.94

Number of Fisher Scoring iterations: 7

Logistic vs probit

The parameter estimates of the probit model are quite different from those of the logit model, because they don't have the same meaning:

$$\log\left(\frac{p_i}{1-p_i}\right) = x_i^T \beta^L \quad \text{versus} \quad \Phi^{-1}(p_i) = x_i^T \beta^P.$$

The fitted values (prediction for in-sample observations) are obtained from the model formulation.

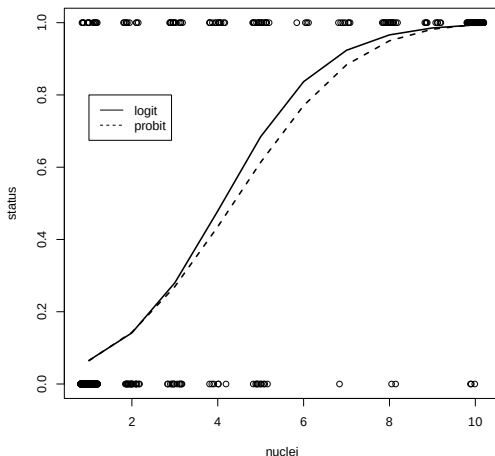
For the logistic model, $\log(p_i/(1 - p_i)) = x_i^T \beta^L$ implies $p_i = \exp(x_i^T \beta^L)/(1 + \exp(x_i^T \beta^L))$ and therefore

$$\hat{y}_i = \hat{p}_i = \frac{\exp(x_i^T \hat{\beta}^L)}{(1 + \exp(x_i^T \hat{\beta}^L))}.$$

For the probit model, $\Phi^{-1}(p_i) = x_i^T \beta^P$ implies $p_i = \Phi(x_i^T \beta^P)$ and therefore

$$\hat{y}_i = \hat{p}_i = \Phi(x_i^T \hat{\beta}^P).$$

Fitted values with logit and probit model



Model validation

One way of assessing the adequacy of a model is to compare it to a more general model with the maximum number of parameters: a **saturated model**, which has the same number of parameters than observations and leads to a perfect fit. The corresponding log-likelihood is denoted by $l(y; y)$.

The **scaled deviance** function is defined:

$$\begin{aligned} D^*(y, \mu) &= 2[l(y; y) - l(\hat{\beta}; y)] = \\ &= \sum_{i=1}^n 2[l_i(y_i; y_i) - l_i(\hat{\beta}; y_i)] = \sum_{i=1}^n d_i, \end{aligned}$$

and the **deviance** $D(y, \mu) = \phi D^*(y, \mu)$.

The deviance can be regarded as the likelihood ratio statistics for testing a specific model within the saturated model, assuming $\phi = 1$. This hypothesis is true for binomial and Poisson models, for which $D(y, \mu) = D^*(y, \mu)$. For other distributions, e.g. normal or Gamma, the deviance is not directly related to a likelihood ratio statistic.

Large values of D indicate that the model is not good. On the other hand, small values of D (or D^*) arise when the log-likelihood $l(\beta; y)$ (from a model with p parameters) is close to the log-likelihood $l(y; y)$ (equivalent to a model with n parameters).

We have the following results:

- ▶ The distribution of D^* is exactly $\chi^2_{(n-(p+1))}$ if Y_i is normally distributed (D^* is the resid. sum of squares divided by σ^2).
- ▶ For certain other distributions of Y_i the $\chi^2_{(n-(p+1))}$ can be taken as an approximation for D^* .
- ▶ $D (=D^*)$ is unusable when Y_i is Bernoulli(p_i). In this case, the distribution of D is degenerate (in fact, D doesn't depend of Y_i).
- ▶ The $\chi^2_{(n-(p+1))}$ approximation works badly for $Y_i \sim \mathcal{B}(m_i, p_i)$ if m_i small (e.g. < 5).
- ▶ Except for the Bernoulli case, a bootstrap procedure could be used to approximate the distribution of D^* .
- ▶ In the case of Bernoulli, there is an ad-hoc fix, see ★
Hosmer and Lemeshow (1980).

An alternative goodness of fit statistic is the Pearson X^2 statistic:

$$X^2 = \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)}.$$

The distribution of X^2 is $\chi^2_{n-(p+1)}$ for normally distributed Y_i and this approximation can be used for other distributions (see the discussion about D).

It can be proven that D and X^2 are related.

Global measure of quality of the fit.

- Proportion of deviance explained

$$(D_{null} - D)/D_{null},$$

where D_{null} is the deviance of an intercept only fit.

- An extension of the R^2 measure (Nagelkerke 1991):

$$\tilde{R}^2 = \frac{1 - (L_{null}/L)^{2/n}}{1 - L_{null}^{2/n}}$$

where L_{null} is the likelihood of an intercept only fit.

Residual analysis

Several type of residuals can be defined for GLM models. In particular,

- ▶ **Pearson residuals:** $r_{iP} = \frac{y_i - \hat{\mu}_i}{\sqrt{v(\hat{\mu}_i)}}$,
- ▶ **deviance residuals:** $r_{iD} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$, where d_i are the individual deviance components to D (see page 92).
- ▶ **deviance standardized residuals:** $r_{iDS} = \frac{r_{iD}}{\sqrt{1 - h_{ii}}}$, where h_{ii} are the diagonal elements of the hat matrix $H = X(X^T X)^{-1} X^T$.

Note:

$$\sum_{i=1}^n r_{iP}^2 = X^2 \text{ and } \sum_{i=1}^n r_{iD}^2 = \sum_{i=1}^n d_i = D.$$

Residuals are usually plotted:

- ▶ against the fitted values to check for lack of fit (departures/structure).
- ▶ against the normal quantiles (QQ-plot).
- ▶ in the order of measurements to check for serial correlation.
- ▶ against each continuous explanatory variable to check the linearity assumption.

Remarks:

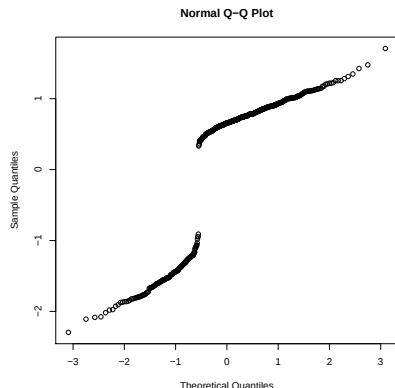
- ▶ Structures very often appears for $Y_i \sim \mathcal{B}(m_i, p_i)$ for small values of m_i .
- ▶ Factors in the covariates can induce particular structures in the residuals.

Note: Normality of the residuals is not expected, and therefore the normal QQ-plot have to be interpreted with caution.

Illustration with simulated data from the model

$$Y_i \sim \text{Bernoulli}(p_i),$$

with $p_i = \exp(x_i^T \beta) / (1 + \exp(x_i^T \beta))$, where $x_i^T = (1, z_i)$ with $z_i \sim N(0, 1)$ and $\beta^T = (1, 0.5)$.



Alternative: randomized quantile residuals

Let $F = F_{\theta_i}$ be the cumulative distribution function of Y_i and Φ be the cumulative distribution function of a standard normal random variable.

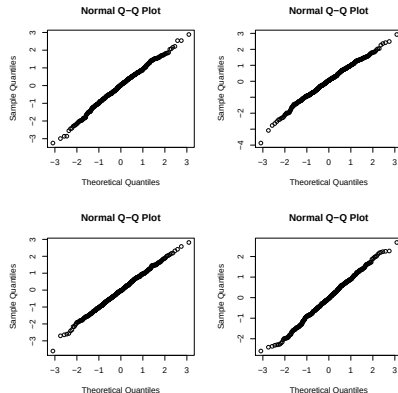
The “randomized quantile residuals” are defined by:

- ▶ if F is continuous: $r_{iQ} = \Phi^{-1}(F(y_i; \hat{\mu}_i, \hat{\phi}))$.
- ▶ if F is discrete: $r_{iQ} = \Phi^{-1}(u_i)$, with u_i a uniform random variable on the interval $(a_i, b_i]$, with $a_i = \lim_{y \rightarrow y_i} F(y; \hat{\mu}_i, \hat{\phi})$ and $b_i = F(y_i; \hat{\mu}_i, \hat{\phi})$,

The distribution of the “randomized quantile residuals” is exactly standard normal, apart from sampling variability in $\hat{\mu}_i$ and $\hat{\phi}$ (thanks to the probability integral transform).

Note: When Y_i is discrete, the definition includes a randomization step: useful to look at several realizations.

On the previous simulated logistic example:



```
> require(statmod)
> qqnorm(qresid(glm.sim))
```

Example 4 of page 28

```
Call: glm(formula = CouTot ~ log(LOS) + Typadm + Typass + age + Sexe +
      dest, family = Gamma(link = log), data = MYdata)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.2338121	0.1468364	49.264	< 2e-16 ***
log(LOS)	0.8222203	0.0279641	29.403	< 2e-16 ***
Typadm	0.2136148	0.0500028	4.272	4.67e-05 ***
Typass	0.0932763	0.0791100	1.179	0.2414
age	-0.0005335	0.0012852	-0.415	0.6790
Sexe	0.0951009	0.0499814	1.903	0.0602 .
dest	-0.1043560	0.0692713	-1.506	0.1353

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.04961642)

Null deviance: 47.9536 on 99 degrees of freedom
Residual deviance: 5.0718 on 93 degrees of freedom
AIC: 1817.9

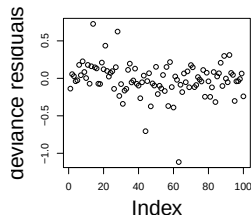
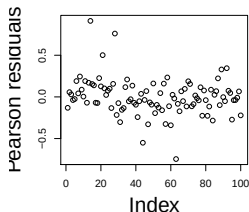
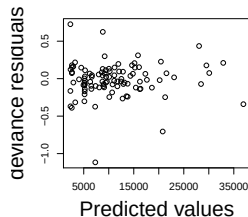
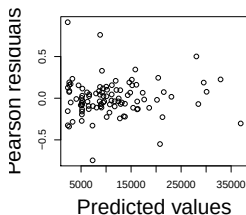
Number of Fisher Scoring iterations: 5

The deviance is $D = 5.07$ in this example see `Residual Deviance` in `summary(MYglm)`.

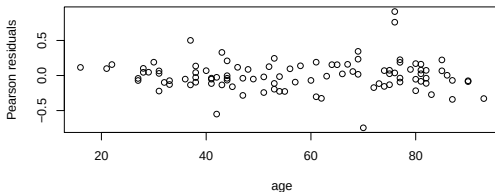
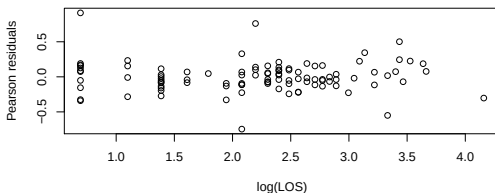
We use it to test the hypothesis H_0 that our model fits the data well.

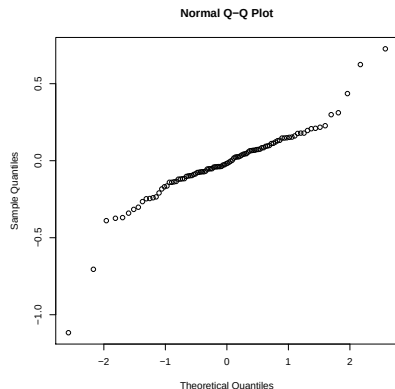
The p-value is $P(D > 5.07)$, which has to be computed according to a χ^2_{93} distribution. This probability is very close to 1, providing no evidence against H_0 .

Pearson and deviance residuals

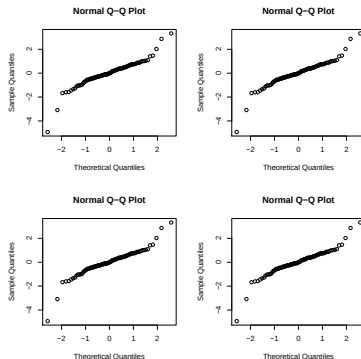


Pearson residuals against continuous covariates





Note: `plot()` in R produces similar plots, plus the Cook distances plot that allows to identify influential observations.



The plots are the same! (no randomization involved).

There seem to be a few outliers, this dataset will be reanalyzed in Chapter 6.

Confidence intervals

The estimator of β in a GLM is usually obtained by maximum likelihood. This means that $\hat{\beta}$ inherits of the general maximum likelihood properties, in particular the asymptotic distribution. Based on

$$(\hat{\beta} - \beta) \sim \mathcal{N}(0, \tilde{J}^{-1}),$$

where $\tilde{J}$ is the information matrix computed from the joint likelihood, the computation of an asymptotic approximation to the standard error of $\hat{\beta}$ is possible.

From the asymptotic normality result, it follows that $(1 - \alpha)$ -confidence intervals for each parameter can be constructed as:

$$[\hat{\beta}_i - z_{1-\alpha/2} \sqrt{(\tilde{\mathcal{J}}^{-1})_{ii}}; \hat{\beta}_i + z_{1-\alpha/2} \sqrt{(\tilde{\mathcal{J}}^{-1})_{ii}}],$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of a $\mathcal{N}(0, 1)$ -distribution.

These intervals are symmetric by construction.

These confidence intervals may work poorly, in particular if the distribution of the parameter estimator is markedly skewed.

In contrast, **profiled likelihood confidence intervals** don't assume normality of the estimator and perform better for small sample sizes. They are obtained by inverting the likelihood ratio statistics, and as such they are still based on asymptotic approximation (the χ^2 distribution of the log-likelihood ratio statistic).

For a model with parameter θ of interest and δ (extra parameters), the profile likelihood is

$$L_1(\theta) = \max_{\delta} L(\theta, \delta).$$

A $(1 - \alpha)$ profiled likelihood confidence intervals is the set of all values θ_0 such that a two-sided test of the null hypothesis $H_0 : \theta = \theta_0$ would not be rejected at the α level. The likelihood ratio test statistic is used.

These confidence intervals are not symmetric, although the result is usually close to the confidence intervals obtained from the maximum likelihood theory.

z-values for testing ($H_0 : \beta_i = 0$ against $H_A : \beta_i \neq 0$), which correspond to a test on the variable x_i :

$$\text{z-value} = \frac{\hat{\beta}_i}{\text{s.e.}(\hat{\beta}_i)} = \frac{\hat{\beta}_i}{\sqrt{(\mathcal{J}^{-1})_{ii}}} \stackrel{H_0}{\sim} \mathcal{N}(0, 1).$$

Then compute the p-value

$$\begin{aligned} P(|\text{z-value}| > |z_{\text{obs}}|) &= \\ &= P(\text{z-value} < -|z_{\text{obs}}|) + P(\text{z-value} > |z_{\text{obs}}|) = \\ &= 2(1 - P(\text{z-value} > |z_{\text{obs}}|)) = 2(1 - \Phi^{-1}(|z_{\text{obs}}|)). \end{aligned}$$

Note: If the distribution depends also on a nuisance or extra parameter (normal, Gamma), the $t_{n-(p+1)}$ distribution is used instead of the normal distribution as reference distribution of the z-statistic under H_0 . The z-values are called t-values in this case, see for example the output of the Gamma example on page 102.

Hypothesis testing

To test a parametric hypothesis on β , e.g. $H_0 : \beta = \tilde{\beta}$, the three classical tests are available:

Wald statistic $(\hat{\beta} - \tilde{\beta})^T \mathcal{J}(\tilde{\beta})(\hat{\beta} - \tilde{\beta})$ which follows a χ^2_{p+1} distribution under H_0 .

score statistic $U(\tilde{\beta})^T \mathcal{J}(\tilde{\beta})^{-1} U(\tilde{\beta})$ which follows a χ^2_{p+1} distribution under H_0 , (U is the score function defined on page 77).

likelihood-ratio statistic $2[l(\hat{\beta}; y) - l(\tilde{\beta}; y)]$ which is in fact a difference of deviances, see below.

Variable selection

Compare two models: a $\mathcal{M}_{p+1}$ with $(p+1)$ parameters and a **nested** model $\mathcal{M}_{p+1-q}$ with $(p+1-q)$ parameters (that is a model that contains a subset of the variables in $\mathcal{M}_{p+1}$).

Common practice in GLM is to use difference of deviances for variable (model) selection.

To test the null hypothesis

$$H_0 : \beta = (\cdot, \dots, \cdot, 0_{p+2-q}, \dots, 0_{p+1})$$

(the model with $(p+1-q)$ parameters is enough), we define the difference of scaled deviances

$$\begin{aligned} \Delta D^* &= D^*(y, \hat{\mu}^{p+1-q}) - D^*(y, \hat{\mu}^{p+1}) = \\ &= 2 \left[l(\hat{\beta}^{p+1}; y) - l(\hat{\beta}^{p+1-q}; y) \right] = \frac{D(y, \hat{\mu}^{p+1-q}) - D(y, \hat{\mu}^{p+1})}{\phi} \end{aligned}$$

If ϕ is known and under H_0 , the asymptotic distribution of ΔD^* is χ_q^2 (it is the likelihood ratio statistics). This approximation is much better than the approximation of the deviance itself.

When ϕ is not known (e.g. Normal, Gamma, “quasi” type of models) the usual approximation under H_0 uses an F statistic:

$$\frac{(D(y, \hat{\mu}^{p+1-q}) - D(y, \hat{\mu}^{p+1})) / q}{\hat{\phi}} \sim F_{q, n-(p+1)},$$

where $\hat{\phi} = \frac{D(y, \hat{\mu}^{p+1})}{n-(p+1)}$.

Note that for the normal-identity case this is an exact result, but for the Gamma the accuracy of this approximation is not well known.

The difference of deviances is often used in a sequential approach, either forward (starting from the null/constant model) or backward (starting from the full model).

Drawback of this approach: the final model depends on the (a priori) arbitrary “path” of the procedure.

A full approach would be to use a general criterion to compare all possible models. One of such criteria is the **Akaike information criterion (AIC)**, which for a general model $\mathcal{M}_d$ with d parameters is defined by:

$$\begin{aligned} -2l(\hat{\beta}^d; y) + 2d &\propto D^*(y, \hat{\beta}^d) + 2d \\ &= \frac{D(y, \hat{\beta}^d)}{\phi} + 2d \propto D(y, \hat{\beta}^d) + 2\phi d. \end{aligned}$$

Need to estimate ϕ .

This criterion takes into account the complexity of the model (measured by the number of parameters). Models with low AIC have to be preferred.

Not feasible to compare all possible models for large p (2^p models). Use stepwise AIC.

Breast cancer dataset

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-10.10394	1.17488	-8.600	< 2e-16	***
clump	0.53501	0.14202	3.767	0.000165	***
cellsize	-0.00628	0.20908	-0.030	0.976039	
cellshape	0.32271	0.23060	1.399	0.161688	
adhesion	0.33064	0.12345	2.678	0.007400	**
singlesize	0.09663	0.15659	0.617	0.537159	
nuclei	0.38303	0.09384	4.082	4.47e-05	***
chromatin	0.44719	0.17138	2.609	0.009073	**
nucleoli	0.21303	0.11287	1.887	0.059115	.
mitoses	0.53484	0.32877	1.627	0.103788	

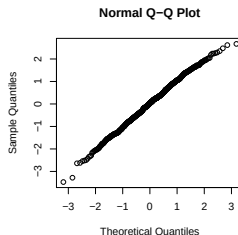
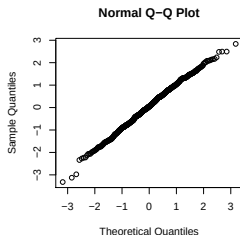
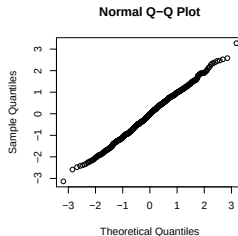
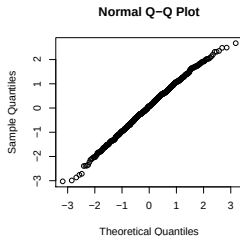
 Signif. codes: 0 *** 0.001 ** 0.01 * 0.05 . 0.1 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 884.35 on 682 degrees of freedom
 Residual deviance: 102.89 on 673 degrees of freedom
 (16 observations deleted due to missingness)
 AIC: 122.89

Number of Fisher Scoring iterations: 8

Randomized residuals



Profiled 95% confidence intervals:

```
> round(confint(breast.fit , level=0.95),4)
```

```
Waiting for profiling to be done...
```

	2.5 %	97.5 %
(Intercept)	-12.7581	-8.0851
clump	0.2742	0.8377
cellsize	-0.3948	0.4379
cellshape	-0.1482	0.7684
adhesion	0.0930	0.5869
singlesize	-0.2169	0.4049
nuclei	0.2069	0.5787
chromatin	0.1232	0.7998
nucleoli	-0.0018	0.4456
mitoses	-0.0067	1.1065

and 95% confidence intervals based on asymptotic normality:

```
> round(confint.default(breast.fit , level=0.95),4)
```

	2.5 %	97.5 %
(Intercept)	-12.4067	-7.8012
clump	0.2567	0.8134
cellsize	-0.4161	0.4035
cellshape	-0.1293	0.7747
adhesion	0.0887	0.5726
singlesize	-0.2103	0.4036
nuclei	0.1991	0.5670
chromatin	0.1113	0.7831
nucleoli	-0.0082	0.4343
mitoses	-0.1095	1.1792

Analysis of deviance (sequential):

```
> anova(breast.fit, test="Chisq")
```

Terms added sequentially (first to last)

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL			682	884.35	
clump	1	425.87	681	458.48	< 2.2e-16 ***
cellsize	1	261.91	680	196.58	< 2.2e-16 ***
cellshape	1	20.08	679	176.50	7.427e-06 ***
adhesion	1	21.39	678	155.11	3.750e-06 ***
singlesize	1	6.45	677	148.66	0.01111 *
nuclei	1	28.97	676	119.69	7.348e-08 ***
chromatin	1	9.21	675	110.48	0.00241 **
nucleoli	1	3.87	674	106.61	0.04906 *
mitoses	1	3.72	673	102.89	0.05378 .

Signif. codes: 0 *** 0.001 ** 0.01 * 0.05 . 0.1 1

The intercept only model is first compared to the model with `clump`. The deviance to test $H_0 : \beta_{\text{clump}} = 0$ drops from 884.35 to 458.48 ($\Delta D = 425.87$) (p-value, according to χ_1^2 , close to 0). Reject H_0 . Then we test $H_0 : \beta_{\text{cellsize}} = 0$: $\Delta D = 261.91$ and p-value $\simeq 0$. H_0 is rejected.

The analysis depends on the order the variables are entered in the model:

```
> breast.fit2 <- glm(status~clump+cellshape+adhesion+singlesize+
  nuclei+chromatin+nucleoli+mitoses+cellsize ,data=breast ,
  family=binomial ,na.action=na.omit)
```

```
> anova(breast.fit2 , test="Chisq")
```

Terms added sequentially (first to last)

	Df	Deviance	Resid.	Df	Resid. Dev	Pr(>Chi)	
NULL				682	884.35		
clump	1	425.87		681	458.48	< 2.2e-16	***
cellshape	1	258.09		680	200.39	< 2.2e-16	***
adhesion	1	40.17		679	160.23	2.330e-10	***
singlesize	1	9.44		678	150.79	0.002127	**
nuclei	1	29.38		677	121.41	5.954e-08	***
chromatin	1	10.58		676	110.83	0.001145	**
nucleoli	1	4.18		675	106.66	0.041022	*
mitoses	1	3.77		674	102.89	0.052188	.
cellsize	1	0.00		673	102.89	0.976067	

AIC on all models. Here $2^9 = 512$ possible models. When too many models, can be used in a stepwise manner.

```
> step(breast.fit)
Start:  AIC=122.89
status ~ clump + cellsize + cellshape + adhesion + singlesize +
        nuclei + chromatin + nucleoli + mitoses
```

	Df	Deviance	AIC
- cellsize	1	102.89	120.89
- singlesize	1	103.27	121.27
- cellshape	1	104.74	122.74
<none>		102.89	122.89
- mitoses	1	106.61	124.61
- nucleoli	1	106.66	124.66
- adhesion	1	110.31	128.31
- chromatin	1	110.33	128.33
- clump	1	120.72	138.72
- nuclei	1	122.07	140.07

Step: AIC=120.89

status ~ clump + cellshape + adhesion + singlesize + nuclei + chromatin + nucleoli + mitoses

	Df	Deviance	AIC
- singlesize	1	103.27	119.27
<none>		102.89	120.89
- mitoses	1	106.66	122.66
- cellshape	1	106.66	122.66
- nucleoli	1	106.76	122.76
- adhesion	1	110.64	126.64
- chromatin	1	110.70	126.70
- clump	1	121.10	137.10
- nuclei	1	122.07	138.07

Step: AIC=119.27

status ~ clump + cellshape + adhesion + nuclei + chromatin + nucleoli + mitoses

	Df	Deviance	AIC
<none>		103.27	119.27
- mitoses	1	107.14	121.14
- nucleoli	1	107.72	121.72
- cellshape	1	107.90	121.90
- chromatin	1	111.69	125.69
- adhesion	1	112.17	126.17
- clump	1	121.55	135.55
- nuclei	1	123.15	137.15

```
Call: glm(formula = status ~ clump + cellshape + adhesion + nuclei +
  chromatin + nucleoli + mitoses, family = binomial, data = breast,
  na.action = na.omit)
```

Coefficients:

(Intercept)	clump	cellshape	adhesion	nuclei	chromatin	nucleoli	mitoses
-9.9828	0.5340	0.3453	0.3425	0.3883	0.4619	0.2261	0.5312

Degrees of Freedom: 682 Total (i.e. Null); 675 Residual

(16 observations deleted due to missingness)

Null Deviance: 884.4

Residual Deviance: 103.3 AIC: 119.3

At each step, all the submodels with one less variable are considered, and the one with the smallest AIC is chosen. The procedure is stopped when no improvement in AIC is observed.

A full search, would give:

```
> require(MuMIn)
> options(na.action = "na.fail")
> breast.woNA <- na.omit(breast)
> breast.fit.woNA <- glm(status~clump+cellsize+cellshape+adhesion+singlesize+nuclei+chromatin,
> breast.all <- dredge(breast.fit.woNA,rank = "AIC")
> head(breast.all)
```

Model selection table

	(Intrc)	adhsn	cllsh	cllsz	chrmt	clump	mitss	nucle
252	-9.983	0.3425	0.3453		0.4619	0.5340	0.5312	0.3883
508	-10.100	0.3299	0.3182		0.4465	0.5346	0.5341	0.3831
220	-9.767	0.3375	0.3495		0.4713	0.6225		0.3786
256	-9.975	0.3415	0.3396	0.007847	0.4610	0.5334	0.5302	0.3883
254	-10.010	0.3450		0.233300	0.4794	0.5759	0.5310	0.4203
124	-9.990	0.3502	0.4655		0.5203	0.5478	0.5690	0.3871

	nuc1l	sngls	df	logLik	AIC	delta	weight
252	0.2261		8	-51.633	119.3	0.00	0.351
508	0.2125	0.09612	9	-51.445	120.9	1.62	0.156
220	0.2432		7	-53.572	121.1	1.88	0.137
256	0.2252		9	-51.633	121.3	2.00	0.129
254	0.2500		8	-52.677	121.4	2.09	0.124
124			7	-53.861	121.7	2.46	0.103

Models ranked by AIC(x)

Cargo vessels

Damages caused by waves to the forward section of certain cargo-carrying vessels (from McCullagh and Nelder (1989), pages 204-208).

The output is the number of damage incidents. The explanatory variables are the ship type (from *A* to *E*), the year of construction (classes 1960-64, 1965-69, 1970-74, 1975-79), the period of operation (1960-74, 1975-79) and the aggregate number of months service.

```
> head(ship)
  Ship   Year  Period Aggregate Incidents
1    A 1960-64 1960-74         127         0
2    A 1960-64 1975-79          63         0
3    A 1965-69 1960-74        1095         3
4    A 1965-69 1975-79        1095         4
5    A 1970-74 1960-74        1512         6
6    A 1970-74 1975-79        3353        18
```

```
> ftable(xtabs(Incidents ~ Ship+Year+Period, data=ship))
      Period 1960-74 1975-79
```

Ship	Year	Period 1960-74	Period 1975-79
A	1960-64	0	0
	1965-69	3	4
	1970-74	6	18
	1975-79	0	11
B	1960-64	39	29
	1965-69	58	53
	1970-74	12	44
	1975-79	0	18
C	1960-64	1	1
	1965-69	0	1
	1970-74	6	2
	1975-79	0	1
D	1960-64	0	0
	1965-69	0	0
	1970-74	2	11
	1975-79	0	4
E	1960-64	0	0
	1965-69	7	7
	1970-74	5	12
	1975-79	0	1

The postulated log-link Poisson model is:

$$\begin{aligned}\log(E(\text{Incidents})) &= \\ &= \beta_0 + \log(\text{Aggregate}) + \beta_1 \text{Ship} + \beta_2 \text{Year} \\ &\quad + \beta_3 \text{Period}.\end{aligned}$$

Remark: β_1 , β_2 and β_3 are vectors, whose length is equal to the number of levels of the corresponding factor minus one.

Rate model

The term $\log(\text{Aggregate})$ has a coefficient $\beta_s = 1$, known a priori (an **offset** in the GLM terminology). This variable is included in the model to control for the number of opportunities for the event to occur (effort).

We note the equivalence between:

$$\log(E(\text{Incidents})) = \beta_0 + \log(\text{Aggregate}) + \dots$$

and

$$\log(E(\text{Incidents})/\text{Aggregate}) = \beta_0 + \dots$$

In cases like the above, where the effort variable is a count, the modelling can be seen as a Poisson approximation to a Binomial model. In other situation (effort variable is time, for example) this is not possible.

The pertinence of fixing $\beta_s = 1$ can be checked by fitting a model where β_s is also estimated from the data

```
> ship.glm <- glm(Incidents ~ Ship+Year+Period+offset(log(Aggregate)),
family=poisson, data=ship)
> summary(ship.glm)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-6.40590	0.21744	-29.460	< 2e-16	***
ShipB	-0.54334	0.17759	-3.060	0.00222	**
ShipC	-0.68740	0.32904	-2.089	0.03670	*
ShipD	-0.07596	0.29058	-0.261	0.79377	
ShipE	0.32558	0.23588	1.380	0.16750	
Year1965-69	0.69714	0.14964	4.659	3.18e-06	***
Year1970-74	0.81843	0.16977	4.821	1.43e-06	***
Year1975-79	0.45343	0.23317	1.945	0.05182	.
Period1975-79	0.38447	0.11827	3.251	0.00115	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

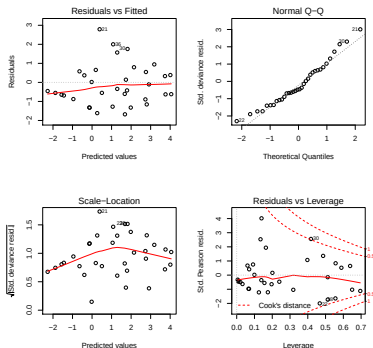
Null deviance: 146.328 on 33 degrees of freedom
 Residual deviance: 38.695 on 25 degrees of freedom
 AIC: 154.56

Goodness of fit test with the deviance statistic:

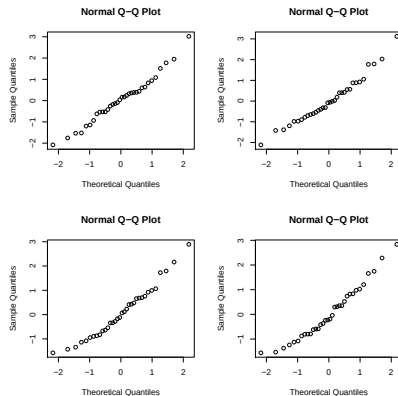
$$P(D > 38.695) = 0.0395,$$

where we have used a χ^2_{25} distribution for D .

We therefore reject (at the 5% level) the null hypothesis that the model fits the data well.



Randomized residuals



Reasons for lack of fit?

If $\text{Var}(Y_i) > E(Y_i)$ the Poisson model does not hold anymore and this phenomenon is called **overdispersion**. (If $\text{Var}(Y_i) < E(Y_i)$ we have **underdispersion** but this latter is less common).

One of the main reasons for overdispersion is the clustering in the population (the parameter vary from cluster to cluster, as a function of cluster size for example). Statistically speaking, the parameter is regarded as random rather than fixed.

A large value of the deviance may indicate overdispersion.

A first check on the data is to assume that $\text{Var}(Y_i) = \sigma^2 v_i = \sigma^2 \mu_i$ and look at the fitted value for σ^2 . This is an *ad-hoc* adjustment and doesn't necessarily correspond to a likelihood.

The model cannot be fitted anymore by maximum likelihood, but via a set of different estimating equations:

$$\sum_{i=1}^n \left(\frac{y_i - \mu_i}{\sigma^2 v(\mu_i)} \right) \frac{\partial \mu_i}{\partial \beta} = 0,$$

that correspond to so called quasi-likelihood functions

$$\sum_{i=1}^n \int_{y_i}^{\mu_i} \frac{y_i - t}{\sigma^2 v(t)} dt.$$

Remarks:

- ▶ The maximum likelihood and the quasi-likelihood approaches give the same estimators for all the models of the one-parameter exponential family (binomial, Poisson, for example).
- ▶ Note that σ^2 does not impact the score equations, but have a multiplicative impact on the standard error of the coefficients (and on the distribution of the difference of deviances).

see ★ Ganio and Schafer (1992), Lambert and Roeder (1995) and Dean and Lawless (1989)

```
> ship.glm.quasi <- glm(Incidents ~ Ship+Year+Period+offset(log(AggregateIncidents)),
family=quasipoisson, data=ship)
> summary(ship.glm.quasi)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-6.40590	0.28276	-22.655	< 2e-16 ***
ShipB	-0.54334	0.23094	-2.353	0.02681 *
ShipC	-0.68740	0.42789	-1.607	0.12072
ShipD	-0.07596	0.37787	-0.201	0.84230
ShipE	0.32558	0.30674	1.061	0.29864
Year1965-69	0.69714	0.19459	3.583	0.00143 **
Year1970-74	0.81843	0.22077	3.707	0.00105 **
Year1975-79	0.45343	0.30321	1.495	0.14733
Period1975-79	0.38447	0.15380	2.500	0.01935 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 1.691028)

Null deviance: 146.328 on 33 degrees of freedom
 Residual deviance: 38.695 on 25 degrees of freedom
 AIC: NA

Parameters interpretation

Recall the model:

$$\begin{aligned}\log(E(\text{Incidents})) &= \\ &= \beta_0 + \log(\text{Aggregate}) + \beta_1 \text{ship} + \beta_2 \text{year} + \beta_3 \text{period}.\end{aligned}$$

In fact, each factor correspond to several parameters (number of levels -1). For example, if we consider that ships of type A are the reference, $\beta_1 \text{ship}$ is:

$$\beta_{1B} \iota(\text{shipB}) + \beta_{1C} \iota(\text{shipC}) + \beta_{1D} \iota(\text{shipD}) + \beta_{1E} \iota(\text{shipE}),$$

where $\iota(\text{ship}i)$ is a dummy variable taking the value 1 for ships of type i and 0 otherwise.

Therefore, β_{1B} is the effect on $\log(\lambda)$ for ships of type B with respect to ships of type A. The estimated value of $\log(\lambda)$ for type B ships is -0.54334 less than that for ships of type A (reference). Equivalently, one can say that the estimated expected number of damages λ is multiplied by $\exp(-0.54334) = 0.58$ for type B ships with respect to type A ships, and therefore reduced.

Ships of type B and C have the lowest risk, Ship of type E the highest.

The oldest ships seems to be the safest, with those built between 1970-74 having the highest risk.

Remark on factor covariates:

In the dummy approach used above, a reference has to be defined (sometimes in a very arbitrary way). The effect of this reference level is melt into the intercept. Interpretation is with respect to the reference. (Here the constraint is $\beta_{1A} = 0$). The default in R is to take the first category (alphabetical order) as the reference. This can be changed by redefining the factor with the function `relevel`.

Alternatively, one could use a different set of **contrasts**, for example those where the effect of each variable is contrasted against an overall mean. The interpretation of the coefficients in this case is different. (Here the constraint is $\sum_{i=1}^5 \beta_{1i} = 0$).

Alternative approach to address overdispersion

Use a different model, for example the **negative binomial** distribution (for $y = 0, 1, 2, \dots$):

$$P(Y_i = y_i) = \frac{\Gamma(y_i + \vartheta)}{\Gamma(\vartheta)\Gamma(y_i + 1)} \left(\frac{\vartheta}{\mu_i + \vartheta} \right)^\vartheta \left(\frac{\mu_i}{\mu_i + \vartheta} \right)^{y_i},$$

where $\Gamma(t) = \int_0^\infty e^{-x} x^{t-1} dx$ (for integer x , $\Gamma(x) = (x-1)!$).
Under this model $E(Y_i) = \mu_i$ and $\text{Var}(Y_i) = \mu_i(\mu_i + 1/\vartheta)$.

Overdispersion can also occur with binomial data. Either use a “quasi-type” model with $\text{Var}(Y_i) = \sigma^2 v(\mu_i)$ or the **beta-binomial** distribution ($y = 0, 1, \dots, m_i$):

$$P(Y_i = y_i) = \binom{m_i}{y_i} \frac{B(\alpha + y_i, m_i + \beta - y_i)}{B(\alpha, \beta)},$$

where $B(u, v) = \int_0^1 x^{u-1} (1-x)^{v-1} dx$.

Here $E(Y_i) = m_i p_i$ and $\text{Var}(Y_i) = m_i p_i (1 - p_i) (1 + (m_i - 1) \tau^2)$, where p_i and τ depends on α and β .

The functions `glm.nb` (package `MASS`) and `negbin`, `betabin` (package `aod`) allows to fit these models.

(Quasi)-separation in binary regression

If the 0 and 1 responses can be (nearly) separated on the basis of their covariates values (**no overlap, or separation**), then there is no solution to the score equations for the binary regression (or the solution is $\beta = \infty$).

Example: Analysis of recreational trips to evaluate the sensitivity of resource usage to entrance charges and costs (Gurmu and Trivedi 1996).

Available variables:

- ▶ TRIPS: Number of boating trips to Lake Somerville, East Texas, in 1980
- ▶ SO: Facility's subjective quality ranking
- ▶ SKI: Respondent's taste for water-skiing
- ▶ I: Income - categorical variable
- ▶ FC3: Cost dummy variable; 1 if an annual user fee is paid at Lake Somerville
- ▶ C1: Travel cost to Lake Conroe
- ▶ C3: Travel cost to Lake Somerville
- ▶ C4: Travel cost to Lake Houston

```
> summary(glm((TRIPS>0)~SO+SKI+FC3+I+C1+C3+C4, family=binomial,
data=recreat))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.881479	0.435663	-6.614	3.74e-11	***
SO	1.437079	0.105799	13.583	< 2e-16	***
SKI	0.405517	0.329754	1.230	0.21879	
FC3	15.660215	907.987771	0.017	0.98624	
I	0.026479	0.084000	0.315	0.75259	
C1	0.008255	0.027099	0.305	0.76064	
C3	-0.080926	0.017134	-4.723	2.32e-06	***
C4	0.071124	0.021697	3.278	0.00105	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

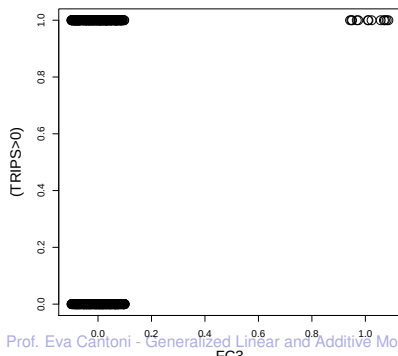
Null deviance: 866.53 on 658 degrees of freedom
Residual deviance: 299.49 on 651 degrees of freedom
AIC: 315.49

Number of Fisher Scoring iterations: 16

Why so?

```
> with(recreat, table(TRIPS > 0, FC3))
```

	FC3	
	0	1
FALSE	417	0
TRUE	229	13



Possible fixes:

- ▶ exact logistic regression (Mehta and Patel 1995).
- ▶ robust regression approach (Rousseeuw and Christmann 2003).
- ▶ Bias reduction method (Firth 1993; Kosmidis and Firth 2009) to reduce finite sample bias in the GLM family. Unaffected by lack of overlap.

A corrected score function is defined

$U^*(\beta) = U(\beta) - \mathcal{J}(\beta) \frac{b_1(\beta)}{n}$ where $\mathcal{J}(\beta)$ is the Fisher information and $b_1(\beta)/n$ is the first term in the asymptotic bias expression of the ML estimator. It corresponds to a penalized log-likelihood (canonical specification)

$$l^*(\beta; y) = l(\beta; y) + \frac{1}{2} \log(|\mathcal{J}(\beta)|).$$

The use of AIC for model selection is controversial.

```
> summary(brglm((TRIPS>0)~SO+SKI+FC3+I+C1+C3+C4, family=binomial,
data=recreat))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.76325	0.41610	-6.641	3.12e-11	***
SO	1.39492	0.10187	13.693	< 2e-16	***
SKI	0.40403	0.32265	1.252	0.21048	
FC3	1.79847	1.71410	1.049	0.29408	
I	0.02348	0.08187	0.287	0.77424	
C1	0.01878	0.02180	0.861	0.38910	
C3	-0.08293	0.01594	-5.202	1.97e-07	***
C4	0.06244	0.01877	3.326	0.00088	***

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 836.18 on 658 degrees of freedom
 Residual deviance: 300.67 on 651 degrees of freedom
 Penalized deviance: 258.0502
 AIC: 316.67

Hauck-Donner phenomenon for logistic regression

The z/t -value is a Wald approximation of the loglikelihood to test $H_0 : \beta_i = 0$ and is sometimes misleading with binomial GLM. In fact, a small value for the z/t -value can either correspond to a small likelihood ratio statistic or to a situation where $|\hat{\beta}_i|$ is large, the Wald approximation is poor and the likelihood ratio statistic is large. These problems can occur in cases when the fitted probabilities are extremely close to 0 or 1.

★ Hauck and Donner (1977)

Excess of zeros in count data

In many applications with count data, the response contains more zeros than expected based on the Poisson (or negative binomial) distribution. We say that there is “excess of zeros”.

Typical examples are: number of visits to the doctor in a given period, number of observed animals (sharks, cods, birds) in a given habitat, number of insurance claims in a year, etc.

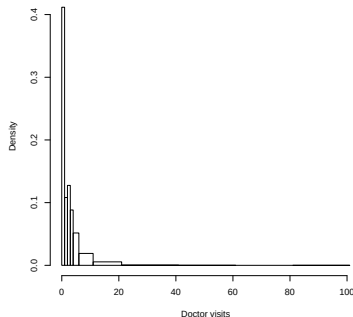
Example

The data are a sample of German male individuals in 1994 and are taken from Riphahn, Wambach, and Million (2003). They have been used to study the demand for health care.

There are 1812 observations. The outcome is `docvisits`, the number of doctor visits in the last three months.

Available covariates are:

- ▶ `age` age
- ▶ `agesq` age squared / 1000
- ▶ `health` health satisfaction, 0 (low) - 10 (high)
- ▶ `handicap` 1 if handicapped, 0 otherwise
- ▶ `hdegree` degree of handicap in percentage points
- ▶ `married` 1 if married, 0 otherwise
- ▶ `schooling` years of schooling
- ▶ `hhincome` household monthly income (Ger. marks / 1000)
- ▶ `children` 1 if children under 16, 0 otherwise
- ▶ `self` 1 if self employed, 0 otherwise
- ▶ `civil` 1 if civil servant, 0 otherwise
- ▶ `bluec` 1 if blue collar employee, 0 otherwise
- ▶ `employed` 1 if employed, 0 otherwise
- ▶ `public` 1 if public health insurance, 0 otherwise
- ▶ `addon` 1 if add-on insurance, 0 otherwise



Proportion of zeros: $746/1812 = 41.2\%$.

```
> mean(docvisits$docvisits)
[1] 2.958057
> mean(docvisits$docvisits[docvisits$docvisits > 0])
[1] 5.028143
```

Hurdle (conditional, two-part, zero-altered) model

Mullahy (1986), Welsh, Cunningham, Donnelly, and Lindenmayer (1996)

$$\begin{aligned} P(Y = y) &= \begin{cases} y = 0 & \text{with prob. } 1 - p(x) \\ y \sim \text{trunc.}\mathcal{P}(\lambda) & \text{with prob. } p(x) \end{cases} \\ &= \begin{cases} (1 - p(x)) & y = 0 \\ p(x) \frac{\exp(-\lambda(z))\lambda(z)^y}{y!(1 - \exp(-\lambda(z)))} & y = 1, 2, \dots \end{cases} \end{aligned}$$

where x and z are two sets of covariates that may or may not be the same.

One can for example use a logistic model for $p(x)$

$$\log \left(\frac{p(x)}{1 - p(x)} \right) = x^T \beta,$$

and a log-linear type of model for $\lambda(z)$

$$\log(\lambda(z)) = z^T \gamma.$$

Note: A truncated negative binomial distribution can be used instead of the truncated Poisson.

Zero-inflated Poisson (ZIP) model

Lambert (1992)

$$\begin{aligned} P(Y = y) &= \begin{cases} y = 0 & \text{with prob. } \pi(x) \\ y \sim \mathcal{P}(\lambda) & \text{with prob. } 1 - \pi(x) \end{cases} \\ &= \begin{cases} \pi(x) + (1 - \pi(x)) \exp(-\lambda(z)) & y = 0 \\ (1 - \pi(x)) \frac{\exp(-\lambda(z)) \lambda(z)^y}{y!} & y = 1, 2, \dots \end{cases} \end{aligned}$$

where x and z are two sets of covariates that may or may not be the same.

One can for example use a logistic model for $1 - \pi(x)$

$$\log \left(\frac{\pi(x)}{1 - \pi(x)} \right) = x^T \alpha,$$

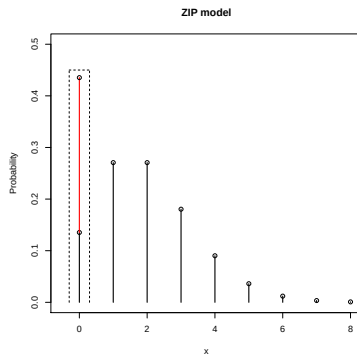
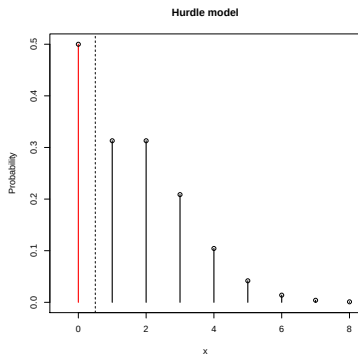
and a log-linear type of model for $\lambda(z)$

$$\log(\lambda(z)) = z^T \delta.$$

A negative binomial distribution can be used instead $\rightarrow$ ZINB.

Note:

- ▶ $p(x)$ in the hurdle model is the probability of “crossing the hurdle” (i.e. $y_i > 0$).
- ▶ $\pi(x)$ in the ZIP model is the probability of observing a zero from the spike at zero.



Moments

Recall that λ_i is the expectation (and the variance) of a Poisson distribution.

Hurdle model

The expectation of a truncated Poisson variable is $\frac{\lambda_i}{1 - \exp(-\lambda_i)}$, and therefore:

$$E(Y_i) = p(x_i) \frac{\lambda_i}{1 - \exp(-\lambda_i)} = p(x_i) E(Y_i | Y_i > 0)$$

$$\text{Var}(Y_i) = p(x_i) \frac{\lambda_i + \lambda_i^2}{1 - \exp(-\lambda_i)} - \left(p(x_i) \frac{\lambda_i}{1 - \exp(-\lambda_i)} \right)^2$$

In this model $\text{Var}(Y_i)$ can be either larger or smaller than $E(Y_i)$ accounting for overdispersion or underdispersion.

ZIP model

$$E(Y_i) = (1 - \pi(x_i))\lambda_i$$

$$\text{Var}(Y_i) = (1 - \pi(x_i))(\lambda_i + \pi(x_i)\lambda_i^2)$$

For this model $\text{Var}(Y_i) > E(Y_i)$ (overdispersion) if $\pi(x_i) > 0$.

In both models, the quantity are estimated by plugging-in the estimates of the parameters.

Estimation - Hurdle models

With a logistic model for $p(x)$

$$\text{logit}(p(x)) = \log\left(\frac{p(x)}{1 - p(x)}\right) = x^T \beta,$$

and a log-linear model for $\lambda(z)$

$$\log(\lambda(z)) = z^T \gamma,$$

the likelihood $L(\beta, \gamma)$ for the hurdle Poisson model is

$$L(\beta, \gamma) = \prod_{i=1}^n P(Y_i = y_i) = \prod_{i=1}^n (1 - p(x_i))^{\iota(y_i=0)} \left(p(x_i) \frac{\exp(-\lambda(z_i)) \lambda(z_i)^{y_i}}{y_i! (1 - \exp(-\lambda(z_i)))} \right)^{\iota(y_i>0)},$$

where $\iota(A) = 1$ if A is true and 0 otherwise.

The log-likelihood $l(\beta, \gamma)$ is:

$$\begin{aligned}
 l(\beta, \gamma) &= \sum_{i=1}^n \iota(y_i = 0) \log(1 - p(x_i)) + \\
 &\quad \sum_{i=1}^n \iota(y_i > 0) [\log(p(x_i)) - \lambda(z_i) + y_i \log(\lambda(z_i)) \\
 &\quad - \log(y_i!) - \log(1 - \exp(-\lambda(z_i)))] \\
 &= \sum_{y_i=0} \log \left(\frac{1}{1 + \exp(x_i^T \beta)} \right) + \sum_{y_i>0} \log \left(\frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right) + \\
 &\quad \sum_{y_i>0} \left(y_i z_i^T \gamma - \exp(z_i^T \gamma) - \log(1 - \exp(-\exp(z_i^T \gamma))) - \log(y_i!) \right) \\
 &= l(\beta) + l(\gamma)
 \end{aligned}$$

The components in $l(\beta, \gamma)$ can be fitted separately (orthogonal parameters).

Note: The logistic link can be replaced by other links for the binomial family, and the expressions updated accordingly.

Note: The log-likelihood for the conditional negative binomial model is obtained similarly.

The two independent parts of a Poisson hurdle models can also be fitted by exploiting the GLM framework (Barry and Welsh 2002):

1. A binary (logistic) regression to separate the zero from the positive (using all the data, and coding 1 all the positive values).
2. A truncated Poisson GLM for the positive (using only the portion of data corresponding to the positive responses).

The expectation of a truncated Poisson variable is

$$\mu = h(\lambda) = \frac{\lambda}{1 - \exp(-\lambda)},$$

so $\lambda = h^{-1}(\mu)$.

With $\log(\lambda) = z^T \gamma$, it gives $\log(h^{-1}(\mu)) = z^T \gamma$, so the link function is $g(x) = \log(h^{-1}(x))$.

Also, the variance of a truncated Poisson variable is

$$v(\mu) = \mu(1 + h^{-1}(\mu) - \mu),$$

and the deviance

$$D(y; \mu) = 2 \left[y \log(h^{-1}(\mu)) - h^{-1}(\mu) \right. \\ \left. \log(1 - \exp(h^{-1}(\mu))) - y \log(h^{-1}(y)) \right. \\ \left. h^{-1}(y) + \log(1 - \exp(h^{-1}(y))) \right]$$

Use the GLM setting with the above specifications.

Remark: Cannot invert h explicitly.

Estimation - ZIP models

With a logistic model for $\pi(x)$

$$\log \left(\frac{\pi(x)}{1 - \pi(x)} \right) = x^T \alpha,$$

and a log-linear model for $\lambda(z)$

$$\log(\lambda(z)) = z^T \delta,$$

the likelihood $L(\alpha, \delta)$ for the ZIP model is

$$\begin{aligned} L(\alpha, \delta) &= \prod_{i=1}^n P(Y_i = y_i) = \\ &\prod_{i=1}^n (\pi(x_i) + (1 - \pi(x_i)) \exp(-\lambda(z_i)))^{\iota(y_i=0)} \\ &\times \left((1 - \pi(x_i)) \frac{\exp(-\lambda(z_i)) \lambda(z_i)^{y_i}}{y_i!} \right)^{\iota(y_i>0)} \end{aligned}$$

This likelihood has to be maximized jointly for (α, δ) .

The ZIP model can also be fitted using the EM algorithm (Dempster, Laird, and Rubin 1977).

Let $W_i = 1$ when y_i is from the spike at zero, and 0 otherwise (W_i is unknown). The likelihood becomes

$$L(\alpha, \delta | W_i) = \prod_{i=1}^n P(Y_i = y_i) = \prod_{i=1}^n \pi(x_i)^{W_i \mathbb{I}(y_i=0)} \times (1 - \pi(x_i))^{1-W_i} \times \left(\frac{\exp(-\lambda(z_i)) \lambda(z_i)^{y_i}}{y_i!} \right)^{1-W_i}$$

Conditioning on W_i and using the EM algorithm, has the effect to separate the components.

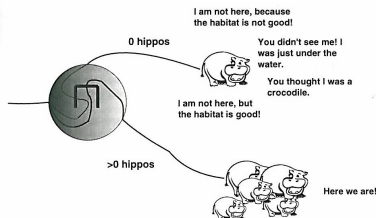
Sources of zeros

Sometimes one likes to discriminate between various type of zeros. There are at least 4 sorts of zeros, see Zuur, Ieno, Walker, Saveliev, and Smith (2009):

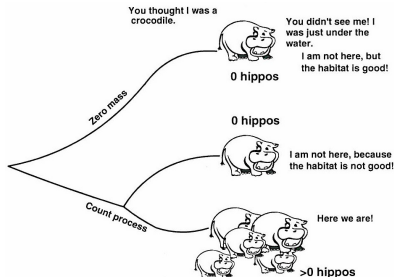
1. structural zeros (e.g. animal not present, because habitat not suitable).
2. design error, poor experimental design or sampling practices are the reason (e.g. sampling at the wrong time, for too short a time period, in too small an area).
3. observer error (e.g. species that are difficult to identify).
4. “bird” error: the habitat is suitable, but the site is not used.

- └ Modeling excess of zeros in count data
- └ Comparison of ZIP and hurdle models

Hurdle model



ZIP model



Figures from Zuur et al. (2009).

Model	Hurdle	ZIP
Parametrisation	orthogonal	not orthogonal
Estim. equations	separate	involve all parameters
Fit components	separately	simultaneously or use EM to separate
Interpret param.	separately	simultaneously
	$p_i()$ = presence prob.	$\pi()$ mixing prob.
	cond. abundance	null abundance

Hurdle fit of doctor visits data: abundance.

Count model coefficients (truncated poisson with log link):					
	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.711909	0.274745	9.871	< 2e-16	***
age	-0.019347	0.012227	-1.582	0.113573	
agesq	0.340346	0.138224	2.462	0.013805	*
health	-0.165889	0.006030	-27.510	< 2e-16	***
handicap	0.360922	0.073755	4.893	9.91e-07	***
hdegree	-0.004873	0.001263	-3.859	0.000114	***
married	-0.150048	0.039991	-3.752	0.000175	***
schooling	-0.001591	0.007247	-0.220	0.826214	
hhincome	0.005181	0.007607	0.681	0.495784	
children	0.106423	0.038281	2.780	0.005435	**
self	-0.183134	0.071802	-2.551	0.010755	*
civil	-0.123018	0.079012	-1.557	0.119481	
bluec	0.077810	0.039401	1.975	0.048285	*
employed	-0.111501	0.046732	-2.386	0.017034	*
public	0.109596	0.069555	1.576	0.115100	
addon	0.261478	0.094248	2.774	0.005531	**

Hurdle fit of doctor visits data: presence/absence.

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	4.228318	0.970927	4.355	1.33e-05	***
age	-0.100415	0.045999	-2.183	0.0290	*
agesq	1.252381	0.537255	2.331	0.0197	*
health	-0.314382	0.028633	-10.980	< 2e-16	***
handicap	0.125226	0.427901	0.293	0.7698	
hdegree	0.003857	0.008130	0.474	0.6352	
married	0.227869	0.147236	1.548	0.1217	
schooling	-0.004350	0.023410	-0.186	0.8526	
hhincome	0.032699	0.026482	1.235	0.2169	
children	-0.329635	0.134636	-2.448	0.0144	*
self	-0.245749	0.199960	-1.229	0.2191	
civil	-0.235280	0.214703	-1.096	0.2731	
bluec	0.029496	0.135280	0.218	0.8274	
employed	0.036584	0.182606	0.200	0.8412	
public	0.051585	0.187571	0.275	0.7833	
addon	-0.034627	0.379243	-0.091	0.9272	

ZIP fit of doctor visits data: abundance.

Count model coefficients (poisson with log link):					
	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.696432	0.275366	9.792	< 2e-16	***
age	-0.018650	0.012254	-1.522	0.128015	
agesq	0.332125	0.138565	2.397	0.016535	*
health	-0.165427	0.006021	-27.473	< 2e-16	***
handicap	0.361505	0.073723	4.904	9.41e-07	***
hdegree	-0.004890	0.001263	-3.872	0.000108	***
married	-0.149734	0.039960	-3.747	0.000179	***
schooling	-0.001747	0.007231	-0.242	0.809124	
hhincome	0.005360	0.007530	0.712	0.476584	
children	0.102861	0.038276	2.687	0.007202	**
self	-0.180518	0.072126	-2.503	0.012321	*
civil	-0.124073	0.079062	-1.569	0.116575	
bluec	0.077995	0.039393	1.980	0.047710	*
employed	-0.111584	0.046757	-2.386	0.017012	*
public	0.111776	0.069614	1.606	0.108349	
addon	0.258275	0.094552	2.732	0.006303	**

ZIP fit of doctor visits data: presence/absence.

Zero-inflation model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.248435	1.016598	-4.179	2.93e-05	***
age	0.103531	0.048036	2.155	0.0311	*
agesq	-1.264186	0.560348	-2.256	0.0241	*
health	0.292041	0.029418	9.927	< 2e-16	***
handicap	-0.065816	0.435600	-0.151	0.8799	
hdegree	-0.005067	0.008355	-0.607	0.5442	
married	-0.274227	0.154044	-1.780	0.0750	.
schooling	0.003972	0.024582	0.162	0.8716	
hhincome	-0.032490	0.027708	-1.173	0.2410	
children	0.361796	0.141202	2.562	0.0104	*
self	0.202949	0.213576	0.950	0.3420	
civil	0.201870	0.229784	0.879	0.3797	
bluec	-0.013367	0.140868	-0.095	0.9244	
employed	-0.057557	0.188931	-0.305	0.7606	
public	-0.012178	0.200784	-0.061	0.9516	
addon	0.077311	0.385901	0.200	0.8412	

- └ Modeling excess of zeros in count data
- └ Comparison of ZIP and hurdle models

Model validation and inference

Both the hurdle and the ZIP model are fitted by using the likelihood. All the inferential methods therefore apply: likelihood ratio tests, Akaike criterion (AIC), etc.

The Poisson hurdle model being separable, the inference on its two components can be carried over separately. In addition, the GLM framework fully applies and one can look at the deviance as well.

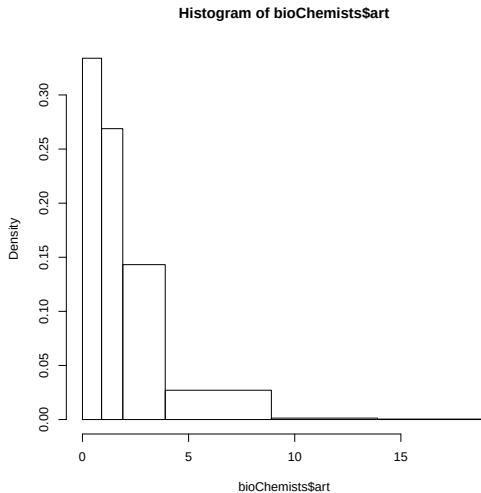
Graphical validation tools are not implemented in the package `pscl`.

Gender differences in science

Study analyzing the established presence of gender differences in science, see Long (1990).

A sample of 915 biochemistry graduate students. The response is `art`, that is the count of articles produced during the last 3 years of Ph.D.

- ▶ `fem` factor indicating student gender (Men or Women)
- ▶ `mar` factor indicating student marital status (Single or Married)
- ▶ `kid5` number of children aged 5 or younger
- ▶ `phd` prestige of Ph.D. program, based on Cartter, Roose and Andersen, and Jones et al studies.
- ▶ `ment` count of articles produced by Ph.D. mentor during last 3 years



Count model coefficients (truncated poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.67114	0.12246	5.481	4.24e-08	***
femWomen	-0.22858	0.06522	-3.505	0.000457	***
marMarried	0.09649	0.07283	1.325	0.185209	
kid5	-0.14219	0.04845	-2.934	0.003341	**
phd	-0.01273	0.03130	-0.407	0.684343	
ment	0.01875	0.00228	8.222	< 2e-16	***

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.23680	0.29552	0.801	0.4230	
femWomen	-0.25115	0.15911	-1.579	0.1144	
marMarried	0.32623	0.18082	1.804	0.0712	.
kid5	-0.28525	0.11113	-2.567	0.0103	*
phd	0.02222	0.07956	0.279	0.7800	
ment	0.08012	0.01302	6.155	7.52e-10	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> extractAIC(bioChurdle)
```

```
[1] 12.000 3234.623
```

Count model coefficients (truncated poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.65835	0.12440	5.292	1.21e-07	***
femWomen	-0.20919	0.07283	-2.872	0.00408	**
kid5	-0.12937	0.05276	-2.452	0.01421	*
marMarried	0.09998	0.07304	1.369	0.17106	
phd	-0.01209	0.03130	-0.386	0.69922	
ment	0.01879	0.00228	8.242	< 2e-16	***
femWomen: kid5	-0.06613	0.11152	-0.593	0.55323	

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.25256	0.30450	0.829	0.4069	
femWomen	-0.27119	0.18436	-1.471	0.1413	
kid5	-0.29739	0.12455	-2.388	0.0170	*
marMarried	0.32078	0.18258	1.757	0.0789	.
phd	0.02140	0.07965	0.269	0.7882	
ment	0.08015	0.01302	6.155	7.52e-10	***
femWomen: kid5	0.04861	0.22555	0.216	0.8293	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> extractAIC(bioChurdlewithint)
```

```
[1] 14.000 3238.217
```

Submodel 1

Count model coefficients (truncated poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.67114	0.12246	5.481	4.24e-08	***
femWomen	-0.22858	0.06522	-3.505	0.000457	***
marMarried	0.09649	0.07283	1.325	0.185209	
kid5	-0.14219	0.04845	-2.934	0.003341	**
phd	-0.01273	0.03130	-0.407	0.684343	
ment	0.01875	0.00228	8.222	< 2e-16	***

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.32116	0.11814	2.718	0.00656	**
kid5	-0.14306	0.09484	-1.508	0.13144	
ment	0.08086	0.01248	6.479	9.25e-11	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> extractAIC(bioChurdlesub1)
[1] 9.000 3235.348
```

Submodel 2

Count model coefficients (truncated poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.691554	0.054696	12.644	< 2e-16	***
femWomen	-0.238888	0.064766	-3.688	0.000226	***
kid5	-0.114713	0.044280	-2.591	0.009580	**
ment	0.018433	0.002228	8.275	< 2e-16	***

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.24871	0.10766	2.310	0.0209	*
ment	0.08092	0.01249	6.477	9.39e-11	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 9

Log-likelihood: -1611 on 6 Df

```
> extractAIC(bioChurdlesub2)
```

```
[1] 6.000 3233.664
```


ZIP fit

Count model coefficients (poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.640838	0.121307	5.283	1.27e-07	***
femWomen	-0.209145	0.063405	-3.299	0.000972	***
marMarried	0.103751	0.071111	1.459	0.144565	
kid5	-0.143320	0.047429	-3.022	0.002513	**
phd	-0.006166	0.031008	-0.199	0.842378	
ment	0.018098	0.002294	7.888	3.07e-15	***

Zero-inflation model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-0.577059	0.509386	-1.133	0.25728	
femWomen	0.109746	0.280082	0.392	0.69518	
marMarried	-0.354014	0.317611	-1.115	0.26502	
kid5	0.217097	0.196482	1.105	0.26919	
phd	0.001274	0.145263	0.009	0.99300	
ment	-0.134114	0.045243	-2.964	0.00303	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Longitudinal data

In **longitudinal studies** (in contrast with cross-sectional studies) individuals are measured repeatedly over time. Longitudinal are also called **panel data** (mostly in economics).

Independence is assumed between individuals (as in GLM), but not between measurements of the same individual.

Longitudinal studies can distinguish changes over time within individuals.

Sometime **clusters** are defined by other groups than an individual (medical practices, cities, etc.).

Notation: see page 43.

Possible approaches to repeated measurements

Model Y_{it} by means of x_{it} :

- ▶ **Marginal models** (as in cross-sectional studies): not only $g(E(Y_{it})) = x_{it}^T \beta$ ($t = 1, \dots, n_i$), but also $Var(Y_i) = V_i(\phi, \alpha)$.
- ▶ **Random (or mixed) effects model**: the correlation among repeated responses is implied by varying regression coefficients across individuals:
 $g(E(Y_{it}|\gamma_i)) = x_{it}^T \beta + z_{it}^T \gamma_i$, and γ_i are iid from a distribution with $E(\gamma_i) = 0$ and $Var(\gamma_i) = \sigma_\gamma^2 I$.
A large literature exists for these models in the Gaussian, identity-link case, much less for other distributions of the exponential family (GLMM models).

Marginal models I

The inference drawn from the marginal model is **population average** versus the **subject specific** that can be drawn from the random effect models.

Ingredients of the model:

- ▶ The marginal expectation of the response $E(Y_{it}) = \mu_{it}$ depends on a set of explanatory variables x_{it} via $g(\mu_{it}) = x_{it}^T \beta$, where g is the link function (same or similar choices as for GLM).
- ▶ The marginal variance depends on the marginal mean $\text{Var}(Y_{it}) = \phi v(\mu_{it})$.
- ▶ The correlation between Y_{it} and $Y_{it'}$ is a function of the marginal means and perhaps of additional parameters α .

Marginal models II

β is the parameter of interest, ϕ and α are nuisance parameters.

The parameters β are interpreted as for GLM.

A marginal distribution for Y_{it} is postulated (e.g. Bernoulli, Poisson). Note, however, that this does not define a joint multivariate distribution for Y_i ($\rightsquigarrow$ no likelihood).

Marginal models III

The regression parameters β are estimated by the **generalized estimating equations (GEE)** approach. By analogy with the score equations of a GLM (see p. 81), we solve (for β):

$$\sum_{i=1}^n D_i^T V_i(\alpha)^{-1} (Y_i - \mu_i) = 0,$$

with $D_i = d\mu_i/d\beta$, $V_i(\alpha) = \phi A_i^{1/2} R(\alpha) A_i^{1/2}$, where $A_i = \text{diag}(v(\mu_{it}))$ and $R(\alpha)$ is a “**working**” **correlation** matrix (in contrast to the true correlation matrix $\text{Corr}(Y_i)$).

The GEE can be written in the form of an iterative reweighted least squares algorithm.

Choices for the working correlation matrix $R(\alpha)$

independence $R(\alpha) = I$, and we have in fact a GLM model.

fixed $R(\alpha)$ have a predefined known form.

exchangeable all the correlations $R(\alpha)_{t,t'}$ are equal to α .

autoregressive (AR) correlation decreases with time difference,
e.g. $R(\alpha)_{t,t'} = \alpha^{|t-t'|}$.

m-dependence Observations are correlated up to time distance m .

unstructured/unspecified $R(\alpha)$ is completely free (beside a diagonal of 1's and the symmetry constraint).

Estimators of ϕ and α

A procedure that iterates between a modified Fisher scoring for β and (moment) estimation of α and ϕ is used.

If we define $r_{it} = (y_{it} - \mu_{it}) / \sqrt{v(\mu_{it})}$, we have that $\text{Var}(r_{it}) = \phi$.
An estimator of ϕ can be obtained by ($N = \sum_{i=1}^n n_i$)

$$\hat{\phi} = \sum_{i=1}^n \sum_{t=1}^{n_i} \frac{\hat{r}_{it}^2}{N - (p + 1)}.$$

The specific estimator of $R(\alpha)$ depends upon the choice of the correlation structure. The general approach is to estimate α by a simple function of couples of residuals $\hat{r}_{iu}, \hat{r}_{iv}$.

Some of the correlation structure require clusters of same size, that is, $n_i = n_{\text{same}}$ for all i .

Examples I

- ▶ If $R(\alpha)_{t,t'} = \alpha$ (exchangeable) for all $t \neq t'$, then given $\hat{\phi}$, we have

$$\hat{\alpha} = \frac{1}{\hat{\phi}} \sum_{i=1}^n \sum_{t > t'} \hat{r}_{it} \hat{r}_{it'} / (K - (p + 1)),$$

where $K = \sum_{i=1}^n 1/2 n_i(n_i - 1)$.

- ▶ If $(R_{\alpha,i})_{tt'} = \alpha^{|t-t'|}$ (autoregressive) an option is to use

$$\hat{\alpha}_{t,t'} = \sum_{i=1}^n \frac{\sum_{t=1}^{n_i-(t-t')} \hat{r}_{it} \hat{r}_{it'}}{n_i}.$$

Examples II

- $\alpha = (\alpha_1, \dots, \alpha_{n_{\text{same}}-1})$, where $\alpha_t = R(\alpha)_{t,t+1}$. Then

$$\hat{\alpha}_t = \frac{1}{\hat{\phi}} \sum_{i=1}^n \hat{r}_{it} \hat{r}_{i(t+1)} / (n - (p + 1)).$$

If $R(\alpha)$ is tridiagonal with $R(\alpha)_{t,t+1} = \alpha_t$ (one-dependent model), then if we let $\alpha_t = \alpha$, we can estimate it by

$$\hat{\alpha} = \sum_{t=1}^{n_{\text{same}}-1} \hat{\alpha}_t / (n_{\text{same}} - 1).$$

Extension to m -dependence possible.

Examples III

- If $R(\alpha)$ is totally unspecified, use

$$\hat{R} = \frac{1}{\hat{\phi}n} \sum_{i=1}^n \hat{A}_i^{-1/2} (Y_i - \hat{\mu}_i) (Y_i - \hat{\mu}_i)^T \hat{A}_i^{-1/2}.$$

Remarks

- ▶ For the independence, exchangeable and m -dependence correlation structure, ϕ does not need to be computed to solve the estimating equations (it cancels out). It is needed for the AR correlation structure.
- ▶ The exchangeable correlation choice allows for different numbers of observations and observations times.
- ▶ The AR-correlation can accommodate an arbitrary number and spacing of observations.

Extension of GEE to include random effects: ★ **Mixed models for non normal data, see Zeger, Liang, and Albert (1988).**

Asymptotic distribution of the GEE estimator

If a $\sqrt{n}$ -consistent estimator is used to estimate α and ϕ , it can be proven that $\sqrt{n}(\hat{\beta} - \beta)$ is asymptotically normal with variance

$$\Omega = \lim_{n \rightarrow \infty} nM^{-1}QM^{-1},$$

where

$$M = \sum_{i=1}^n D_i^T V_i^{-1} D_i,$$

and

$$Q = \sum_{i=1}^n D_i^T V_i^{-1} \text{Var}(Y_i) V_i^{-1} D_i.$$

Remarks

- ▶ The asymptotic variance of $\hat{\beta}$ does not depend on the choice of the estimators for α and ϕ among those $\sqrt{n}$ -consistent.
- ▶ The consistency of $\hat{\beta}$ (and $\hat{\Omega}$, see page 200) depends only on the correct specification of the mean μ_i and not on the correct specification of the correlation structure.
- ▶ Choosing R close to the true correlation matrix increases efficiency.
- ▶ Inference and diagnostic are limited.

Diagnostic

As in GLM, we define the Pearson residuals

$$\hat{r}_{it} = \frac{y_{it} - \hat{\mu}_{it}}{\sqrt{v(\hat{\mu}_{it})}}.$$

They can be plotted to identify outliers and other violation of the assumption.

Inference

Use the asymptotic theory to derive (approximate) confidence intervals and z-tests.

An estimator of the asymptotic variance Ω is needed. This is $\hat{\Omega} = \hat{M}^{-1} \hat{Q} \hat{M}^{-1}$, where

$$\hat{M} = \sum_{i=1}^n \hat{D}_i^T \hat{V}_i^{-1} \hat{D}_i,$$

and

$$\hat{Q} = \sum_{i=1}^n \hat{D}_i^T \hat{V}_i^{-1} (Y_i - \hat{\mu}_i)(Y_i - \hat{\mu}_i)^T \hat{V}_i^{-1} \hat{D}_i.$$

$\hat{D}_i$ and $\hat{V}_i$ are obtained by plugging in $\hat{\beta}$, $\hat{\phi}$ and $\hat{\alpha}$.

The lack of a likelihood function for these models makes inference limited.

★ A C_p -like criterion for variable selection for marginal longitudinal models, see Cantoni, Mills Flemming, and Ronchetti (2005).

★ An Akaike-type criterion for GEE, see Pan (2001).

Analysis of the CARDIA dataset (see p. 33)

```
Call: gee(formula = smoke ~ age + factor(birth) + factor(education) +
  factor(racesex), id = id, data = CARDIA.sub, family = binomial,
  constr = "exchangeable")
```

Coefficients:

	Estimate	Naive S.E.	Naive z	Robust S.E.	Robust z
(Intercept)	-1.33707777	0.54231807	-2.465486	0.54510289	-2.452891
age	0.06480087	0.02646754	2.448315	0.02669744	2.427231
factor(birth)2	-0.19856466	0.14606724	-1.359406	0.14563751	-1.363417
factor(birth)3	-0.40380459	0.23689643	-1.704562	0.23858001	-1.692533
factor(education)2	-0.68712005	0.08313860	-8.264754	0.08490269	-8.093030
factor(education)3	-1.96140806	0.10016299	-19.582164	0.10054908	-19.506972
factor(racesex)2	-0.17668132	0.09763816	-1.809552	0.09900202	-1.784623
factor(racesex)3	-0.13445604	0.10634023	-1.264395	0.10833799	-1.241079
factor(racesex)4	-0.12730906	0.10530815	-1.208919	0.10599195	-1.201120

Estimated Scale Parameter: 1.000362

Number of Iterations: 1

Working Correlation

	[,1]	[,2]	[,3]	[,4]
[1,]	1.000000	0.727139	0.727139	0.727139
[2,]	0.727139	1.000000	0.727139	0.727139
[3,]	0.727139	0.727139	1.000000	0.727139
[4,]	0.727139	0.727139	0.727139	1.000000

```
Call: gee(formula = smoke ~ age + factor(birth) + factor(education) +
  factor(racesex), id = id, data = CARDIA.sub, family = binomial,
  corstr = "AR-M", Mv = 1)
```

Coefficients:

	Estimate	Naive S.E.	Naive z	Robust S.E.	Robust z
(Intercept)	-1.46502183	0.51848241	-2.825596	0.54981825	-2.664557
age	0.07058083	0.02530312	2.789412	0.02691652	2.622212
factor(birth)2	-0.21226645	0.13970893	-1.519348	0.14716072	-1.442412
factor(birth)3	-0.41934193	0.22641663	-1.852081	0.24111643	-1.739168
factor(education)2	-0.70331577	0.07936239	-8.862080	0.08536346	-8.239072
factor(education)3	-1.98023517	0.09576186	-20.678745	0.10143031	-19.523112
factor(racesex)2	-0.19210488	0.09334897	-2.057922	0.09978203	-1.925245
factor(racesex)3	-0.14273599	0.10162928	-1.404477	0.10912066	-1.308056
factor(racesex)4	-0.11428211	0.10047526	-1.137415	0.10674042	-1.070654

Estimated Scale Parameter: 1.006728

Number of Iterations: 2

Working Correlation

	[,1]	[,2]	[,3]	[,4]
[1,]	1.0000000	0.7684591	0.5905294	0.4537977
[2,]	0.7684591	1.0000000	0.7684591	0.5905294
[3,]	0.5905294	0.7684591	1.0000000	0.7684591
[4,]	0.4537977	0.5905294	0.7684591	1.0000000

```
Call: gee(formula = smoke ~ age + factor(education) + factor(racesex),
  id = id, data = CARDIA.sub, family = binomial, corstr = "exchangeable")
```

Coefficients:

	Estimate	Naive S.E.	Naive z	Robust S.E.	Robust z
(Intercept)	-0.51416495	0.251656751	-2.043120	0.253634983	-2.027185
age	0.02272524	0.009743051	2.332456	0.009868721	2.302754
factor(education)2	-0.68987727	0.082992723	-8.312503	0.084840099	-8.131500
factor(education)3	-1.96816994	0.100008335	-19.680059	0.100455198	-19.592515
factor(racesex)2	-0.17769210	0.097552550	-1.821501	0.098969587	-1.795421
factor(racesex)3	-0.13115039	0.106197596	-1.234966	0.108213885	-1.211955
factor(racesex)4	-0.12631590	0.105246698	-1.200189	0.106046722	-1.191134

Estimated Scale Parameter: 0.9997882

Number of Iterations: 1

Working Correlation

	[,1]	[,2]	[,3]	[,4]
[1,]	1.0000000	0.7272402	0.7272402	0.7272402
[2,]	0.7272402	1.0000000	0.7272402	0.7272402
[3,]	0.7272402	0.7272402	1.0000000	0.7272402
[4,]	0.7272402	0.7272402	0.7272402	1.0000000

```
Call: gee(formula = smoke ~ age + factor(education), id = id, data = CARDIA.sub,
  family = binomial, corstr = "exchangeable")
```

Coefficients:

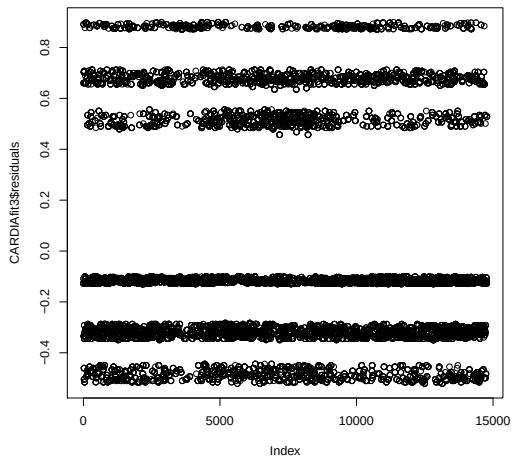
	Estimate	Naive S.E.	Naive z	Robust S.E.	Robust z
(Intercept)	-0.59982654	0.247153965	-2.426935	0.248914981	-2.409765
age	0.02204566	0.009672377	2.279239	0.009739698	2.263485
factor(education)2	-0.70520857	0.082571026	-8.540630	0.084484169	-8.347227
factor(education)3	-1.98709483	0.096223429	-20.650842	0.096193423	-20.657284

Estimated Scale Parameter: 0.9995199

Number of Iterations: 1

Working Correlation

	[,1]	[,2]	[,3]	[,4]
[1,]	1.0000000	0.7275345	0.7275345	0.7275345
[2,]	0.7275345	1.0000000	0.7275345	0.7275345
[3,]	0.7275345	0.7275345	1.0000000	0.7275345
[4,]	0.7275345	0.7275345	0.7275345	1.0000000



Anylsis of the epileptic dataset

Clinical trial with 59 epileptic patients, randomized to either the anti-epileptic drug progabide or to placebo (from Thall and Vail 1990).

Before treatment, the number of seizures in an eight-week period were counted for each patient as a baseline measurement. Is the new treatment effective at reducing the rate of seizures among epileptic patients?

Start with a Poisson model, but...Possible overdispersion...

	Visit			
Treatment	1	2	3	4
Treated	38.7	16.8	23.8	18.8
Placebo	10.8	7.5	24.5	7.3

Also, the observations time are: after 8 (baseline), 10, 12, 14 and 16 weeks. Need to take into account the different length of the observation period. We use an offset of the form $\log(t_{it})$, where $t_{it} = 8$ for the first visit and $t_{it} = 2$ for the consequent ones.

A `post` variable to account for the beginning of the study, which takes the value 1 for visits 1, 2, 3 and 4 and the value 0 for the baseline visit.

Allow for interaction between `post` and `group`.


```
Call: gee(formula = y ~ post * factor(group) + offset(log(rep(c(8,
  2, 2, 2, 2), 59))), id = Subject, data = seizure, family = poisson,
  corstr = "exchangeable")
```

Coefficients:

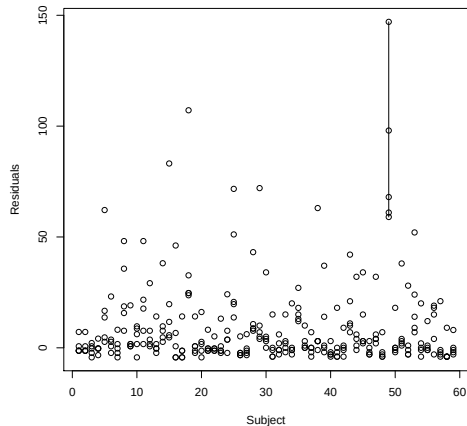
	Estimate	Naive S.E.	Naive z	Robust S.E.	Robust z
(Intercept)	1.34760922	0.1510969	8.9188397	0.1573571	8.5640166
post	0.11183602	0.1545145	0.7237900	0.1159304	0.9646821
factor(group)progabide	0.02753449	0.2071018	0.1329515	0.2217878	0.1241479
post:factor(group)progabide	-0.10472579	0.2197052	-0.4766650	0.2134448	-0.4906459

Estimated Scale Parameter: 19.6797

Number of Iterations: 1

Working Correlation

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	1.0000000	0.7713861	0.7713861	0.7713861	0.7713861
[2,]	0.7713861	1.0000000	0.7713861	0.7713861	0.7713861
[3,]	0.7713861	0.7713861	1.0000000	0.7713861	0.7713861
[4,]	0.7713861	0.7713861	0.7713861	1.0000000	0.7713861
[5,]	0.7713861	0.7713861	0.7713861	0.7713861	1.0000000



A very large outlier!

Without patient 49

```
Call: gee(formula = y ~ post * factor(group) + offset(log(rep(c(8,
  2, 2, 2, 2), 59))), id = Subject, data = seizure, subset = Subject !=
  49, family = poisson, corstr = "exchangeable")
```

Coefficients:

	Estimate	Naive S.E.	Naive z	Robust S.E.	Robust z
(Intercept)	1.3476092	0.1105249	12.1928162	0.1573571	8.5640166
post	0.1118360	0.1231346	0.9082421	0.1159304	0.9646821
factor(group)progabide	-0.1068224	0.1578051	-0.6769263	0.1936977	-0.5514904
post:factor(group)progabide	-0.3023841	0.1933863	-1.5636272	0.1710601	-1.7677071

Estimated Scale Parameter: 10.52997

Number of Iterations: 1

Working Correlation

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	1.000000	0.593689	0.593689	0.593689	0.593689
[2,]	0.593689	1.000000	0.593689	0.593689	0.593689
[3,]	0.593689	0.593689	1.000000	0.593689	0.593689
[4,]	0.593689	0.593689	0.593689	1.000000	0.593689
[5,]	0.593689	0.593689	0.593689	0.593689	1.000000

Large differences in (some) estimated parameters and standard errors!

Generalized linear mixed models (GLMM)

With longitudinal/clustered data one need to take into account the within cluster correlation.

GLM is not adequate. GLMM extends GLM by allowing coefficients to vary with cluster, allowing for within cluster correlation.

GLMM is a compromise between the GLM population-average model and a (over-parametrized) modelling approach that would fit each cluster separately.

Ingredients:

- ▶ response vectors $Y_i = (Y_{i1}, \dots, Y_{in_i})$, for $i = 1, \dots, n$.
- ▶ design matrix X_i for the fixed effects
- ▶ design matrix Z_i for the random effects (possibly overlapping with X_i).
- ▶ random effects γ_i for each cluster i . These are latent (non-observed) random variables.

Assumptions:

- ▶ $Y_{it} \mid \gamma_i$ are independently distributed according to a distribution from the exponential family.
- ▶ γ_i are independently distributed according to $\mathcal{N}(0, \Psi)$.

Two sources of error: the sampling process and the measurement process within the sample itself.

The Gaussian-identity link case

In the Gaussian-identity link case, the model can be written as:

$$Y_i = X_i\beta + Z_i\gamma_i + \epsilon_i \quad (\text{or } Y_{it} = x_{it}^T\beta + z_{it}^T\gamma_i + \epsilon_{it}),$$

with $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2 I)$ and $\gamma_i \sim \mathcal{N}(0, \Psi)$, for example $\Psi = \sigma_\gamma^2 I$.

Equivalently: $E(Y_i | \gamma_i) = X_i\beta + Z_i\gamma_i$.

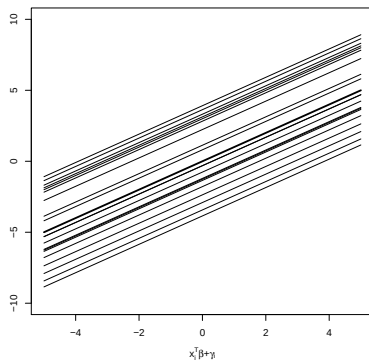
It follows that

$$Y_i \sim \mathcal{N}(X_i\beta, Z_i\Psi Z_i^T + \sigma_\epsilon^2 I),$$

and the (log-)likelihood can be written explicitly for this multivariate normal distribution (but still needs an optimization algorithm because of the parameters in Ψ).

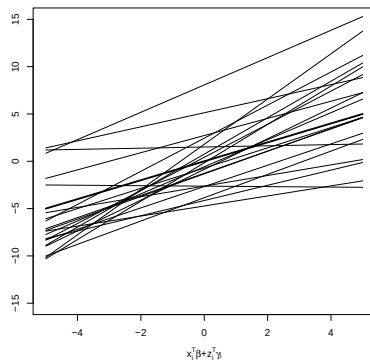
Random intercept model

$$Y_{it} = \mathbf{x}_{it}^T \boldsymbol{\beta} + \gamma_i + \epsilon_{it}$$



Random slope model

$$Y_{it} = \mathbf{x}_{it}^T \boldsymbol{\beta} + \mathbf{z}_{it}^T \boldsymbol{\gamma}_i + \epsilon_{it}$$



The generalized linear mixed model

Denote $\mu_{it} = E(Y_{it} \mid \gamma_i) (\neq E(Y_{it}))$.

Model (conditional on γ_i , a standard GLM):

- ▶ $g(E(Y_{it} \mid \gamma_i)) = g(\mu_{it}) = x_{it}^T \beta + z_{it}^T \gamma_i$ for a link function g .
- ▶ $Var(Y_{it} \mid \gamma_i) = v(\mu_{it})$.

Two sources of error: the sampling process and the measurement process within the sample itself.

Estimation of GLMM

Combine the information from

- ▶ the conditional distribution of $Y_{it} \mid \gamma_i$: $f(y_{it} \mid \gamma_i, \beta, \phi)$;
- ▶ the marginal distribution of the random effects: $f(\gamma_i \mid \phi, \Psi)$.

The γ_i are unobservable, we rely on the marginal likelihood:

$$\begin{aligned} L(\beta, \phi, \Psi) &= f(y_1, \dots, y_n) = \prod_{i=1}^n f(y_i) \\ &= \prod_{i=1}^n \int \cdots \int f(y_i \mid \gamma_i, \beta, \phi) f(\gamma_i \mid \phi, \Psi) d\gamma_i \\ &= \prod_{i=1}^n \int \cdots \int \prod_{t=1}^{n_i} f(y_{it} \mid \gamma_i, \beta, \phi) f(\gamma_i \mid \phi, \Psi) d\gamma_i. \end{aligned}$$

As in GLM, use the model relationship to introduce the data information.

Approximations of $L(\beta, \phi, \Psi)$

The likelihood cannot – in general – be expressed in closed form. Numerical methods and approximations to compute/approximate the integrals are needed:

- ▶ Monte-Carlo integrations (MCMC) (approximate the integrals by simulation)
- ▶ Gauss-Hermite quadrature approximation
- ▶ Laplace approximation
- ▶ Penalized Quasi-Likelihood (PQL)
- ▶ Marginal Quasi-Likelihood (MQL)

Note: PQL and MQL can produce biased estimators. Better to use Gaussian quadrature or Laplace approximation, even if they can be computationally quite intensive.

For elaborate specifications of random effects, the only effective integration schemes seem to be those based on MCMC.

Some available options in R

- ▶ package `nlme`: functions `lme` and `nlme` for linear and nonlinear mixed effects models. Allow for nested random effects.
- ▶ package `lme4`: function `lmer`. Allows Laplace approximation (default) and adaptive Gaussian quadrature approximation. Can define random intercepts and random slopes, as well as nested and crossing grouping factors for random effects.
- ▶ package `ADMB`: function `glmmadmb` based on a Laplace approximation with automatic differentiation. Wide range of families, wide range of link functions, single or multiple random effects, including both nested and crossed effects.
- ▶ package `glmmTMB`, function `glmmTMB`. Similar to `glmmadmb`, but some different extensions.

Interpretation of the fixed effects

A GLMM is a subject specific analysis: the regression coefficients apply to each individual, but not necessarily to the population (in contrast to a marginal analysis, e.g. GEE).

We have that:

$$\mu_{it} = E(Y_{it} | \gamma_i) = g^{-1}(x_{it}^T \beta + z_{it}^T \gamma_i),$$

so that the conditional mean varies with the predictors according to β with in addition a subject specific contribution due to γ_i .

The population mean $E(Y_{it})$ is defined by

$$E(Y_{it}) = E_{\gamma_i}(E(Y_{it} | \gamma_i)) = \int g^{-1}(x_{it}^T \beta + z_{it}^T \gamma_i) dF(\gamma_i),$$

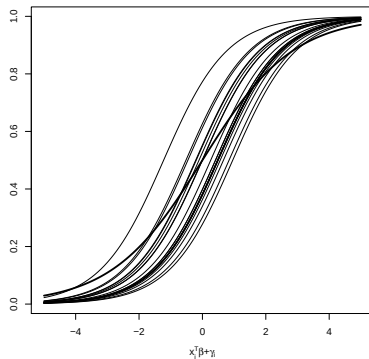
which is different from $g^{-1}(x_{it}^T \beta)$, except in the Gaussian-identity link case.

In some cases (e.g. the probit model) it holds that

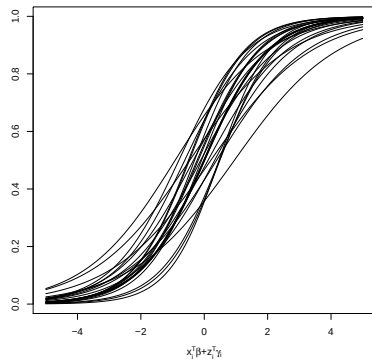
$$E(Y_{it}) = E_{\gamma_i}(E(Y_{it} | \gamma_i)) = g^{-1}(x_{it}^T \beta^*),$$

with $|\beta^*| \leq |\beta|$.

Random intercept model



Random slope model



Prediction of the random effects

For linear mixed models theory exists to define BP (best predictors minimizing MSE, $\hat{\gamma}_i = E(\gamma_i | y_i)$), BLP (best linear predictors) and BLUE (best linear unbiased predictors). They all go down to the same.

In GLMM software, usually the conditional modes are implemented, that is the values maximising the density of $\gamma_i | y_1, \dots, y_n$. They are not BLUP, because they are not linear, they might not be unbiased and we don't know in which sense they are best.

The best predictors would be the one satisfying $\hat{\gamma}_i = E(\gamma_i | y_i)$. Still an open field of research (in particular to define the properties of $\hat{\gamma}_i$).

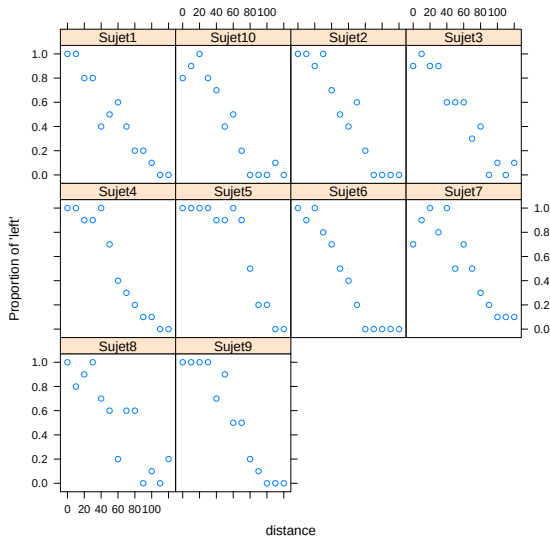
In fact, the question to ask is what is the prediction for:

- ▶ a new data point in an existing group: use $\hat{\mu}_{it} = g^{-1}(x_{it}^T \hat{\beta} + z_{it}^T \hat{\gamma}_i)$;
- ▶ a new data point for a new group: use $\hat{\mu}_i = g^{-1}(x_{it}^T \hat{\beta})$.

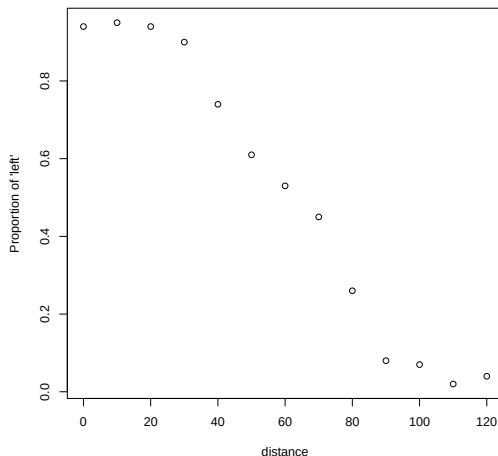
Analysis on data on the pseudoneglect

The effects of hemispace on a tactile line bisection task are measured. Data come from a diploma thesis (Sabina Catalano, FAPSE, 2003). We have observations on 10 subjects measured 10 times for each of 13th distances (from 0 to 120, 60 being the center of the line). The subject is presented (in a random sequence) with a position on a line and has to decide whether it lies on the left or on the right of the center. The number of "left" answers (for each distance) is recorded. In this experiment the scientific question is to see whether there is a systematic bias to the left from the center line (as suggested by some psychologists).

The "equality subjective point" (ESP), i.e. the distance for which the probability of "left" is $1/2$, is the parameter of interest.



Average over individuals



We consider a binomial GLMM model with random intercept:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 \text{distance} + \gamma_i,$$

with $p_i = P(\text{"left" for individual } i)$ and $\gamma_i \sim \mathcal{N}(0, \sigma_\gamma^2)$ is a random intercept.

For each individual, the ESP is the distance such that

$$\beta_0 + \beta_1 \text{distance} + \gamma_i = 0,$$

that is

$$ESP_i = -\frac{\beta_0 + \gamma_i}{\beta_1}.$$

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) ['glmerMod']

Family: binomial (logit)

Formula: cbind(left, 10 - left) ~ distance + (1 | sujet)

Data: hemi

AIC	BIC	logLik	deviance	df.resid
371.4	380.0	-182.7	365.4	127

Scaled residuals:

Min	1Q	Median	3Q	Max
-6.6927	-0.6634	-0.1685	0.6606	3.5747

Random effects:

Groups Name	Variance	Std.Dev.
sujet (Intercept)	0.2469	0.4969

Number of obs: 130, groups: sujet, 10

Fixed effects:

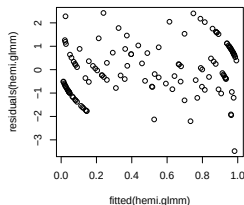
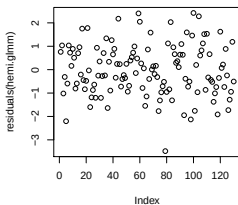
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.804820	0.264566	14.38	<2e-16 ***
distance	-0.063078	0.003267	-19.31	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1

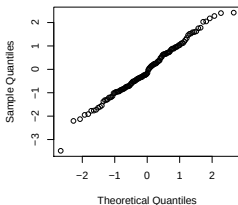
Correlation of Fixed Effects:

(Intr)
distance -0.744

With caution:



Normal Q-Q Plot



Are the random effects worth the trouble? Comparison with a GLM fit (which is nested within a GLMM model):

```
Call: glm(formula = cbind(gauche, 10 - gauche) ~ item, family = binomial,
data = hemi)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-3.1932	-0.9789	-0.2511	0.9206	3.6980

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.650109	0.202573	18.02	<2e-16 ***
item	-0.060521	0.003093	-19.57	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 988.40 on 129 degrees of freedom
Residual deviance: 198.04 on 128 degrees of freedom
AIC: 390.75

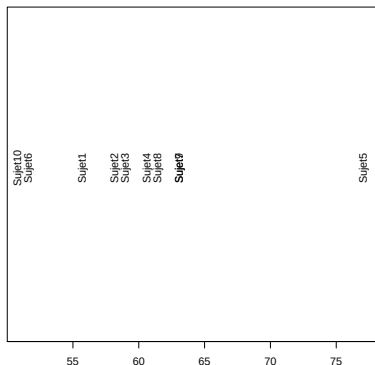
Number of Fisher Scoring iterations: 5

In GLMM: AIC=371.4 and the deviance=365.4.

Prediction of ESP

```
-(fixef(hemi.glm)[1] + ranef(hemi.glm)$sujet) / fixef(hemi.glm)[2]
(Intercept)
```

Sujet1	55.97849
Sujet10	51.06305
Sujet2	58.43176
Sujet3	59.24924
Sujet4	60.88410
Sujet5	77.32295
Sujet6	51.88350
Sujet7	63.33696
Sujet8	61.70159
Sujet9	63.33696



With random slope?

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) ['glmerMod']

Family: binomial (logit)

Formula: cbind(left, 10 - left) ~ item + (1 + distance | sujet)

Data: hemi

AIC	BIC	logLik	deviance	df.resid
368.4	382.8	-179.2	358.4	125

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.8813	-0.5607	-0.0854	0.5181	2.3284

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
sujet	(Intercept)	1.2714312	1.12758	
	distance	0.0001859	0.01363	-0.87

Number of obs: 130, groups: sujet, 10

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	4.02204	0.43108	9.33	<2e-16 ***
distance	-0.06629	0.00565	-11.73	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)
item	-0.885

Robust statistics

Statistical analyses can be very much affected by outliers. We do not want the analysis (and therefore the conclusions) to be driven by a single or very few observations.

Rather: model the majority (bulk) of the data.

With the **robust statistics** approach (Huber (1981) and Hampel, Ronchetti, Rousseeuw, and Stahel (1986)) we consider that:

- ▶ models are at best ideal approximations of the underlying process
- ▶ deviations from the distributional assumptions are (almost) always present in real data.

Basic idea: assume that the distribution of the data comes from a **neighborhood of the postulated model**.

Construct robust estimators (and test statistics) such that the estimated parameters (inferences) are consistent at the postulated model and stable in a neighborhood of it.

Correct estimation and inference is obtained for the parameters of the postulated model (the one corresponding to the **majority of the data**) by limiting the influence of (a small fraction of) data points which are thought of as coming from a different population.

The **stability** of the robust technique is achieved at the price of a slight loss of **efficiency** at the model. This can be viewed as an insurance premium one is willing to pay to protect against biases and losses of efficiency due to deviations from the assumed model.

Robust statistics, as a collection of related theories, is the statistics of approximate parametric models.

from Hampel, Ronchetti, Rousseeuw, and Stahel (1986), p. 7.

Neighborhood of the postulated model: $F_\epsilon = (1 - \epsilon)F_\theta + \epsilon G$, where F_θ is the postulated model, θ is a set of parameters of interest, G is an arbitrary distribution and $0 \leq \epsilon \leq 1$.

Inference		Classical		
G	$0 < \varepsilon < 1$	$0 < \varepsilon < 1$	$\varepsilon = 0$	
arbitrary	?	?	F_θ	
$G = \Delta_z$	F_ε	F_ε	F_θ	
$G = F_{(\theta, \theta')}$	F_ε	F_ε	F_θ	
G such that $F_\varepsilon = F_{(\theta, \theta')}$	$F_{(\theta, \theta')}$	$F_{(\theta, \theta')}$	F_θ	
Robust				
arbitrary	?	F_θ	F_θ	
$G = \Delta_z$	F_ε	F_θ	F_θ	
$G = F_{(\theta, \theta')}$	F_ε	F_θ	F_θ	
G such that $F_\varepsilon = F_{(\theta, \theta')}$	$F_{(\theta, \theta')}$	F_θ	F_θ	

Table from Heritier, Cantoni, Copt, and Victoria-Feser (2009).

Tools to measure robustness

- **Influence function** defined by

$$IF(\mathbf{z}; T, F) = \lim_{\epsilon \rightarrow 0} \left(\frac{T(F_\epsilon) - T(F)}{\epsilon} \right),$$

where $T(F)$ is a functional that defines the estimator $T(F^{(n)})$, $F^{(n)}$ is the empirical distribution function, $F_\epsilon = (1 - \epsilon)F + \epsilon\Delta_{\mathbf{z}}$, and $\Delta_{\mathbf{z}}$ is a distribution that puts all its mass at $\mathbf{z}$.

IF measures the effect on the estimate of an infinitesimal contamination at the point $\mathbf{z}$, standardized by the amount of contamination. The maximal marginal effect of an observation $\mathbf{z}$ on T is approximately $\epsilon \cdot IF(\mathbf{z}; T, F)$.

A **bounded influence function** is a desirable property for an estimator.

- **Breakdown point** It measures the maximum permitted percentage of the minority such that it has only limited influence on the estimator. The **larger** the breakdown point, the more robust the estimator.

Robust statistics versus diagnostic?

Why not performing diagnostic to identify outlying observations on the basis of a classical analysis and then remove the unusual data points from the sample?

Can be unreliable because a **masking effect** can occur, where a single large outlier may mask others. This means that the distorted data appear to be the norm rather than the exception.

Removal of outlying points base on a classical analysis is a binary decision (in or out), whereas the robust approach is a smoother process.

If the removal approach is used, the subsequent inference should take it into account (often not the case).

M-estimation

A large number of estimators fall in the class of M-estimators defined by Huber (1964). To estimate a parameter θ , for a given function Ψ , they are defined by the estimating equations:

$$\sum_{i=1}^n \Psi(y_i, \theta) = 0.$$

Under certain regularity conditions:

- ▶ $\sqrt{n}(\hat{\theta} - \theta)$ is asymptotically normally distributed with mean 0 and variance $\Omega = M^{-1}QM^{-1}$, where $M = -E(d/d\theta \Psi(y, \theta))$ and $Q = E(\Psi(y, \theta)\Psi^T(y, \theta))$.
- ▶ the influence function of an M-estimator is $M^{-1}\Psi(y, \theta)$, that is proportional to $\Psi(y, \theta)$.

Example

The **ML estimator for GLM** (p. 81) is an M-estimator with

$$\psi(y_i, \mu_i) = \frac{y_i - \mu_i}{\text{Var}(Y_i)} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) = \frac{r_i}{v^{1/2}(\mu_i)} \mu'_i.$$

This function is unbounded with respect to y_i and with respect to x_j . The GLM estimator is **not robust**!

Robust GLM estimator

Introduce a function ψ to control large deviations in the y -space and a set of weights $w(\mathbf{x}_i)$ to downweight leverage points:

$$\sum_{i=1}^n \left[\psi(r_i) w(\mathbf{x}_i) \frac{1}{v^{1/2}(\mu_i)} \mu'_i - a(\beta) \right] = \mathbf{0},$$

where $r_i = (y_i - \mu_i)/v^{1/2}(\mu_i)$ are the Pearson residuals (Cantoni and Ronchetti 2001).

The correction term $a(\beta)$ ensures Fisher consistency.

The classical estimating equations are a special case: ψ is the identity function and $w(\mathbf{x}_i) \equiv 1$, in which case it holds that $a(\beta) = 0$.

The set of estimating equations for robust GLM can be rewritten as :

$$\sum_{i=1}^n \left[\tilde{w}(r_i) r_i w(\mathbf{x}_i) \frac{1}{v^{1/2}(\mu_i)} \mu'_i - a(\beta) \right] = \mathbf{0},$$

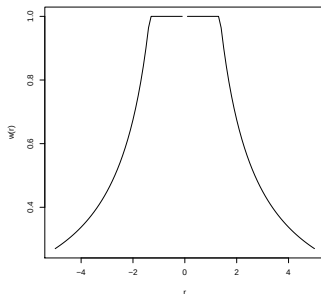
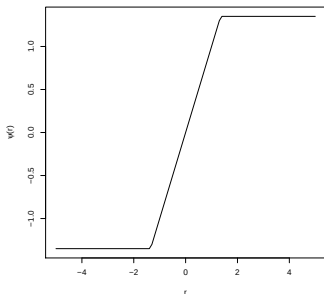
where $\tilde{w}(r) = \psi(r)/r$.

Interpretation: these are GLM estimating equations weighted to ensure robustness and recentered to ensure consistency.

Choices for ψ

A common choice for ψ to ensure robustness is the so-called Huber's function defined by $\psi_c(r) = r \cdot \min(1, c/|r|)$.

c allows the tuning of the robustness-efficiency compromise. (In practice, take c between 1 and 2).



Choices for $w(\mathbf{x}_i)$

Examples of weights $w(\mathbf{x}_i)$:

- ▶ as a function of the diagonal elements of the hat matrix $H = X(X^T X)^{-1} X^T$ (e.g. $w(\mathbf{x}_i) = \sqrt{1 - H_{ii}}$)
- ▶ based on the Mahalanobis distances of the design matrix (with the center and the covariance matrix estimated robustly).

$\tilde{w}$ and w can be used for diagnostic: they give information on how each observation is handled.

Properties of the robust GLM

The distributional and robustness properties of the robust GLM estimator follow from general results on M-estimation, with

$$M = \frac{1}{n} X^T B X \quad \text{and} \quad Q = \frac{1}{n} X^T A X - a(\beta) a(\beta)^T,$$

where B and A are diagonal matrices containing elements

$$b_i = E[\psi_c(r_i) \frac{\partial}{\partial \mu_i} \log h(y_i | \mathbf{x}_i, \mu_i)] \frac{1}{v^{1/2}(\mu_i)} w(\mathbf{x}_i) \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2,$$

and

$$a_i = E[\psi_c(r_i)^2] w^2(\mathbf{x}_i) \frac{1}{v(\mu_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2.$$

The influence function is bounded with respect to y for a bounded choice of ψ , and the effect of outliers in the design is controlled with appropriate weights $w(\mathbf{x})$.

Variable selection

The robust estimating equations of page 244 can be seen as the derivatives with respect to β of the **robust quasi-likelihood** function $\sum_{i=1}^n Q_M(y_i, \mu_i)$, where

$$Q_M(y_i, \mu_i) = \int_{\tilde{s}}^{\mu_i} \phi(y_i, t) w(\mathbf{x}_i) dt - \frac{1}{n} \sum_{j=1}^n \int_{\tilde{t}}^{\mu_j} E[\phi(y_j, t) w(\mathbf{x}_j)] dt,$$

where $\phi(y_i, t) = \psi((y_i - t)/v^{1/2}(t))/v^{1/2}(t)$, $\tilde{s}$ and $\tilde{t}$ such that $\phi(y_i, \tilde{s}) = 0$, $E[\phi(y_j, \tilde{t})] = 0$.

The above integrals are usually computed numerically.

Compare with the classical quasi-likelihood function on page 136.

To compare a model $\mathcal{M}_{p+1}$ with $(p+1)$ variables to a nested model $\mathcal{M}_{p+1-q}$ with only $(p+1-q)$ variables, a test statistic can be constructed based on twice the **difference of quasi-likelihood** functions

$$\Lambda_{QM} = 2 \left[\sum_{i=1}^n Q_M(y_i, \hat{\mu}_i^{p+1}) - \sum_{i=1}^n Q_M(y_i, \hat{\mu}_i^{p+1-q}) \right].$$

Note that Λ_{QM} is independent of $\tilde{s}$ and $\tilde{t}$.

Under the null hypothesis that

$$H_0 : \beta = (\cdot, \dots, \cdot, 0_{p+2-q}, \dots, 0_{p+q})$$

and under quite general conditions, Λ_{QM} is asymptotically distributed as

$$\sum_{i=1}^q \lambda_i N_i^2,$$

where $N_1, \dots, N_q \sim \mathcal{N}(0, 1)$ and $\lambda_1, \dots, \lambda_q$ are the q positive eigenvalues of the matrix $Q(\psi, F_\beta)(M^{-1}(\psi, F_\beta) - \tilde{M}^+(\psi, F_\beta))$.

It can be proven that the asymptotic level and power under small deviations from the model are stable as long as an estimator of β with bounded influence function is used.

Example: Hospital costs of stay

The Gamma distribution involves a second parameter ν that has also to be estimated robustly. In fact this parameter is such

$$\text{Var}((Y_i - \mu_i)/\nu^{1/2}(\mu_i)) = 1/\nu,$$

and therefore any robust estimator of the variance of $(Y_i - \mu_i)/\nu^{1/2}(\mu_i)$ can be used, for example a simple M-estimator (Huber's Proposal 2), which solves

$$\sum_{i=1}^n \left[\psi_c \left(\frac{y_i - \mu_i}{\nu^{1/2}(\mu_i)/\sqrt{\nu}} \right)^2 - E \left(\psi_c \left(\frac{y_i - \mu_i}{\nu^{1/2}(\mu_i)/\sqrt{\nu}} \right)^2 \right) \right] = 0,$$

The asymptotic results on β still hold (for $\sqrt{n}$ -consistent estimator for ν).

```
Call: glmrob(formula = CouTot ~ log(LOS) + Typadm + Typass + age +
Sexe + dest, family = Gamma(link = log), data = MYdata)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	7.2541684	0.1038707	69.838	< 2e-16	***
log(LOS)	0.8391397	0.0197817	42.420	< 2e-16	***
Typadm	0.2232685	0.0353717	6.312	2.75e-10	***
Typass	0.0027594	0.0559621	0.049	0.9607	
age	-0.0010104	0.0009091	-1.111	0.2664	
Sexe	0.0711542	0.0353567	2.012	0.0442	*
dest	-0.1243614	0.0490022	-2.538	0.0112	*

```
---
Signif. codes:  0      ***      0.001    **      0.01    *      0.05    .      0.1      1
```

Robustness weights w.r * w.x:

82 weights are ~ 1. The remaining 18 ones are summarized as

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2048	0.4902	0.6399	0.6208	0.7696	0.9925

Number of observations: 100

Fitted by method Mqle (in 4 iterations)

(Dispersion parameter for Gamma family taken to be 0.02349196)

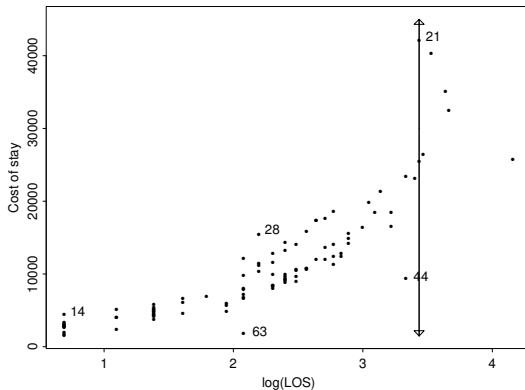
No deviance values available

	Classical		Robust	
variable	coeff.	st. err.	coeff.	st. err.
Intercept	7.2338	0.1469	7.2523	0.1049
log(LOS)	0.8222	0.0280	0.8391	0.0200
ADM	0.2136	0.0500	0.2221	0.0357
INS	0.0933	0.0791	0.0093	0.0565
AGE	-0.0005	0.0013	-0.0010	0.0009
SEX	0.0951	0.0500	0.0727	0.0357
DEST	-0.1043	0.0693	-0.1230	0.0495
	scale: 0.0496		scale: 0.0243	

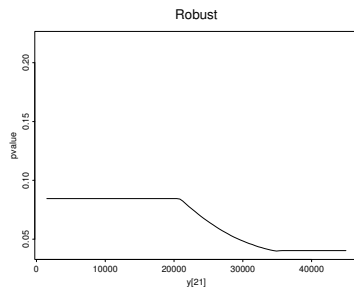
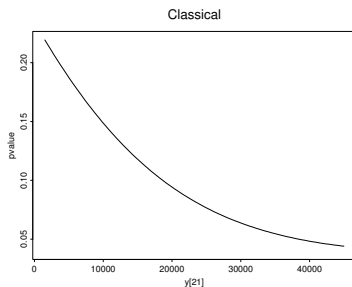
Similar coefficients, but larger standard errors.

Weights less or equal than 0.5: $\tilde{w}_{14} = 0.23$, $\tilde{w}_{21} = 0.50$,
 $\tilde{w}_{28} = 0.24$, $\tilde{w}_{44} = 0.42$ and $\tilde{w}_{63} = 0.32$.

Sensitivity analysis for variable selection



p-values for the test of $H_0 : \beta_{SEX} = 0$.



Robust GEE

A **robust version of GEE** can be defined (Cantoni 2004):

$$\sum_{i=1}^n D_i^T \Gamma_i^T V_i^{-1} (\psi_i - c_i) = 0,$$

where $D_i = \partial \mu_i / \partial \beta$, $V_i = \phi A_i^{1/2} R_i(\alpha) A_i^{1/2}$, with $R_i(\alpha)$ the “working” correlation matrix. Moreover, $\psi_i = W_i \cdot (Y_i - \mu_i)$, where $W_i = W_i(X_i, y_i, \mu_i)$ is a diagonal, and $c_i = E(\psi_i)$. Finally, $\Gamma_i = E(\tilde{\psi}_i - \tilde{c}_i)$ with $\tilde{\psi}_i = \partial \psi_i / \partial \mu_i$ and $\tilde{c}_i = \partial c_i / \partial \mu_i$.

The classical GEE equations are obtained with W_i equal to the identity matrix.

Estimation of ϕ and α has also to be made robust.

Robust hurdle model

A **robust hurdle** model can be defined by building on each of its two parts.

A robust binary regression is used to separate the zero from the positive values.

The robust GLM for truncated Poisson is also available (Cantoni and Zedini 2011). It uses the GLM framework of the truncated Poisson (see page 167) and defines the estimator with the estimating equations of page 243.

Generalized additive models (GAM)

Consider the same kind of models as for GLM (exponential family distributions for continuous and discrete variables, link functions, etc), but replace the linear part of GLM with nonparametric functions:

$$g(\mu_i) = \alpha + f_1(x_{i1}) + \dots + f_p(x_{ip}).$$

This new model is particularly useful when no functional form is known a priori. It is also a way to check whether the linear (or other parametric) assumption is sensible.

Let the data show us the appropriate functional form.

from Hastie and Tibshirani (1990), p. 1.

Univariate smoothing

Data: (x_i, y_i) for individuals $i = 1, \dots, n$.

We start with the simple univariate model

$$Y_i = f(x_i) + \epsilon_i,$$

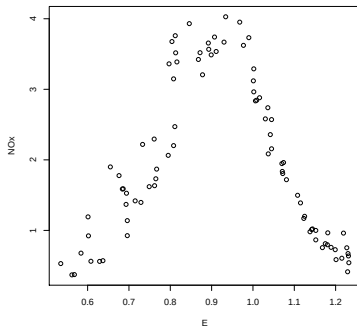
with the assumption that the ϵ_i are independently drawn from $N(0, \sigma^2)$.

It is the univariate nonparametric analogue to

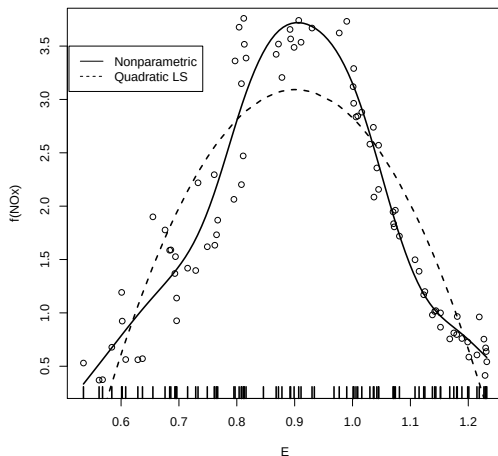
$$Y_i = \alpha + \beta x_i + \epsilon_i.$$

Ethanol example

Engine exhaust for 88 burnings of ethanol in a single-cylinder automobile test engine. Outcome: concentration of nitric oxides (NO_x). Explanatory variable: equivalence ratio, that is a measure of the richness of the air/ethanol mix (E).



Nonparametric vs quadratic fit



Basis functions

How to represent $f(x)$? With an approximation with known functions:

$$f(x) = \sum_{k=1}^K b_k(x) \beta_k,$$

where the $b_k(x)$ are basis functions. The model is therefore

$$Y_i = \sum_{k=1}^K b_k(x_i) \beta_k + \epsilon_i,$$

for $i = 1, \dots, n$.

Model fitting

For $Y = (Y_1, \dots, Y_n)^T$, $\beta = (\beta_1, \dots, \beta_K)^T$ and $\epsilon = (\epsilon_1, \dots, \epsilon_n)$, the model can be equivalently expressed as

$$Y = X\beta + \epsilon,$$

with

$$X = \begin{pmatrix} b_1(x_1) & b_2(x_1) & \dots & b_K(x_1) \\ \vdots & \vdots & \vdots & \vdots \\ b_1(x_n) & b_2(x_n) & \dots & b_K(x_n) \end{pmatrix}.$$

It can be fitted by least squares:

$$\hat{\beta} = \operatorname{argmin}_{\beta} \|y - X\beta\|^2.$$

Polynomial basis

If f is believed to be a 4th order polynomial, a possible basis is

$$b_1(x) = 1, b_2(x) = x, b_3(x) = x^2, b_4(x) = x^3, b_5(x) = x^4,$$

so that

$$f(x) = \beta_1 + x\beta_2 + x^2\beta_3 + x^3\beta_4 + x^4\beta_5$$

and the model

$$y_i = \beta_1 + x_i\beta_2 + x_i^2\beta_3 + x_i^3\beta_4 + x_i^4\beta_5 + \epsilon_i,$$

The piecewise linear (or tent) basis - formula

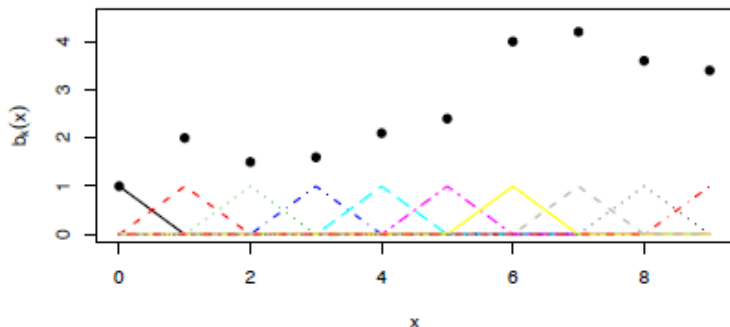
Given knots x_j^* , for $j = 1, \dots, k$, with $x_j^* > x_{j-1}^*$, we have for $j = 2, \dots, k-1$:

$$b_j(x) = \begin{cases} (x - x_{j-1}^*) / (x_j^* - x_{j-1}^*) & x_{j-1}^* < x < x_j^* \\ (x_{j+1}^* - x) / (x_{j+1}^* - x_j^*) & x_j^* < x < x_{j+1}^* \\ 0 & \text{otherwise} \end{cases}$$

$$b_1(x) = \begin{cases} (x_2^* - x) / (x_2^* - x_1^*) & x < x_2^* \\ 0 & \text{otherwise} \end{cases}$$

$$b_k(x) = \begin{cases} (x - x_{k-1}^*) / (x_k^* - x_{k-1}^*) & x > x_{k-1}^* \\ 0 & \text{otherwise} \end{cases}$$

The piecewise linear (or tent) basis - graphical representation

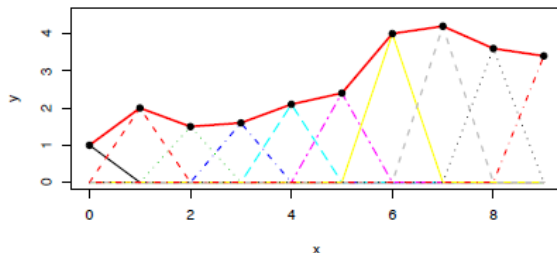


1

¹Figure from S. Wood slides at the *Ecole doctorale d'hiver CUSO*, 2014,

Approximation of f

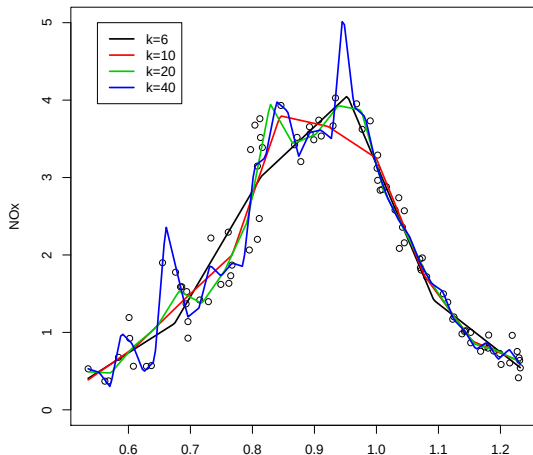
Data (x_k^*, y_k^*) are interpolated by just setting $\beta_k = y_k^*$. The function f is represented by multiplying each tent function by its coefficient β_k and summing the result.



2

²Figure from S. Wood slides at the *Ecole doctorale d'hiver CUSO*, 2014,

In practice: choose some x_k^* values spread through the range of the observed x_i 's.



Other bases: splines

The polynomial and the tent bases are easy to understand, but improvement is possible. In particular, spline bases reduce function approximation error for a given dimension of the smoothing basis.

A cubic smoothing spline solves (with respect to f)

$$\sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \int (f''(x))^2 dx.$$

Cubic splines arise naturally from the specification of the smoothing objective function above (defined in a basis independent way).

The drawback of smoothing splines is that they have n free parameters. To retain their good properties, but improve on computational efficiency, penalized regression splines are used instead.

It implies constructing a spline basis for a much smaller data set and then using that basis (plus penalties) to model the original dataset. The covariates values in the smaller dataset should be chosen to nicely cover the range of the covariates in the original dataset.

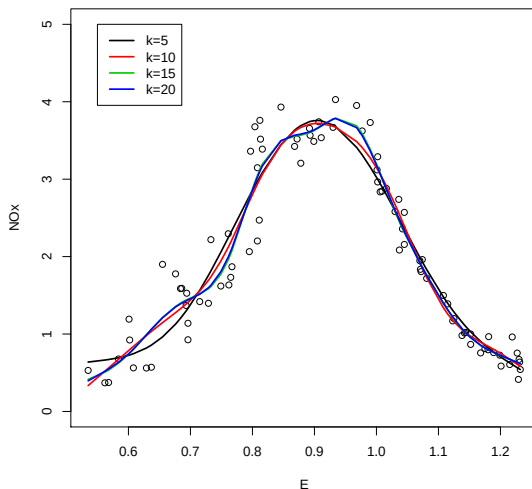
Informal basis dimension check

Perform a check to see whether k is not restrictively low.

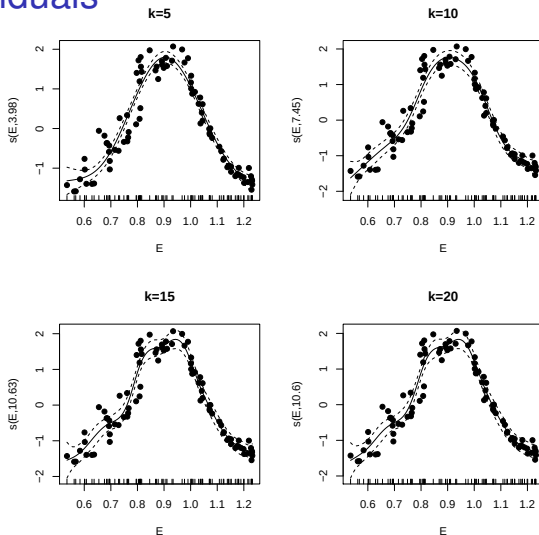
1. Look at partial residuals against term estimates, looking for systematic departures
2. Use a residual randomization test to test for residuals pattern.

Idea: estimate the scale parameter σ^2 by differencing residuals. Differencing residuals that are neighboring according to x_i should give an estimate of σ^2 that is indistinguishable from a differencing estimate obtained with any random ordering of residuals, under the null hypothesis that there is no residual pattern.

Ethanol example with regression splines



Partial residuals



```
ethanol1 <- gam(NOx~s(E,k=5),data=ethanol)
```

```
> gam.check(ethanol1)
```

```
Method: GCV   Optimizer: magic
```

```
Smoothing parameter selection converged after 10 iterations.
```

```
The RMS GCV score gradient at convergence was 3.154738e-06 .
```

```
The Hessian was positive definite.
```

```
Model rank = 5 / 5
```

Basis dimension (k) checking results. Low p-value ($k\text{-index} < 1$) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(ethanol\$E)	4.000	3.982	0.682	0

```
ethanol2 <- gam(NOx~s(E,k=10),data=ethanol)
```

```
> gam.check(ethanol2)
```

```
Method: GCV   Optimizer: magic
```

```
Smoothing parameter selection converged after 8 iterations.
```

```
The RMS GCV score gradient at convergence was 2.421344e-06 .
```

```
The Hessian was positive definite .
```

```
Model rank = 10 / 10
```

Basis dimension (k) checking results. Low p-value ($k\text{-index} < 1$) may indicate that k is too low, especially if edf is close to k' .

	k'	edf	$k\text{-index}$	p-value
s(ethanol\$E)	9.000	7.452	0.766	0

```
ethanol3 <- gam(NOx~s(E,k=15),data=ethanol)
```

```
> gam.check(ethanol3)
```

```
Method: GCV   Optimizer: magic
```

```
Smoothing parameter selection converged after 8 iterations.
```

```
The RMS GCV score gradient at convergence was 1.551372e-05 .
```

```
The Hessian was positive definite.
```

```
Model rank = 15 / 15
```

Basis dimension (k) checking results. Low p-value ($k\text{-index} < 1$) may indicate that k is too low, especially if edf is close to k' .

	k'	edf	$k\text{-index}$	p-value
s(ethanol\$E)	14.000	10.633	0.884	0.11

```
ethanol4 <- gam(NOx~s(E,k=20),data=ethanol)
```

```
> gam.check(ethanol4)
```

```
Method: GCV   Optimizer: magic
```

```
Smoothing parameter selection converged after 6 iterations.
```

```
The RMS GCV score gradient at convergence was 1.246923e-06 .
```

```
The Hessian was positive definite .
```

```
Model rank = 20 / 20
```

Basis dimension (k) checking results. Low p-value ($k\text{-index} < 1$) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(ethanol\$E)	19.000	10.603	0.872	0.09

More formal choice of K ?

- ▶ Models with different K are not nested: rules out hypothesis testing.
- ▶ Need to fit all possible K value if wants to use e.g. AIC
- ▶ Difficult to generalize to models with more than one function

In addition, where to put the knots?

Smoothing

Instead of selecting K , rather use *smoothing*:

- ▶ Make K 'large enough' (negligible bias)
- ▶ Use even spaced x_k^*
- ▶ To avoid overfit, penalize the wiggleness of f , e.g. with

$$\mathcal{P}(f) = \sum_{j=2}^{k-1} \{f(x_{j-1}^*) - 2f(x_j^*) + f(x_{j+1}^*)\}^2.$$

This penalty can be regarded as a crude approximation of the second derivative of f .

When f is wiggly the penalty is large, when f is smooth the penalty is low. (The penalty is zero for a straight line).

The model is fitted by maximisation of the penalized least squares criterion

$$\hat{\beta} = \operatorname{argmin}_{\beta} \{ \|y - X\beta\|^2 + \lambda \mathcal{P}(f) \}.$$

The *smoothing parameter* λ controls the trade-off between smoothness of the estimated f and fidelity to the data.

$\lambda \rightarrow \infty$ leads to a straight line estimate for f while $\lambda = 0$ results in an unpenalized piecewise linear regression estimate.

Smoothing with the tent basis

For the tent basis, it holds that $\beta_j = f(x_j^*)$ and therefore the penalty can be expressed as a quadratic form, because

$$\begin{pmatrix} \beta_1 - 2\beta_2 + \beta_3 \\ \beta_2 - 2\beta_3 + \beta_4 \\ \beta_3 - 2\beta_4 + \beta_5 \\ \vdots \end{pmatrix} = \begin{bmatrix} 1 & -2 & 1 & 0 & . & . & . \\ 0 & 1 & -2 & 1 & 0 & . & . \\ 0 & 0 & 1 & -2 & 1 & 0 & . \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \vdots \end{bmatrix} = D\beta$$

and therefore

$$\mathcal{P}(f) = \sum_{j=2}^{k-1} \{f(x_{j-1}^*) - 2f(x_j^*) + f(x_{j+1}^*)\}^2 = \beta^T D^T D \beta = \beta^T S \beta$$

Smoothing (general)

We solve

$$\hat{\beta} = \operatorname{argmin}_{\beta} \{ \|y - X\beta\|^2 + \lambda \beta^T S \beta \},$$

which has the explicit solution

$$\hat{\beta} = (X^T X + \lambda S)^{-1} X^T y.$$

Therefore

$$\hat{y} = \hat{f}(x) = X\hat{\beta} = X(X^T X + \lambda S)^{-1} X^T y = Ay,$$

where $A = X(X^T X + \lambda S)^{-1} X^T$ is the influence (hat) matrix.

For computational stability, note that the problem is transformed in an augmented form as follow:

$$\{\|y - X\beta\|^2 + \lambda\beta^T S\beta\} = \left\| \begin{bmatrix} y \\ 0 \end{bmatrix} - \begin{bmatrix} X \\ \sqrt{\lambda}D \end{bmatrix} \beta \right\|^2,$$

which is a least squares problem.

Natural basis

For any model matrix X and penalty matrix S we can consider the reparametrization that makes the penalty matrix diagonal.

Let $X = QR$ (QR decomposition), and $R^{-T}SR^{-1} = U\Lambda U^T$ (SVD decomposition).

Define $P = U^T R$ and reparametrize $\beta' = P\beta$.

In the new parametrization, the model matrix is $X' = QU$, which has orthogonal columns ($X = X'P$). The penalty matrix is now the diagonal matrix Λ .

Degrees of freedom

Penalization restricts the freedom of coefficients to vary: K coefficients have $< K$ *effective degrees of freedom* (edf).

Consider a natural parametrization.

Without penalization: $\tilde{\beta}' = (X')^T y$

With penalization: $\hat{\beta}' = (I + \lambda \Lambda)^{-1} (X')^T y$, therefore

$\hat{\beta}'_j = \tilde{\beta}'_j (1 + \lambda \Lambda_{jj})^{-1}$. So $(1 + \lambda \Lambda_{jj})^{-1}$ is the shrinkage factor of the j^{th} coefficient. It gives the EDF for $\hat{\beta}_j$.

The total EDF is $\sum_j (1 + \lambda \Lambda_{jj})^{-1} = \text{Tr}(F)$ with

$$F = (X^T X + \lambda S)^{-1} X^T X.$$

Smoothing bias

Because $\hat{\beta} = (X^T X + \lambda S)^{-1} X^T y$, it holds that

$$\begin{aligned} E(\hat{\beta}) &= (X^T X + \lambda S)^{-1} X^T E(y) \\ &= (X^T X + \lambda S)^{-1} X^T X \beta \\ &= F \beta \neq \beta \end{aligned}$$

Smooths are biased due to the penalization.

This bias makes frequentist inference difficult.

The Bayesian paradigm in a nutshell

With the distribution $f(y | \beta)$ of $(y | \beta)$ and the **prior distribution** $\pi(\beta)$ on the parameter β , construct a **posterior distribution**, that is the distribution of $\beta | y$, which is (thanks to Bayes theorem):

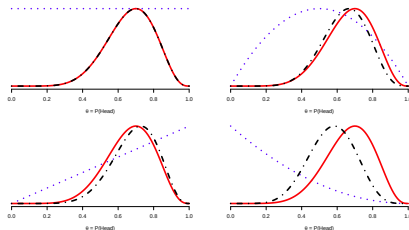
$$\pi(\beta | y) = \frac{f(y | \beta)\pi(\beta)}{f(y)}.$$

Inference is conducted on the basis of the posterior distribution.

Illustration of the Bayesian paradigm

The posterior distribution

- Informally, we update beliefs by combining the **prior** and **likelihood** to obtain the posterior



From:

29

BIO 233, Spring 2015

https://cdn1.sph.harvard.edu/wp-content/uploads/sites/565/2018/08/233Spr15_Part1_Bayes.pdf

Bayesian smoothing model

The penalization can be seen as a prior put on wiggleness in a Bayesian framework:

$$\text{wiggleness prior} \propto \exp\left(-\lambda\beta^T S\beta/(2\sigma^2)\right)$$

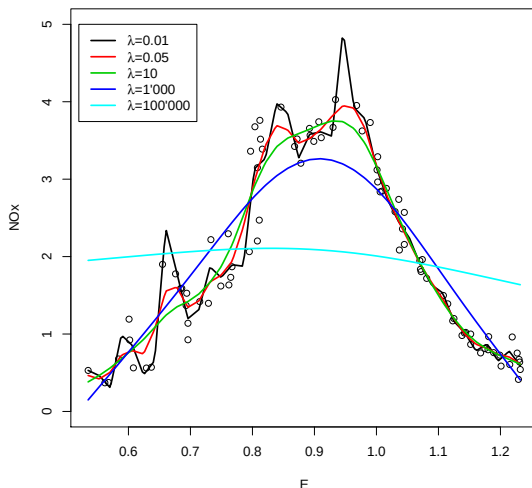
equivalent to a prior $\beta \sim \mathcal{N}(0, S^- \sigma^2/\lambda)$, where S^- is a generalized inverse of S .

From the model we have $y \mid \beta \sim \mathcal{N}(X\beta, \sigma^2 I)$, therefore (Bayes rule)

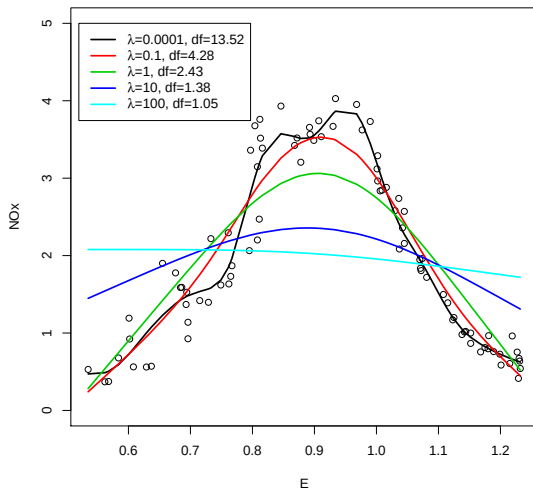
$$\beta \mid y \sim \mathcal{N}(\hat{\beta}, \sigma^2(X^T X + \lambda S)^{-1}).$$

Also: $\hat{\sigma}^2 = \|y - X\hat{\beta}\|^2/(n - \text{Tr}(F))$.

Ethanol example with different λ (tent basis)



Ethanol example with different λ (spline basis)



Choice of λ ?

Ideally, choose λ so that $\hat{f}$ is as close as possible to f , by minimizing, for example,

$$M = \frac{1}{n} \sum_{i=1}^n \left(\hat{f}(x_i) - f(x_i) \right)^2.$$

But f is unknown and M cannot be used directly.

Estimation of M by cross-validation

Define the ordinary cross-validation criterion

$$OCV(\lambda) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}_{\lambda}^{-i}(x_i))^2,$$

where $\hat{f}_{\lambda}^{-i}(x_i)$ indicates the fit at x_i leaving out the i th data point. It implies the fit of n submodels.

It can be proven that $E(OCV) \approx E(M) + \sigma^2$.

Because $\hat{y} = Ay$ (see page 282)

$$OCV(\lambda) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{f}_\lambda(x_i)}{1 - A_{ii}} \right)^2,$$

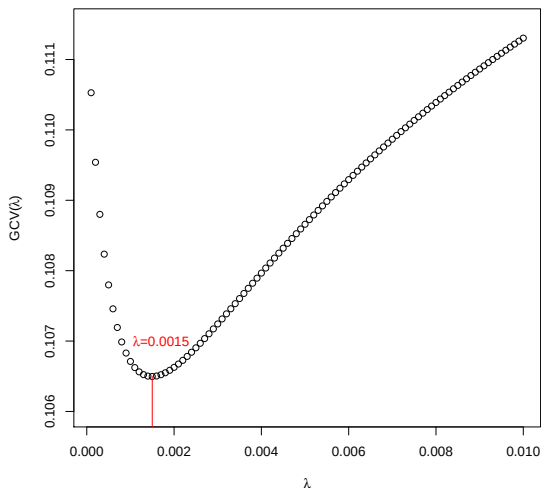
which can be obtained from the fit of the full dataset only.

In practice, the A_{ii} are often replaced by their average, giving the generalized cross-validation criterion

$$GCV(\lambda) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{f}_\lambda(x_i)}{1 - \text{Tr}(A)/n} \right)^2 = \frac{n \sum_{i=1}^n (y_i - \hat{f}_\lambda(x_i))^2}{(n - \text{Tr}(A))^2}.$$

Note: $\text{Tr}(A) = \text{Tr}(F)$ because of the circularity of the trace.

GCV for the ethanol dataset (spline basis)



Ethanol automatic fit

```
> ethanol.gam <- gam(NOx ~ s(E, k=15), data=ethanol)
```

```
> ethanol.gam
```

```
Family: gaussian
```

```
Link function: identity
```

```
Formula:
```

```
NOx ~ s(E, k = 15)
```

```
Estimated degrees of freedom:
```

```
10.6 total = 11.63
```

```
GCV score: 0.1064968
```

```
> ethanol.gam$sp
```

```
s(E)
```

```
0.001490695
```


Ethanol automatic fit

```
> summary(ethanol.gam)
```

Family: gaussian

Link function: identity

Formula:

NOx ~ s(E, k = 15)

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.95737	0.03241	60.4	<2e-16 ***

Signif.	codes:	0	***	0.001	**	0.01	*	0.05	.	0.1	1
---------	--------	---	-----	-------	----	------	---	------	---	-----	---

Approximate significance of smooth terms:

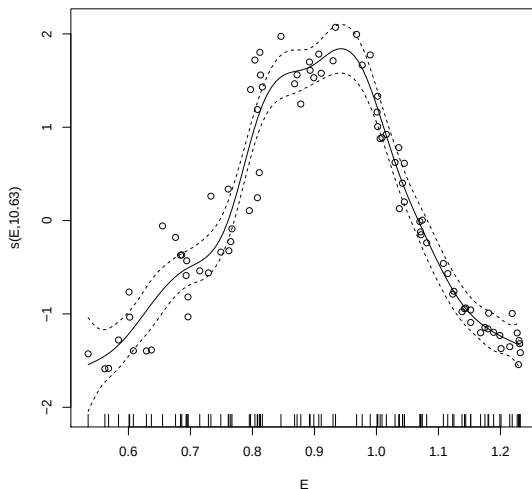
	edf	Ref.df	F	p-value
s(E)	10.63	12.31	91.39	<2e-16 ***

Signif.	codes:	0	***	0.001	**	0.01	*	0.05	.	0.1	1
---------	--------	---	-----	-------	----	------	---	------	---	-----	---

R-sq.(adj) = 0.928 Deviance explained = 93.7%

GCV = 0.1065 Scale est. = 0.092419 n = 88

Final fit



Additive models - AM

How do we go from 1 to p predictors?

Ideally one should consider surface smoothers

$$y = f(x_1, \dots, x_p) + \epsilon,$$

but with large p this goal is out of reach.

Additive models are an approximation:

$$y = \alpha + f_1(x_1) + f_2(x_2) + \dots + f_p(x_p) + \epsilon,$$

with $E(\epsilon) = 0$, $\text{Var}(\epsilon) = \sigma^2$ and ϵ independent of x_j .

With more than one component, there is an identifiability problem: each function f_j is estimable only up to an additive constant. The constraint that

$$\sum_{i=1}^n f_j(x_{ji}) = 0$$

for all j is therefore added.

Extensions of the simple additive model above can include component functions with two or more dimensions, categorical variable terms and their interactions with continuous variables.

Remarks:

- ▶ The additive form of the model allows the plotting of the p functions separately (better interpretability).
- ▶ Additive models are more general approximations than linear regression models.
- ▶ Additive fits can be helpful to test linearity.

Fitting additive models

We represent each function with basis functions such that $f_j(x_j) = \sum_{k=1}^K b_k(x_j) \delta_k^{(j)}$. The model is

$$y_i = \alpha + \sum_{j=1}^p f_j(x_{ji}) + \epsilon_i = \alpha + \sum_{j=1}^p X^{(j)} \delta^{(j)} + \epsilon_i,$$

where $X^{(j)}$ contains the basis function for covariate j .

Define the associated penalties $(\delta^{(j)})^T \bar{S}_j \delta^{(j)}$ and $X = [1 \ X^{(1)} \ \dots \ X^{(p)}]$. Consider also the vector of all the parameters $\beta = (\alpha, \delta^{(1)}, \dots, \delta^{(p)})^T$.

We now solve

$$\hat{\beta} = \operatorname{argmin}_{\beta} \{ \|y - X\beta\|^2 + \sum_{j=1}^p \lambda_j \beta^T S_j \beta \},$$

where S_j is constructed with $\bar{S}_j$ and padded out with zeros such that $(\delta^{(j)})^T \bar{S}_j \delta^{(j)} = \beta^T S_j \beta$.

Similarly to the univariate case, we have

$$\hat{\beta} = (X^T X + \lambda_1 S_1 + \dots + \lambda_p S_p)^{-1} X^T y,$$

$$A = X(X^T X + \lambda_1 S_1 + \dots + \lambda_p S_p)^{-1} X^T,$$

and

$$F = (X^T X + \lambda_1 S_1 + \dots + \lambda_p S_p)^{-1} X^T X.$$

For computational stability, the corresponding augmented problem is considered:

$$\begin{aligned} & \|y - X\beta\|^2 + \beta^T(\lambda_1 S_1 + \dots + \lambda_p S_p)\beta \\ &= \left\| \begin{pmatrix} y \\ 0 \end{pmatrix} - \begin{pmatrix} X \\ B \end{pmatrix} \beta \right\|^2, \end{aligned}$$

where B is such that $B^T B = \lambda_1 S_1 + \dots + \lambda_p S_p$.

It is a least squares problem.

Choice of $(\lambda_1, \dots, \lambda_p)$ by GCV

Find $(\lambda_1, \dots, \lambda_p)$ that minimizes

$$GCV(\lambda_1, \dots, \lambda_p) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{f}_{\lambda_1, \dots, \lambda_p}(x_i)}{1 - \text{Tr}(\mathbf{A})/n} \right)^2 \quad (16)$$

$$= \frac{n \sum_{i=1}^n \left(y_i - \hat{f}_{\lambda_1, \dots, \lambda_p}(x_i) \right)^2}{(n - \text{Tr}(\mathbf{A}))^2}. \quad (17)$$

Posterior distribution for β

We have

$$\beta|y \sim \mathcal{N}(\hat{\beta}, V_{\beta}),$$

where $V_{\beta} = \sigma^2(X^T X + \lambda_1 S_1 + \dots + \lambda_p S_p)$.

V_{β} is estimated by plugging in $\hat{\sigma}^2 = \|y - X\hat{\beta}\|^2 / (n - \text{Tr}(F))$.

The above result can be used for inference on β , either directly (quantities linear in the model parameters) or by simulation or bootstrap (in other cases).

Hypothesis testing on β

Suppose we want to test $H_0 : \beta_j = 0$, where β_j is a subvector of size p_j of β . Let V_{β_j} denote the block of V_β corresponding to β_j .

Then, under H_0 one can use the test statistics

$$\hat{\beta}_j^T V_{\beta_j}^{-1} \hat{\beta}_j / p_j \sim F_{p_j, n-p},$$

if there is a scale parameter involved (ϕ unknown), and

$$\hat{\beta}_j^T V_{\beta_j}^{-1} \hat{\beta}_j \sim \chi_{p_j}^2,$$

if not (ϕ known).

Confidence intervals for f_j

From the posterior distribution of β , credible intervals can be constructed for f_j .

Let $\tilde{f}_j = \tilde{X}\beta$, where $\tilde{X}$ has zeros in the columns corresponding to the coefficients having nothing to do with f_j , while its other columns contain the basis functions for f_j . Let $v = \text{diag}(\tilde{X}V_\beta\tilde{X}^T)$, then

$$\hat{f}_{ji} \pm z_{\alpha/2} \sqrt{v_i}$$

is an approximate $(1 - \alpha)100\%$ credible interval for f_{ji} .

It turns out that Bayesian credible intervals have good frequentist coverage properties.

Variable selection for smooth terms

A test statistics can be derived to test the null hypothesis

$H_0 : f_j(x_j) = 0$. Let $f_j = \tilde{X}\beta$. Then, from $\hat{\beta} \sim \mathcal{N}(\beta, V_\beta)$, we have $\hat{f}_j \sim \mathcal{N}(f_j, V_{f_j})$, where $V_{f_j} = \tilde{X} V_\beta \tilde{X}^T$.

The test statistics is defined by

$$T_r = \hat{f}_j^T V_{f_j}^{r-} \hat{f}_j,$$

where $V_{f_j}^{r-}$ is a rank r pseudo-inverse of V_{f_j} .

Some care has to be taken for the choice of r . In a nutshell, this choice is done based on the estimated degrees of freedom of the function of f_j .

The distribution of T_r under H_0 is χ_r^2 if r is integer.

Boston housing data analysis

```
> BHnpfit <- gam(log(medv) ~ chas + s(crim) + s(zn) + s(indus) + s(nox) + s(rm) + s(age) + s(dis) +
+ s(rad, k=8) + s(tax) + s(ptratio) + s(b) + s(lstat), data=BostonHousing)
> BHnpfit
```

Family: gaussian

Link function: identity

Formula:

```
log(medv) ~ chas + s(crim) + s(zn) + s(indus) + s(nox) + s(rm) +
+ s(age) + s(dis) + s(rad, k = 8) + s(tax) + s(ptratio) + s(b) +
+ s(lstat)
```

Estimated degrees of freedom:

```
3.12 1.00 7.83 9.00 4.59 1.00 8.61
1.33 3.09 1.00 4.99 5.24 total = 52.8
```

GCV score: 0.02158189

```
> gam.check(BHnpfit)
```

```
Method: GCV   Optimizer: magic
```

```
Smothing parameter selection converged after 20 iterations.
```

```
The RMS GCV score gradient at convergence was 2.97197e-08 .
```

```
The Hessian was positive definite.
```

```
Model rank = 108 / 108
```

Basis dimension (k) checking results. Low p-value ($k\text{-index} < 1$) may indicate that k is too low, especially if edf is close to k' .

	k'	edf	$k\text{-index}$	p-value
s(crim)	9.000	3.123	1.074	0.94
s(zn)	9.000	1.000	0.797	0.00
s(indus)	9.000	7.825	0.790	0.00
s(nox)	9.000	9.000	0.838	0.00
s(rm)	9.000	4.591	0.941	0.10
s(age)	9.000	1.000	0.974	0.23
s(dis)	9.000	8.613	0.936	0.08
s(rad)	7.000	1.330	0.796	0.00
s(tax)	9.000	3.089	0.798	0.00
s(ptratio)	9.000	1.000	0.814	0.00
s(b)	9.000	4.989	0.981	0.34
s(lstat)	9.000	5.240	1.072	0.94

```
> BHnpfit.new <- gam(log(medv)~chas+s(crim)+s(zn)+s(indus)+s(nox,k=20)+s(rm)+s(age)+
  s(dis,k=20)+s(rad,k=8)+s(tax)+s(ptratio)+s(b)+s(lstat),data=BostonHousing)
> BHnpfit.new
```

Family: gaussian

Link function: identity

Formula:

```
log(medv) ~ chas + s(crim) + s(zn) + s(indus) + s(nox, k = 20) +
  s(rm) + s(age) + s(dis, k = 20) + s(rad, k = 8) + s(tax) +
  s(ptratio) + s(b) + s(lstat)
```

Estimated degrees of freedom:

```
3.37  1.00  5.21 13.07  5.10  1.00 13.40
2.26  3.20  1.00  1.94  4.76 total = 57.3
```

GCV score: 0.02116857


```
> gam.check(BHnpfit.new)
```

Method: GCV Optimizer: magic

Smoothing parameter selection converged after 21 iterations.

The RMS GCV score gradient at convergence was 2.186018e-08 .

The Hessian was positive definite.

Model rank = 128 / 128

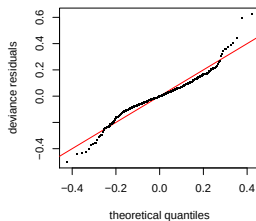
Basis dimension (k) checking results. Low p-value (k-index < 1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(crim)	9.000	3.368	1.078	0.94
s(zn)	9.000	1.000	0.838	0.00
s(indus)	9.000	5.211	0.819	0.00
s(nox)	19.000	13.065	0.875	0.00
s(rm)	9.000	5.101	0.948	0.14
s(age)	9.000	1.000	0.990	0.40
s(dis)	19.000	13.397	0.967	0.20
s(rad)	7.000	2.260	0.835	0.00
s(tax)	9.000	3.202	0.834	0.00
s(ptratio)	9.000	1.000	0.856	0.00
s(b)	9.000	1.945	0.978	0.34
s(lstat)	9.000	4.755	1.077	0.94

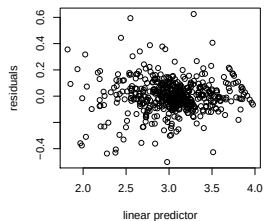
Generalized Linear and Additive Models

- Generalized additive models (GAM)

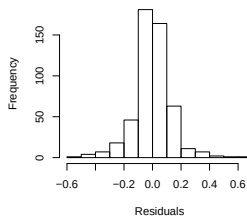
- Additive models



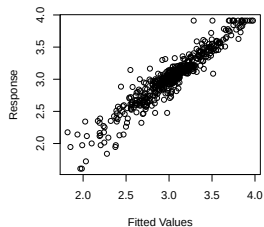
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



Family: gaussian

Link function: identity

Formula: $\log(\text{medv}) \sim \text{chas} + \text{s}(\text{crim}) + \text{s}(\text{zn}) + \text{s}(\text{indus}) + \text{s}(\text{nox}, k = 20) + \text{s}(\text{rm}) + \text{s}(\text{age}) + \text{s}(\text{dis}, k = 20) + \text{s}(\text{rad}, k = 8) + \text{s}(\text{tax}) + \text{s}(\text{ptratio}) + \text{s}(\text{b}) + \text{s}(\text{lstat})$

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.031791	0.006376	475.497	<2e-16 ***
chas	0.039350	0.027264	1.443	0.15

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 1

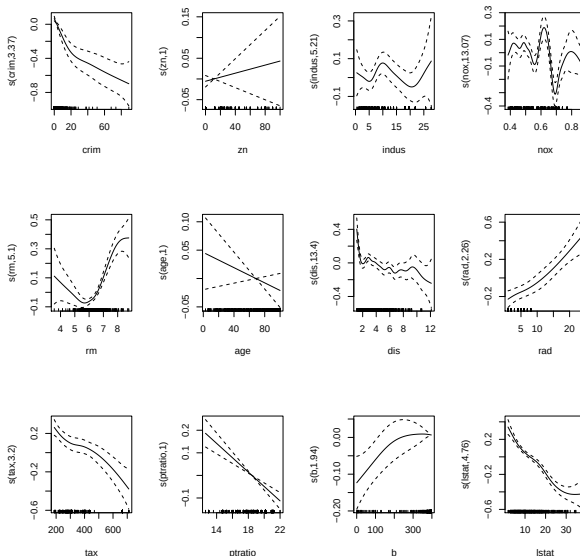
Approximate significance of smooth terms:

	edf	Ref. df	F	p-value
s(crim)	3.368	4.184	19.183	7.60e-15 ***
s(zn)	1.000	1.000	0.639	0.42437
s(indus)	5.211	6.119	1.882	0.08638 .
s(nox)	13.065	15.171	11.476	< 2e-16 ***
s(rm)	5.101	6.262	17.018	< 2e-16 ***
s(age)	1.000	1.000	1.964	0.16174
s(dis)	13.397	15.676	3.900	9.16e-07 ***
s(rad)	2.260	2.627	10.931	4.05e-06 ***
s(tax)	3.202	3.789	11.648	1.24e-08 ***
s(ptratio)	1.000	1.000	37.846	1.63e-09 ***
s(b)	1.945	2.401	5.254	0.00327 **
s(lstat)	4.755	5.864	35.015	< 2e-16 ***

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 1

R-sq.(adj) = 0.888 Deviance explained = 90%

GCV = 0.021169 Scale est. = 0.018771 n = 506



Remove zn, age and chas?

```
> BHnpfit2.new <- gam(log(medv) ~ s(crim) + s(indus) + s(nox, k=20) + s(rm) + s(dis, k=20) +
  s(rad, k=8) + s(tax) + s(ptratio) + s(b) + s(lstat), data=BostonHousing)
> BHnpfit2.new
```

Family: gaussian

Link function: identity

Formula: $\log(\text{medv}) \sim s(\text{crim}) + s(\text{indus}) + s(\text{nox}, k = 20) + s(\text{rm}) + s(\text{dis}, k = 20) + s(\text{rad}, k = 8) + s(\text{tax}) + s(\text{ptratio}) + s(\text{b}) + s(\text{lstat})$

Estimated degrees of freedom:

3.35	5.12	12.79	4.96	13.33	2.35	3.27
1.00	1.91	4.83	total = 53.92			

GCV score: 0.02112012

```
> gam.check(BHnpfit2.new)
```

```
Method: GCV   Optimizer: magic
```

```
Smoothing parameter selection converged after 20 iterations.
```

```
The RMS GCV score gradient at convergence was 4.284208e-08 .
```

```
The Hessian was positive definite.
```

```
Model rank = 109 / 109
```

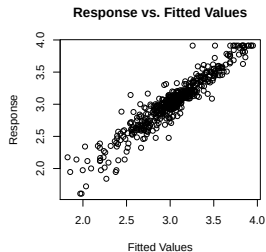
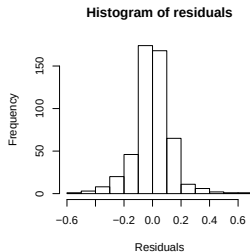
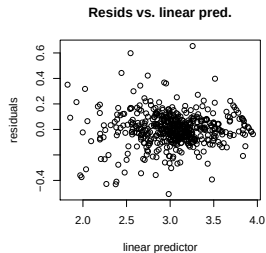
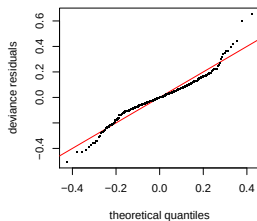
```
Basis dimension (k) checking results. Low p-value (k-index<1) may  
indicate that k is too low, especially if edf is close to k'.
```

	k'	edf	k-index	p-value
s(crim)	9.000	3.352	1.082	0.96
s(indus)	9.000	5.123	0.813	0.00
s(nox)	19.000	12.795	0.876	0.00
s(rm)	9.000	4.961	0.940	0.08
s(dis)	19.000	13.325	0.967	0.22
s(rad)	7.000	2.353	0.828	0.00
s(tax)	9.000	3.268	0.823	0.00
s(ptratio)	9.000	1.000	0.849	0.00
s(b)	9.000	1.915	0.987	0.44
s(lstat)	9.000	4.828	1.076	0.96

Generalized Linear and Additive Models

- Generalized additive models (GAM)

- Additive models



```
> summary(BHnpfit2.new)
```

```
Family: gaussian
```

```
Link function: identity
```

```
Formula: log(medv) ~ s(crim) + s(indus) + s(nox, k = 20) + s(rm) + s(dis, k = 20) +
s(rad, k = 8) + s(tax) + s(ptratio) + s(b) + s(lstat)
```

```
Parametric coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.034513	0.006107	496.9	<2e-16 ***

```
---
Signif. codes:  0      ***      0.001      **      0.01      *      0.05      .      0.1      1
```

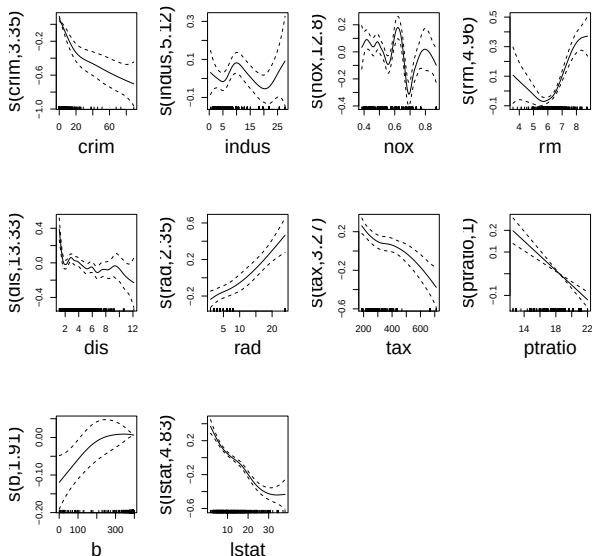
```
Approximate significance of smooth terms:
```

	edf	Ref.df	F	p-value
s(crim)	3.352	4.165	19.885	2.34e-15 ***
s(indus)	5.123	6.030	2.132	0.04962 *
s(nox)	12.795	14.900	11.670	< 2e-16 ***
s(rm)	4.961	6.109	17.497	< 2e-16 ***
s(dis)	13.325	15.611	3.932	6.55e-07 ***
s(rad)	2.353	2.735	11.034	2.13e-06 ***
s(tax)	3.268	3.868	11.418	1.14e-08 ***
s(ptratio)	1.000	1.000	44.876	5.87e-11 ***
s(b)	1.915	2.365	5.030	0.00437 **
s(lstat)	4.828	5.938	43.788	< 2e-16 ***

```
---
Signif. codes:  0      ***      0.001      **      0.01      *      0.05      .      0.1      1
```

```
R-sq.(adj) = 0.887   Deviance explained = 89.9%
```

```
GCV = 0.02112   Scale est. = 0.01887   n = 506
```

Remove `ptratio`?

```
> BHnpfit3.new <- gam(log(medv) ~ s(crim) + s(indus) + s(nox, k=20) + s(rm) + s(dis, k=20) + s(rad, k=8)
+ s(tax) + s(b) + s(lstat), data=BostonHousing)
> BHnpfit3.new
```

```
Family: gaussian
Link function: identity
```

Formula:

```
log(medv) ~ s(crim) + s(indus) + s(nox, k = 20) + s(rm) + s(dis,
k = 20) + s(rad, k = 8) + s(tax) + s(b) + s(lstat)
```

Estimated degrees of freedom:

```
3.35  8.10 13.58  5.30 12.33  1.00  3.27
1.77  4.70 total = 54.39
```

GCV score: 0.02258293

```
> BHnpfit3.new$aic
[1] -486.3607
```

The Akaike value of this model is -486.36, much worst of what we had before.

GAM ingredients

1. Independent response variables $Y_1, \dots, Y_n$ which are assumed to share the same distribution from the **exponential family** (F_{θ_i}).
2. A set of **explanatory variables**
 $x_i^T = (x_{i1}, \dots, x_{ip})$ for $i = 1, \dots, n$.
3. A monotone **link** function g such that

$$g(\mu_i) = \alpha + f_1(x_{i1}) + \dots + f_p(x_{ip}) = \eta_i.$$

η_i is called the **additive predictor**.

We assume that

$$E(Y_i) = \mu_i \text{ and } \text{Var}(Y_i) = v_i = v(\mu_i).$$

The difference with the GLM model is the fact that μ_i is obtained nonparametrically, via $g^{-1}(\alpha + f_1(x_{i1}) + \dots + f_p(x_{ip}))$.

We now have

$$g(\mu_i) = \alpha + \sum_{j=1}^p f_j(x_{ji}) = \alpha + \sum_{j=1}^p X^{(j)} \delta^{(j)},$$

with associated penalties $(\delta^{(j)})^T \bar{S}_j \delta^{(j)}$ and $X = [1 \ X^{(1)} \ \dots \ X^{(p)}]$.

Consider $\beta = (\alpha, \delta^{(1)}, \dots, \delta^{(p)})$. We solve

$$\hat{\beta} = \operatorname{argmin}_{\beta} \left\{ D(y, \mu) + \sum_{j=1}^p \lambda_j \beta^T S_j \beta \right\},$$

where $D(y, \mu)$ is the deviance function (see p. 92) and where S_j is constructed with $\bar{S}_j$ and padded out with zeros such that $(\delta^{(j)})^T \bar{S}_j \delta^{(j)} = \beta^T S_j \beta$.

GAM are fitted with a **local scoring algorithm**.

At step $(m - 1)$, construct an **adjusted dependent variable**

$$z_i^{(m-1)} = \eta_i^{(m-1)} + (y_i - \mu_i^{(m-1)}) \left(\frac{\partial \eta_i}{\partial \mu_i} \right)^{(m-1)},$$

with $\eta_i^{(m-1)} = \alpha^{(m-1)} + \sum_{j=1}^p f_j^{(m-1)}(x_{ij})$, and weights

$$w_i^{(m-1)} = \frac{1}{\text{Var}(Y_i)^{(m-1)}} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2_{(m-1)}.$$

Fit a **weighted additive model** to $z_i^{(m-1)}$ to obtain estimated functions $f_j^{(m)}$, additive predictor $\eta_i^{(m)}$, and fitted values $\mu_i^{(m)}$. Repeat until convergence is reached.

The weighted least squares fit of GLM in the local scoring algorithm is replaced by a weighted additive fit (see page 84).

The weighted additive fit can be written as a weighted least squares fit for the augmented problem as follow:

$$\left\| \begin{pmatrix} \sqrt{W^{(m-1)}} & 0 \\ 0 & I \end{pmatrix} \left[\begin{pmatrix} z^{(m-1)} \\ 0 \end{pmatrix} - \begin{pmatrix} X \\ B \end{pmatrix} \right] \beta \right\|^2,$$

with $W^{(m-1)} = \text{diag}(w_1^{(m-1)}, \dots, w_n^{(m-1)})$ and B is such that $B^T B = \lambda_1 S_1 + \dots + \lambda_p S_p$.

Properties

The effective degrees of freedom matrix becomes

$$F = (X^T W X + \lambda_1 S_1 + \dots + \lambda_p S_p)^{-1} X^T W X.$$

It holds asymptotically that

$$\beta | y \sim \mathcal{N}(\hat{\beta}, V_\beta),$$

where $V_\beta = \sigma^2 (X^T W X + \lambda_1 S_1 + \dots + \lambda_p S_p)^{-1}$.

The scale parameter, if unknown, is estimated by

$$\hat{\phi} = \sum_{i=1}^n (y_i - X_i \hat{\beta})^2 / (n - \text{Tr}(F)).$$

Choice of $\lambda_1, \dots, \lambda_p$ for ϕ known

Minimize mean squared error:

For an additive model with constant variance, it holds

$$\begin{aligned} M(\lambda_1, \dots, \lambda_p) &= E \left(\frac{\| \mu - X\hat{\beta} \|^2}{n} \right) \\ &= E \left(\frac{\| y - Ay \|^2}{n} \right) - \phi + \frac{2\phi \text{Tr}(A)}{n}, \end{aligned}$$

(see p. 52 and p. 255 in Wood (2017) on how to obtain the last expression.)

The form of M suggests the following estimator (Un-Biased Risk Estimator)

$$UBRE(\lambda_1, \dots, \lambda_p) = \frac{\|y - Ay\|^2}{n} - \phi + \frac{2\phi \text{Tr}(A)}{n} = D(y, \hat{\mu}) + \frac{2\phi \text{Tr}(F)}{n}.$$

which is also known as Mallows' C_p .

UBRE has to be minimized.

By extension, it is used for all the families with ϕ known.

Choice of λ for ϕ unknown

If ϕ unknown, it has to be estimated. Plugging-in $\hat{\phi}$ in the UBRE definition makes it unsuitable for model selection.

Therefore, GCV is used. It generalizes to

$$GCV(\lambda_1, \dots, \lambda_p) = \frac{nD(y, \hat{\mu})}{(n - \text{Tr}(F))^2}.$$

Model validation

- ▶ Informal checks on K
- ▶ Residuals analysis

Comparing (nested) models

Being in a likelihood framework, the Akaike criterion can be used to compare nested models:

$$AIC = -2l(\hat{\beta}; y) + 2\phi \text{Tr}(F).$$

Models with smallest AIC have to be preferred.

$GCV(\hat{\lambda}_1, \dots, \hat{\lambda}_p)$ estimates the prediction error. It can be used to compare models. Models with smallest GCV have to be preferred.

The proportion of deviance explained is also a measure of the quality of a fit.

Analysis of the birds dataset

Outcome: Number of birds observed (counts, NUMBER).

Predictors:

- ▶ year (YEAR) - Main interest.
- ▶ day (DAY)
- ▶ number of observer on the island (NO.OBS)
- ▶ wind direction at midnight the night before (WIND.0)
- ▶ wind direction at noon the day before (WIND.1)
- ▶ wind direction at midnight 2 days before (WIND.2)
- ▶ wind direction at noon 2 days before (WIND.3)
- ▶ the speed of the wind that day (WINDSPD)
- ▶ sky condition (1=clear, 2=cloudy, 3=fog, 4=rain) (SKY)
- ▶ proportion of illuminated moon (MOON)
- ▶ bird species (1-5) (BIRD)

Poisson model with logarithmic link

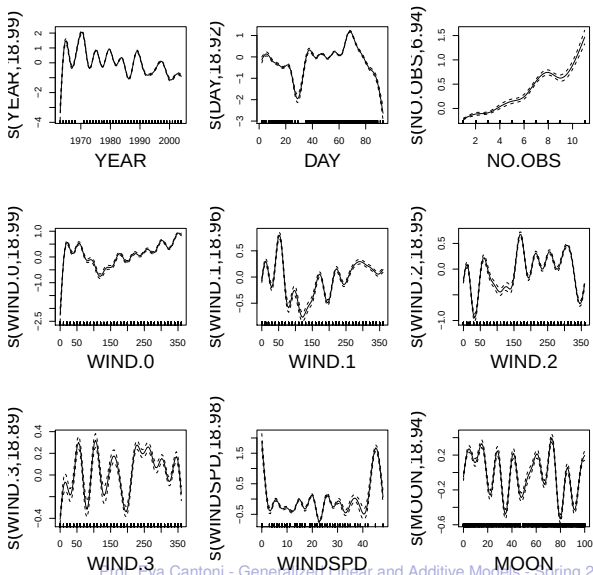
$$\begin{aligned} \text{NUMBER} \sim & \alpha + f_1(\text{YEAR}) + f_2(\text{DAY}) + f_3(\text{NO.OBS}) \\ & + f_4(\text{WIND.0}) + f_5(\text{WIND.1}) + f_6(\text{WIND.2}) \\ & + f_7(\text{WIND.3}) + f_8(\text{WINDSPD}) + \text{factor}(\text{SKY}) \\ & + f_9(\text{MOON}) + \text{factor}(\text{BIRD}) \end{aligned}$$

- > require(mgcv)
- > GrA.fit <- gam(NUMBER ~ s(YEAR, k=20) + s(DAY, k=20) + s(NO.OBS, k=8) +
s(WIND.0, k=20) + s(WIND.1, k=20) + s(WIND.2, k=20) + s(WIND.3, k=20) +
s(WINDSPD, k=20) + factor(SKY) + s(MOON, k=20) + factor(BIRD),
family=poisson, data=GrA, na.action=na.omit)
- > plot(GrA.fit, scale=0, pages=1, pers=T, cex.lab=2)

Generalized Linear and Additive Models

Generalized additive models (GAM)

Analysis of the birds dataset



```
> gam.check(GrA.fit)
```

```
Method: UBRE   Optimizer: outer newton
full convergence after 12 iterations.
Gradient range [-1.543031e-05,2.132528e-05]
(score 38.92044 & scale 1).
eigenvalue range [-1.317038e-05,8.758125e-05].
Model rank = 167 / 167
```

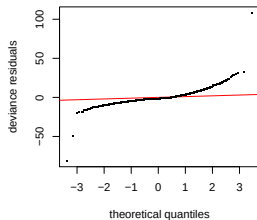
Basis dimension (k) checking results. Low p-value ($k\text{-index} < 1$) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(YEAR)	19.000	18.987	0.676	0.00
s(DAY)	19.000	18.923	0.838	0.00
s(NO.OBS)	7.000	6.941	0.704	0.00
s(WIND.0)	19.000	18.985	0.846	0.00
s(WIND.1)	19.000	18.962	0.894	0.00
s(WIND.2)	19.000	18.951	0.903	0.00
s(WIND.3)	19.000	18.891	0.895	0.00
s(WINDSPD)	19.000	18.982	0.888	0.00
s(MOON)	19.000	18.945	0.959	0.31

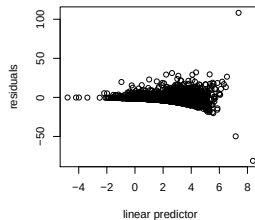
Generalized Linear and Additive Models

└ Generalized additive models (GAM)

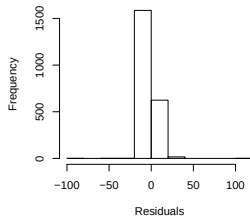
└ Analysis of the birds dataset



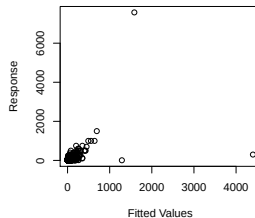
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



```
> summary(GrA. fit)
```

Family: poisson

Link function: log

Formula:

```
NUMBER ~ s(YEAR, k = 20) + s(DAY, k = 20) + s(NO.OBS, k = 8) +  
          s(WIND.0, k = 20) + s(WIND.1, k = 20) + s(WIND.2, k = 20) +  
          s(WIND.3, k = 20) + s(WINDSPD, k = 20) + factor(SKY) + s(MOON,  
          k = 20) + factor(BIRD)
```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.84578	0.01186	239.898	< 2e-16	***
factor(SKY)cloudy	0.03137	0.01300	2.413	0.0158	*
factor(SKY)fog	-0.04209	0.01852	-2.272	0.0231	*
factor(SKY)rain	-0.09024	0.02229	-4.048	5.16e-05	***
factor(BIRD)2	-2.00090	0.02604	-76.849	< 2e-16	***
factor(BIRD)3	-2.60308	0.03415	-76.223	< 2e-16	***
factor(BIRD)4	0.20671	0.01208	17.118	< 2e-16	***
factor(BIRD)5	1.22781	0.01020	120.359	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1

Approximate significance of smooth terms:

	edf	Ref. df	Chi.sq	p-value	
s(YEAR)	18.987	19.000	9207	<2e-16	***
s(DAY)	18.923	18.999	6365	<2e-16	***
s(NO.OBS)	6.941	6.998	1862	<2e-16	***
s(WIND.0)	18.985	19.000	3543	<2e-16	***
s(WIND.1)	18.962	19.000	2172	<2e-16	***
s(WIND.2)	18.951	19.000	3784	<2e-16	***
s(WIND.3)	18.891	18.998	1383	<2e-16	***
s(WINDSPD)	18.982	19.000	3259	<2e-16	***
s(MOON)	18.945	18.999	1746	<2e-16	***

Signif. codes: 0 *** 0.001 ** 0.01 * 0.05 . 0

R-sq.(adj) = 0.0895 Deviance explained = 65.5%

UBRE = 38.92 Scale est. = 1 n = 2230

Quasi-poisson model, cycle basis for WIND

```
> gam.check(GrA.quasifit)
Method: GCV   Optimizer: outer newton
full convergence after 9 iterations.
Gradient range [-3.33374e-06,2.451736e-05]
(score 45.78461 & scale 58.39889).
Hessian positive definite , eigenvalue range [0.003653669,0.04407047].
Model rank = 163 / 163
```

Basis dimension (k) checking results. Low p-value ($k\text{-index} < 1$) may indicate that k is too low, especially if edf is close to k'.

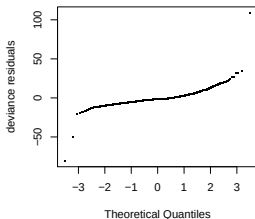
	k'	edf	k-index	p-value
s(YEAR)	19.000	18.819	0.685	0.00
s(DAY)	19.000	12.869	0.831	0.00
s(NO.OBS)	7.000	2.317	0.705	0.00
s(WIND.0)	18.000	9.992	0.840	0.00
s(WIND.1)	18.000	9.003	0.890	0.00
s(WIND.2)	18.000	14.793	0.908	0.00
s(WIND.3)	18.000	4.803	0.887	0.00
s(WINDSPD)	19.000	17.248	0.905	0.00
s(MOON)	19.000	16.376	0.956	0.27

Generalized Linear and Additive Models

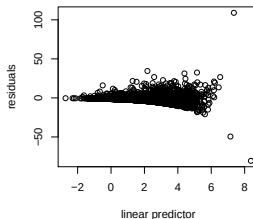
└ Generalized additive models (GAM)

└ Analysis of the birds dataset

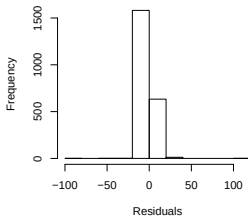
Normal Q-Q Plot



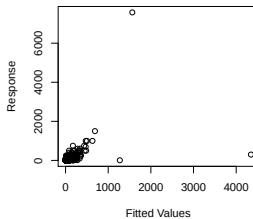
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



```
> summary(GrA. quasifit)
```

Family: quasipoisson

Link function: log

Formula:

```
NUMBER ~ s(YEAR, k = 30) + s(DAY, k = 30) + s(NO.OBS, k = 8) +  
  s(WIND.0, k = 30, bs = "cc") + s(WIND.1, k = 20, bs = "cc") +  
  s(WIND.2, k = 30, bs = "cc") + s(WIND.3, k = 30, bs = "cc") +  
  s(WINDSPD, k = 30) + factor(SKY) + s(MOON, k = 30) + factor(BIRD)
```

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.880e+00	8.486e-02	33.940	<2e-16	***
factor(SKY)cloudy	-4.954e-05	9.023e-02	-0.001	0.9996	
factor(SKY)fog	-1.048e-02	1.310e-01	-0.080	0.9362	
factor(SKY)rain	-2.195e-01	1.541e-01	-1.425	0.1544	
factor(BIRD)2	-2.001e+00	1.916e-01	-10.444	<2e-16	***
factor(BIRD)3	-2.603e+00	2.514e-01	-10.355	<2e-16	***
factor(BIRD)4	2.065e-01	8.888e-02	2.324	0.0202	*
factor(BIRD)5	1.228e+00	7.508e-02	16.356	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value	
s(YEAR)	28.697	28.964	8.007	< 2e-16	***
s(DAY)	17.522	20.852	5.485	2.00e-14	***
s(NO.OBS)	3.534	4.265	9.785	4.15e-08	***
s(WIND.0)	24.510	28.000	3.388	9.87e-12	***
s(WIND.1)	8.080	18.000	2.438	1.30e-08	***
s(WIND.2)	2.820	28.000	0.652	3.03e-05	***
s(WIND.3)	23.924	28.000	1.585	0.00223	**
s(WINDSPD)	8.943	10.887	4.105	5.99e-06	***
s(MOON)	26.886	28.474	2.840	8.02e-07	***

Signif. codes: 0 *** 0.001 ** 0.01 * 0.05 . 0

R-sq.(adj) = 0.103 Deviance explained = 67%

GCV = 43.889 Scale est. = 54.171 n = 2230

Quasi-poisson, cycle basis for WIND, without outlier

```
> gam.check(GrA.quasifit.wo)
Method: GCV   Optimizer: outer newton
full convergence after 9 iterations.
Gradient range [-2.221881e-06,2.419194e-05]
(score 32.65511 & scale 44.22684).
Hessian positive definite , eigenvalue range [0.00213954,0.02999257].
Model rank = 163 / 163
```

Basis dimension (k) checking results. Low p-value ($k\text{-index} < 1$) may indicate that k is too low, especially if edf is close to k'.

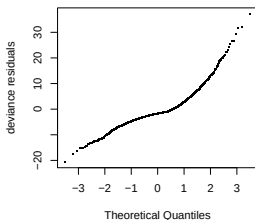
	k'	edf	k-index	p-value
s(YEAR)	19.000	18.826	0.568	0.00
s(DAY)	19.000	16.152	0.841	0.00
s(NO.OBS)	7.000	1.939	0.617	0.00
s(WIND.0)	18.000	8.973	0.858	0.00
s(WIND.1)	18.000	15.711	0.920	0.02
s(WIND.2)	18.000	15.715	0.887	0.00
s(WIND.3)	18.000	15.871	0.904	0.00
s(WINDSPD)	19.000	15.680	0.910	0.00
s(MOON)	19.000	17.462	0.968	0.55

Generalized Linear and Additive Models

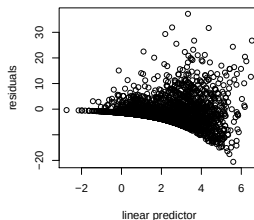
└ Generalized additive models (GAM)

└ Analysis of the birds dataset

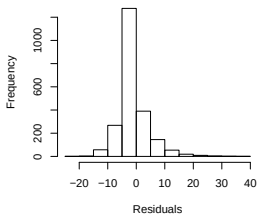
Normal Q-Q Plot



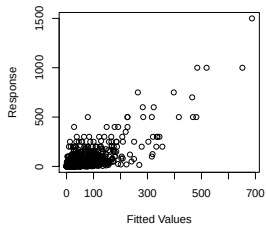
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



Negative binomial, cycle basis for WIND, without outlier

```
> gam.check(GrAFitnegbin.wo)
Method: REML    Optimizer: outer newton
full convergence after 6 iterations.
Gradient range [-0.002532445,0.001774942]
(score 7125.691 & scale 1).
Hessian positive definite , eigenvalue range [0.002520513,740.6654].
Model rank = 153 / 153
```

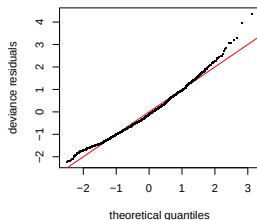
Basis dimension (k) checking results. Low p-value ($k\text{-index} < 1$) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(YEAR)	19.000	14.481	0.356	0.00
s(DAY)	19.000	6.014	0.620	0.00
s(NO.OBS)	7.000	2.457	0.404	0.00
s(WIND.0)	18.000	3.852	0.703	0.00
s(WIND.1)	18.000	2.709	0.734	0.00
s(WIND.2)	18.000	2.182	0.689	0.00
s(WIND.3)	18.000	2.225	0.729	0.00
s(WINDSPD)	19.000	2.202	0.698	0.00
s(MOON)	9.000	1.007	0.778	0.34

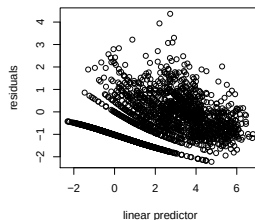
Generalized Linear and Additive Models

└ Generalized additive models (GAM)

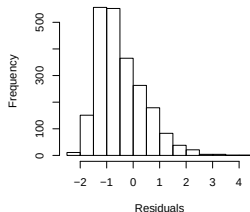
└ Analysis of the birds dataset



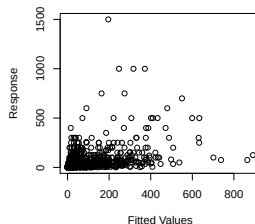
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



```
> summary(GrAFitnegbin.wo)
```

Family: Negative Binomial(0.444)

Link function: log

Formula:

```
NUMBER ~ s(YEAR, k = 20) + s(DAY, k = 20) + s(NO.OBS, k = 8) +  
  s(WIND.0, k = 20, bs = "cc") + s(WIND.1, k = 20, bs = "cc") +  
  s(WIND.2, k = 20, bs = "cc") + s(WIND.3, k = 20, bs = "cc") +  
  s(WINDSPD, k = 20) + factor(SKY) + s(MOON) + factor(BIRD)
```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.15046	0.08405	37.484	<2e-16	***
factor(SKY)cloudy	-0.06807	0.08731	-0.780	0.4356	
factor(SKY)fog	-0.19088	0.11097	-1.720	0.0854	.
factor(SKY)rain	-0.12774	0.13333	-0.958	0.3380	
factor(BIRD)2	-2.42337	0.10868	-22.298	<2e-16	***
factor(BIRD)3	-2.98799	0.11281	-26.487	<2e-16	***
factor(BIRD)4	-1.03015	0.10348	-9.955	<2e-16	***
factor(BIRD)5	1.52882	0.10154	15.057	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1

Approximate significance of smooth terms:

	edf	Ref. df	Chi. sq	p-value	
s(YEAR)	14.481	16.591	188.028	< 2e-16	***
s(DAY)	6.014	7.448	239.399	< 2e-16	***
s(NO.OBS)	2.457	3.033	28.412	3.30e-06	***
s(WIND.0)	3.852	18.000	75.064	< 2e-16	***
s(WIND.1)	2.709	18.000	16.754	8.48e-05	***
s(WIND.2)	2.182	18.000	9.214	0.00399	**
s(WIND.3)	2.225	18.000	9.895	0.00296	**
s(WINDSPD)	2.202	2.796	10.133	0.01661	*
s(MOON)	1.007	1.013	0.067	0.79966	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

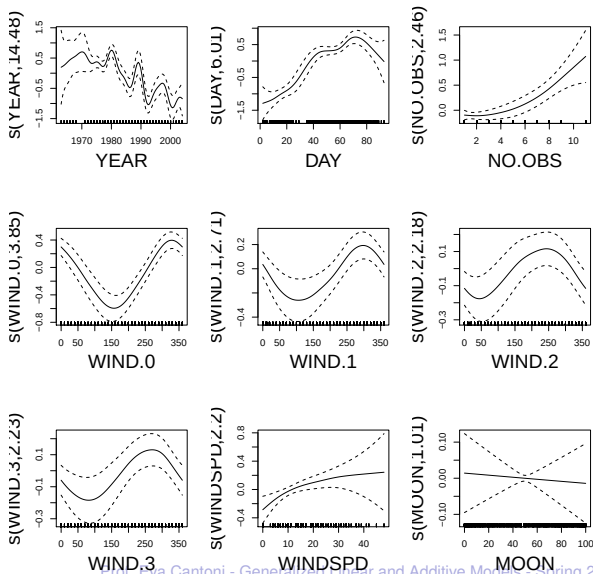
R-sq.(adj) = -0.121 Deviance explained = 52.2%

-REML = 7125.7 Scale est. = 1 n = 2229

Generalized Linear and Additive Models

Generalized additive models (GAM)

Analysis of the birds dataset



- └ Generalized additive models (GAM)
- └ Analysis of the birds dataset

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- └ Generalized additive models (GAM)
- └ Analysis of the birds dataset

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